
Structural Analysis and Design Report

Project Name- LA MARINA

Project:	LA MARINA	
Address:	2211 North Bay Road Miami Beach FL 33140	
Project #:		
Client:		
Issue Date	22 nd SEP 2025	
Revision	P00	Date: 22 nd SEP 2025

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Client:					
Calc.By: SS	Date:22 nd SEP 25	Chk'd By:PD	Date:22 nd SEP 25	Rev: P0	Date:22 nd SEP 25

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1. FORWARD

The intent of this document is to identify and record all the pertinent input requirement, analysis and design criteria for an efficient structural design of the building / project. It is aimed at formulating the basic of the structural analysis, design and detailing work that the Structural Engineer is planning in delivering the structural scheme of the building. The scheme will be compatible with the architectural theme, satisfy the functional /operation's needs, at the same time conforming to the Standards and other applicable and prevalent building norms to achieve safe, stable, strong, and yet optimally economical structures.

The parameters adopted in this report are going to be the basic of the structural design calculations. Hence it is expected that the planning and design team members- architects, services consultants and project management team- give their feedback and approval to the parameters, suggestion, recommendations mentioned in the report.

This report covers the minimum design requirement to establish the unified design basis that will form the overall design philosophy to be adopted in the structural design of the proposed building.

The engineering calculations and structural drawings produced have been based on the drawings provided by the Architecture model file and where information and details are missing Request for Information and questions will be issued.

The design will aim to achieve.

- Structural and functional integrity.
- Desirable Structural performance under characteristic services design loads.
- Resistance to the loads due to natural phenomena i.e., wind, earthquakes, snow.
- Structural durability and maintainability.
- Structural Safety, Performance during Fire and Fire safety measures.
- Compatible with all relevant codes of practice

2. PROJECT DETAIL

The project is LA MARINA. The structure proposed to be design in Light gauge steel.Our scope is to design and detail the light gauge steel.

As per the scope below,

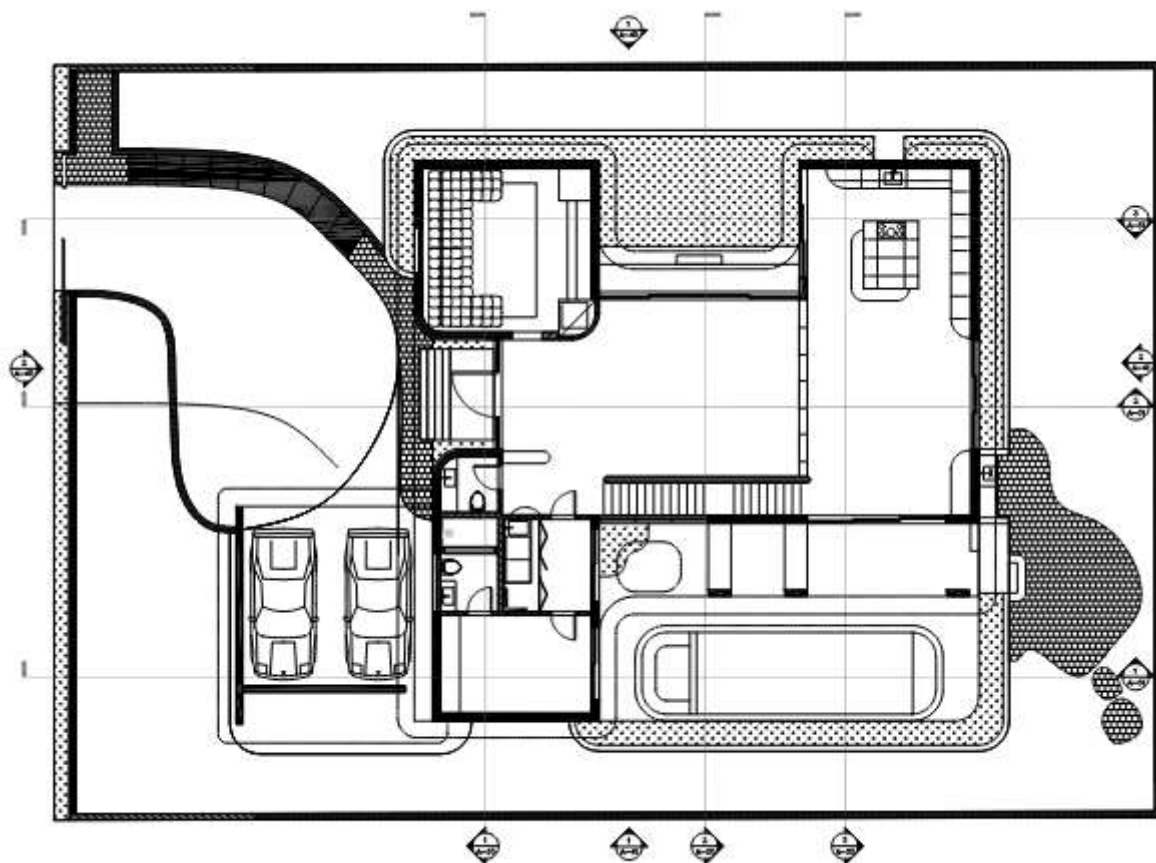
- Light gauge steel Engineering Calculation Report.
- Light gauge steel Structural Drawings and Details.

The Calculation and drawings have been developed based on the architectural and structural drawings and information given. Light gauge steel design is based on the AISI specification.

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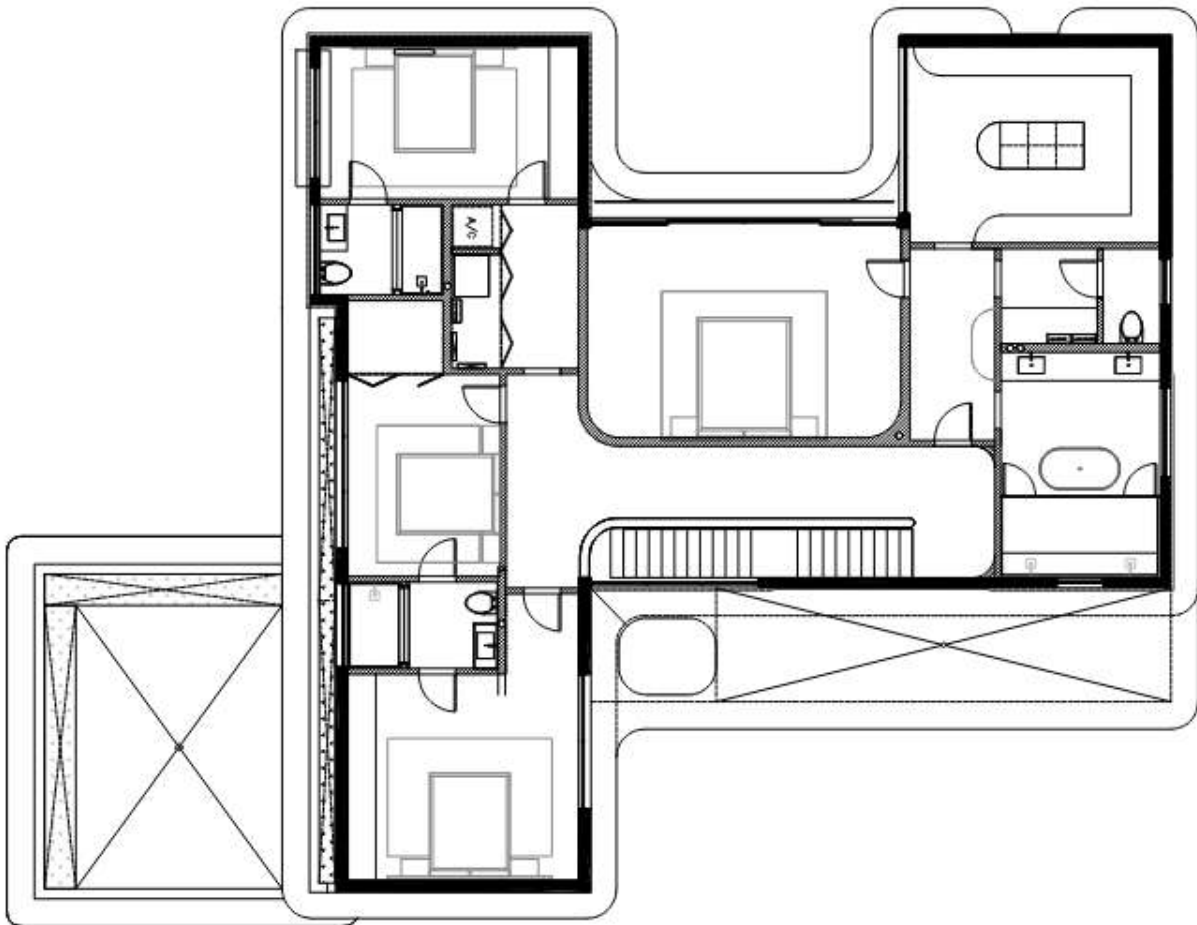
3. DETAIL DESCRIPTION OF BUILDING

Ground Floor Plan(Architectural Plan) below:



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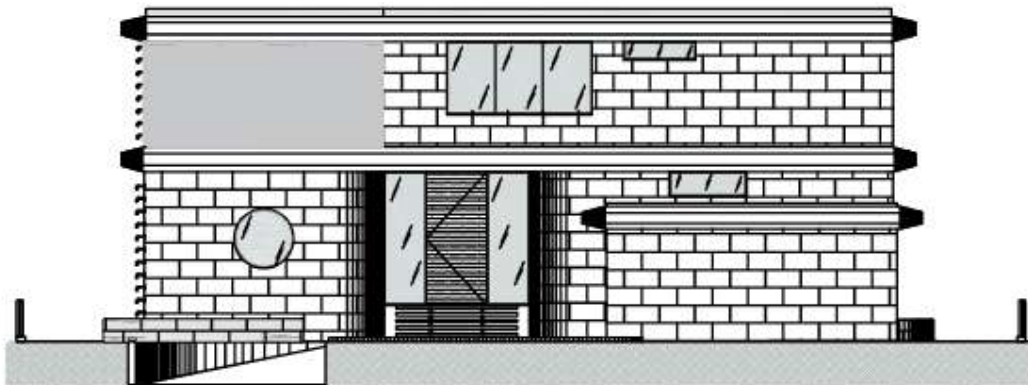
Second Floor Plan(Architectural Plan) below:



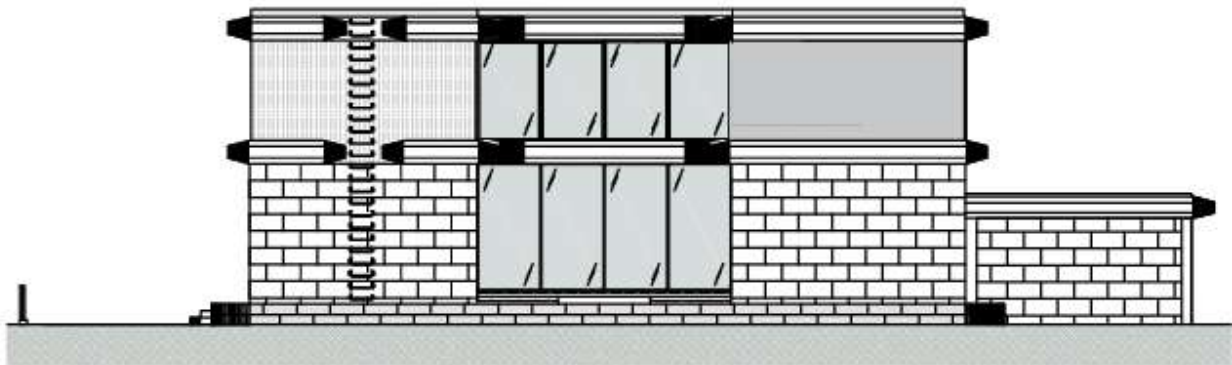
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BUILDING ELEVATION:

West Elevation

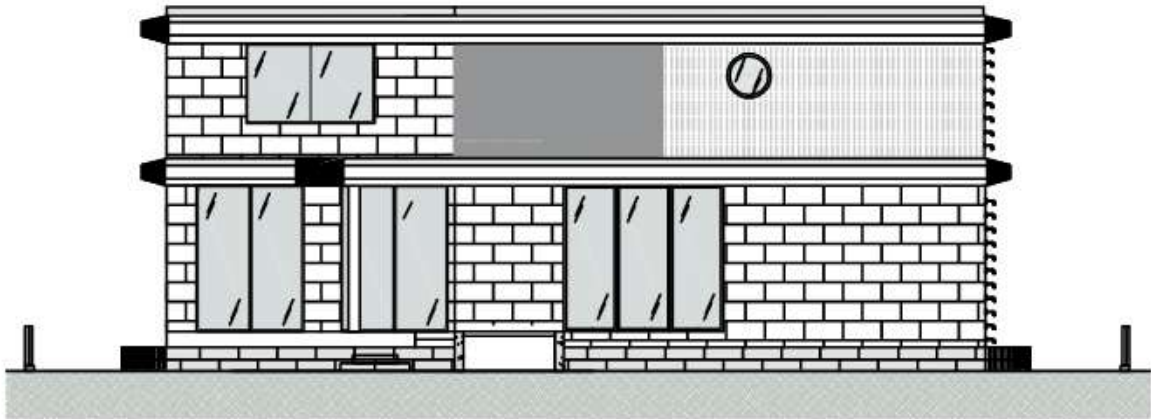


North Elevation

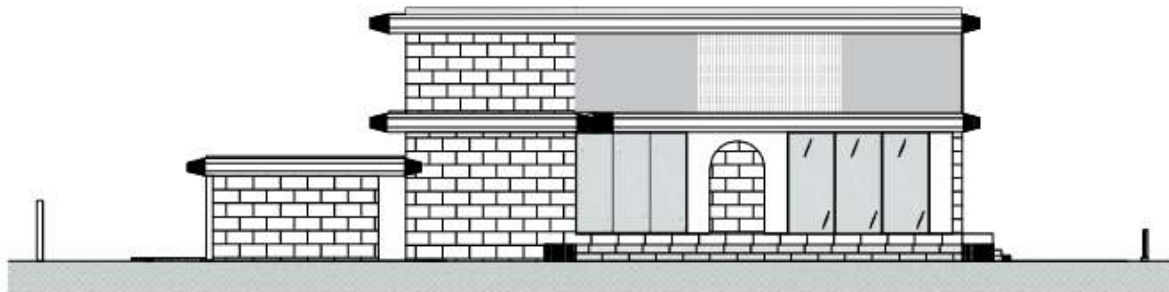


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East Elevation



Right side Elevation



Building Floor Height:

Ground Floor to Second floor	-	9'-0"
Second Floor to Roof	-	11'-0"

The heights are taken from architectural drawings

Opening schedule: The opening schedule followed as per architectural drawings

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4. LAWS AND REGULATIONS

FBC 2023	-	Florida Building Code
IBC 2021	-	International Building Code
ASCE 7-22	-	American Society of civil Engineers (Minimum Design loads for buildings & other structures)
AISC 360-21	-	American Institute of steel construction (Specification for structural steel buildings)
AISI - 2020	-	American Iron and Steel Institute (Design of Cold Formed steel & structural members)
ACI 318-19	-	American Concrete Institute (Building code requirements for structural concrete)
ASTM	-	American Society for Testing and materials.

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5. STRUCTURE LOADING CALCULATION

Floor Dead Loads:

Lewis deck 2'' gypcrete concrete	=	20 psf
1/2'' Gypsum board	=	2 psf
LGS self weight	=	1.41 psf
Misc.	=	1.59 psf
Total dead load	=	25 psf

Roof Dead Loads:

Lewis deck 2'' gypcrete concrete	=	20 psf
1/2'' Gypsum board	=	2 psf
LGS self weight	=	1.41 psf
Solar weight	=	5 psf
Misc.	=	1.59 psf
Total dead load	=	30 psf

External Wall:

3/4" Coral stone tiles	=	10 psf
Skim coat level 5plaster finish	=	2.0 psf
LGS self weight	=	1.41 psf
5/8'' Drywall panel	=	2.5 psf
6'' Rockwool conforbatt	=	2.25 psf
3/4''' Layer of: self-furring metal lath + type S scratch coat + type S setting Mortar	=	5 psf
3/8" Drainage mat	=	0.25 psf
2" OX-IS structural insulated panel	=	2.5 psf
Misc.	=	4.09 psf
Total dead load	=	30 psf

Internal wall:

5/8'' USG Sheetrock panel	=	1.8 psf
LGS self weight	=	1.41 psf
5/8'' USG Sheetrock panel	=	1.8 psf
1/8'' Skim Coat Plaster finish	=	0.75 psf
1/8'' Skim Coat Plaster finish	=	0.75 psf
Misc.	=	5.49 psf
Total dead load	=	12 psf

Live Load:

The following live loads are referred to all buildings (where applicable):

Live Load Roof	=	20 psf (ASCE 7-16 Table 4.3-1)
Live Load Floor	=	40 psf (ASCE 7-16 Table 4.3-1)

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Wind Load:

Building data

Type of roof - Flat
Length of building; b = 62 ft
Width of building; d = 61 ft

General wind load requirements

Basic wind speed;	V = 175 mph
Risk category;	II
Velocity pressure exponent coef (Table 26.6-1);	Kd = 0.85
Exposure category (cl 26.7.3);	D
Enclosure classification (cl.26.12);	Enclosed buildings
Internal pressure coef +ve (Table 26.13-1);	GCpi_p = 0.18
Internal pressure coef -ve (Table 26.13-1);	GCpi_n = -0.18

Component and Cladding design analysis:

Wall Design Wind Pressure Interior	-	114.1 psf
Wall Design Wind Pressure Corner	-	134.1 psf

MWFRS design analysis:

Lateral Design Wind pressure	-	98.97 psf
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For Detail information see Wind Load appendix

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Earthquake Load:

The following seismic load design parameters have been utilized:

Analysis Procedure -Equivalent Static Method:

Site parameters

Site class; D, Soil properties not known

Mapped acceleration parameters (Section 11.4.2)

at short period;

at 1 sec period;

Site coefficient at short period (Table 11.4-1);

at 1 sec period (Table 11.4-2);

Spectral response acceleration parameters

at short period (Eq. 11.4-1);

at 1 sec period (Eq. 11.4-2);

$$S_s = 0.048$$

$$S_1 = 0.021$$

$$F_a = 1.600$$

$$F_v = 2.400$$

$$SMS = F_a \times S_s = 0.154$$

$$SM_1 = F_v \times S_1 = 0.096$$

Design spectral acceleration parameters (Sect 11.4.4)

at short period (Eq. 11.4-3);

at 1 sec period (Eq. 11.4-4);

$$SDS = 2/3 \times SMS = 0.102$$

$$SD_1 = 2/3 \times SM_1 = 0.064$$

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6. LOADS COMBINATION

Load combinations using strength design or load and resistance factor design. Where strength design or load and resistance factor design is used, buildings and other structures, and portions there of, shall be designed to resist the most critical effects resulting from the following combinations of factored loads:

According to FBC-2023

1.4D

1.2D+1.6L+(0.5L_r or 0.3S or 0.5R)

1.2D + (1.6L_r or 1.0S or 1.6R) + (L or 0.5W)

1.2D + 1.0(W or W_T)+ L + (0.5L_r or 0.3S or 0.5R)

0.9D+1.0W

0.9D+1.0E

Notations

D = Dead load.

L = Live load, except roof live load, including any permitted live load reduction.

L_r = Roof live load including any permitted live load reduction.

R = Rain load.

S = Snow load.

W = Load due to wind pressure.

7. SERVICIALBITY DEFLECTION LIMIT

For Wall

Total Load = TL/240

Live Load = L/360

Roof = L/480

= TL/360

Floor = L/360

= TL/240

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8. STRUCTURAL MATERIAL

Light Gauge steel Section:

All Material manufactured to meet the requirements of S136-07 and S136S2-10 (Supplements 2) with minimum Yield strength: 33 ksi (for design thicknesses less than or equal 43 mil) and minimum Yield strength: 50 ksi (for design thicknesses greater than or equal 43 mil).

Material Thicknesses is given as the minimum thickness exclusive of coatings and represent 95% of the design thickness, see S136-07 Section A2.4.

Structural Steel:

The structural steel design as the following material

Rolled W and WT shapes : ASTM A992, Grade 50 ksi

Rolled M, S, C and MC shapes and Angle: ASTM A36, FY = 36 ksi

Plated and Bars : ASTM A36, FY = 36 ksi

Round Section : ASTM A 500, Grade C, FY – 46 ksi

Square and Rectangular sections : ASTM A500, Grade B, FY – 46 ksi

Steel Pipes : ASTM A53, Type E or S, Grade B, FY – 35 ksi

Connectors:

All the connectors and fixing used are Simpson as per the design requirement and its availability.

9. PROPOSED APPROACH TO STRUCTURAL ANALYSIS & DESIGNS

These studs are provided at the load-bearing walls stud shall be designed as compression members by 'elastic theory of design'. These studs are for transferring of lateral forces from X strap bracing. These studs are of light gauge structural steel.

The floor and roof have design considering the dead and live load coming on to it and checking flexure, shear and deflection for the governing load.

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10. ROOF JOIST DESIGN

10.1 Roof Joist design (RJ 1):

Roof span = 19.5 ft
 Roof Dead load = 30 psf
 Roof Live load = 20 psf
 Joist Spacing = 24" O.C

Roof Truss Specification :- Using Roof joist 1000S200-68 (50) @ 24" O.C.

For Additional information refer Appendix 10.1 Roof Joist design.

10.2 Roof Joist design (RJ 2):

Roof span = 22.67 ft
 Roof Dead load = 30 psf
 Roof Live load = 20 psf
 Joist Spacing = 24" O.C

Roof Truss Specification :- Using Roof joist BTB1000S200-54 (50) @ 24" O.C.

For Additional information refer Appendix 10.2 Roof joist design.

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11. FLOOR JOIST DESIGN

11.1 Floor Joist design (FJ 1):

Floor span = 18.67 ft
 Floor Dead load = 25 psf
 Floor Live load = 40 psf
 Joist Spacing = 24" O.C

Floor Joist Specification :- Using Floor joist 1000S200-68(50)@24 O.C

For Additional information refer Appendix 11.1 Floor Joist design.

11.2 Floor Joists design (FJ 2):

Floor span = 8 ft
 Floor Dead load = 25 psf
 Floor Live load = 40 psf
 Joist Spacing = 24" O.C

Floor Joist Specification :- Using Floor joist 1000S200-43(50) @ 24" O.C.

For Additional information refer Appendix 11.2 Floor Joist design.

11.3 Floor Joist design (FJ 3):

Floor span = 22.5 ft
 Floor Dead load = 25 psf
 Floor Live load = 40 psf
 Joist Spacing = 24" O.C

Floor Joist Specification :- Using Floor joist BTB 1000S200-54 (50) @ 24" O.C.

For Additional information refer Appendix 11.3 Floor Joist design.

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12. DESIGN OF WALL STUD

12.1 External Wall Design Calculation GRID-5 (Second floor)

Roof Dead loads, Live loads are transferred from truss to walls

Roof span 1	:	20 feet
Roof span 2	:	2 feet
Roof Dead Load	:	30 psf
Roof Live Load	:	20 psf
Wall height	:	9 ft
Wall Dead load	:	30 psf
Wind Load	:	114 psf

$$\begin{aligned} \text{Max DL at wall} &= (20/2+2) \times 30 + (9 \times 30) \\ &= 630 \text{ plf} \end{aligned}$$

$$\begin{aligned} \text{Load per stud} &= 630 \times 2 \\ &= 1260 \text{ lb} \end{aligned}$$

$$\begin{aligned} \text{Max LL at wall} &= (20/2+2) \times 20 \\ &= 240 \text{ plf} \end{aligned}$$

$$\begin{aligned} \text{Load per stud} &= 240 \times 2 \\ &= 480 \text{ lb} \end{aligned}$$

Wall stud specification:- Using Wall stud of BTB 600S162-43 (50) @24" O.C. mid point noggin

For additional information see Appendix 12.1 External Wall Design Calculation.

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12.2 External Wall Design Calculation GRID-5 (Ground floor)

Roof Dead loads, Live loads are transferred from truss to walls

Floor span 1	:	20 feet
Floor span 2	:	2 feet
Floor Dead Load	:	25 psf
Floor Live Load	:	40 psf
Wall height	:	11 ft
Wall Dead load	:	30 psf
Wind Load	:	114 psf

Load from above,		
Dead load	=	1260 lb
Roof Live load	=	480 lb

Max DL at wall	=	$(20/2+2) \times 25 + (11 \times 30)$
	=	630 plf

Load per stud	=	630x2
	=	1260 lb

Max LL at wall	=	$(20/2+2) \times 40$
	=	480 plf

Load per stud	=	480x2
	=	960 lb

Wall stud specification:- Using Wall stud of BTB 600S162-54 (50) @24" O.C. with third point noggin.

For additional information see Appendix 12.2 External Wall Design Calculation.

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12.3 External Wall Design Calculation GRID-A (Second floor)

Roof Dead loads, Live loads are transferred from Joist to walls

Roof span 1	:	2 ft
Cant. Span	:	2 ft
Roof Dead Load	:	30 psf
Roof Live Load	:	20 psf
Wall height	:	9 ft
Wall Dead load	:	30 psf
Wind Load	:	114 psf

Max DL at wall	=	$(2/2+2) \times 30 + (9 \times 30)$
	=	360 plf

Load per stud	=	360×2
	=	720 lb

Max LL at wall	=	$(2/2+2) \times 20$
	=	60 plf

Load per stud	=	60×2
	=	120 lb

Wall stud specification:- Using Wall stud of 600S162-54 @24" O.C. with midpoint noggin.

For additional information see Appendix 12.3 Internal Wall Design Calculation.

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12.4 External Wall Design Calculation GRID-A (Ground floor)

Roof Dead loads, Live loads are transferred from Joist to walls

Floor span 1	:	2 ft
Cant. Span	:	2 ft
Floor Dead Load	:	25 psf
Floor Live Load	:	40 psf
Wall height	:	11 ft
Wall Dead load	:	30 psf
Wind Load	:	114 psf

Load from above,		
Dead load	=	720 lb
Roof Live load	=	120 lb

Max DL at wall	=	$(2/2+2) \times 25 + (11 \times 30)$
	=	405 plf

Load per stud	=	405x2
	=	810 lb

Max LL at wall	=	$(2/2+2) \times 40$
	=	120 plf

Load per stud	=	120x2
	=	240 lb

Wall stud specification:- Using Wall stud of BTB 600S162-43 @24" O.C. with midpoint noggin.

For additional information see Appendix 12.4 Internal Wall Design Calculation.

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12.5 Internal Wall Design Calculation GRID-F (Second floor)

Roof Dead loads, Live loads are transferred from Joist to walls

Roof span 1	:	2 ft
Cant. Span	:	2 ft
Roof Dead Load	:	30 psf
Roof Live Load	:	20 psf
Wall height	:	9 ft
Wall Dead load	:	12 psf

$$\begin{aligned} \text{Max DL at wall} &= (2/2+2) \times 30 + (9 \times 12) \\ &= 198 \text{ plf} \end{aligned}$$

$$\begin{aligned} \text{Max LL at wall} &= (2/2+2) \times 20 \\ &= 60 \text{ plf} \end{aligned}$$

$$\begin{aligned} \text{Critical Load combination} &= 1.2 \times 198 + 1.6 \times 60 \\ &= 333.6 \text{ plf} \end{aligned}$$

$$\begin{aligned} \text{Load per stud} &= \text{Load} \times \text{Stud spacing} \\ &= 333.6 \times 2 \\ &= 668 \text{ lb} \end{aligned}$$

$$\text{Allowable Load per stud} = 3417 \text{ lb @ 24" spacing} > 668 \text{ lb}$$

Wall stud specification:- Using Wall stud of 362S162-43 @24" O.C. with midpoint noggin.

For additional information see Appendix 12.5 Internal Wall Design Calculation.

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12.6 Internal Wall Design Calculation GRID-F (Ground floor)

Roof Dead loads, Live loads are transferred from Joist to walls

Floor span 1	:	2 ft
Cant. Span	:	2 ft
Floor Dead Load	:	25 psf
Floor Live Load	:	40 psf
Wall height	:	10 ft
Wall Dead load	:	30 psf
Wind Load	:	63.1 psf

Load from above,		
Dead load	=	198 plf
Roof Live load	=	60 plf

Max DL at wall	=	$(2/2+2) \times 25 + (10 \times 12)$
	=	195 plf

Max LL at wall	=	$(2/2+2) \times 40$
	=	120 plf

Critical Load combination	=	$1.2 \times 393 + 1.6 \times 465$
	=	1215 plf

Load per stud	=	Load x Stud spacing
	=	1215×2
	=	2430 lb

Allowable Load per stud	=	$3664 \text{ lb @ } 24'' \text{ spacing}$	>	2430 lb
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Wall stud specification:- Using Wall stud of 362S162-43 @24" O.C. with midpoint noggin.

For additional information see Appendix 12.6 Internal Wall Design Calculation.

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13. DESIGN OF RCC

13.1 Design of RCC wall

Height of wall : 10 ft
 Tributary Span : 3.5ft
 Self weight of wall : 55 psf
 Maximum Axial : 3.4 kips/ft
 Maximum Moment : 5.1 kip_ft/ft
 Maximum Shear : 3.4 kips/ft

Wall Specification:- Provide 6" concrete wall with #4 vertical rebar @12" o.c.

For additional information refer below attached appendix.

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13.2 Design of RCC Staircase Tread

Span length (L) : 3.5 ft
 Dead Load : 55 psf
 Live Load : 60 psf
 Concrete compressive Strength (fc') : 4000 psi
 Yield strength of reinforcement (fy) : 60000 psi

Depth check :-

Minimum depth required = 42/12
 = 3.5"

Hence provide 4" Tread for staircase,

Total Load (w) = $1.2 \times 55 + 1.6 \times 60$ = 162 psf

Maximum Moment (Mu) = $w \times L^2 / 2 = 162 \times (3.5)^2 / 2 = 992$ lb-ft/ft

Reinforcement calculation,

$A_s = Mu / (f_y \times d) = 992 \times 12 / (60000 \times 11) = 0.018$ in²

Deflection check,

Maximum allowable deflection = $L / 240 = 3.5 \times 12 / 240 = 0.175$ in

$E = 3600$ and $I = (1 \times h^3) / 12$

Deflection = $(162 \times (3.5 \times 12)^4) / (8 \times 3600 \times 10^3 \times (1 \times (4)^3) / 12)$
 = 0.022 in

0.022 in < 0.175 in

Hence Ok.

Provide minimum #4 rebar @ 6" O.C per ft

Area of steel provided = 0.2 in² > 0.018 in² Hence OK.

Provide staircase tread of 4" thick of concrete strength minimum 4000 psi with #4 rebar @6" O.C.

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14. DESIGN OF WEBBED BEAM & HEADER

14.1 Webbed Beam Design at Roof (WB1)-GRID-4

Beam span : 13 ft
 Roof span : 20 feet
 Cant. Span : 2 ft
 Roof Dead Load : 30 psf
 Roof Live Load : 20 psf

Max Dead Load = $(20/2+2) \times 30$
 = 360 plf

Max Roof live Load = $(20/2) \times 20$
 = 240 plf

End reaction on beam,
 = 4800 lb

Using LGS post min (2) 600S162-54(50) to support the LGS webbed beam.

Capacity of LGS post = 12545 lb > 4800 lb

Beam Specification: Using 1'-9 5/8" deep webbed beam 600S162-54 (50) with reinforced with L -plate 1-1/2" x 8" -54 mil at top and bottom chord both side with min (2) 600S162-54 (50). Refer "13.1 webbed beam (WB1) Design report" for more information.

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14.2 Webbed Beam Design at Floor (WB 2)-GRID-6

Beam span : 22.5 ft
 Floor span : 2 ft
 Cant. Span : 2 ft
 Floor Dead Load : 25 psf
 Floor Live Load : 40 psf

Max Dead Load = $(2/2+2) \times 25$
 = 75 plf

Max Floor live Load = $(2/2+2) \times 40$
 = 120 plf

End reaction on beam,
 = 3280 lb

Using LGS post min (2) 600S162-54(50) to support the LGS webbed beam.

Capacity of LGS post = 12545 lb > 3280 lb

Beam Specification: Using 1'-9 5/8" deep webbed beam 600S162-43(33) with reinforced with L-plate 1-1/2" x 8" -54 mil at top and bottom chord both side with min (2)600S162-54(50). Refer "13.2 webbed beam (WB2) Design report" for more information.

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14.3 Webbed Beam Design at canopy (WB3)

Beam span : 19 ft
 Floor span : 2 ft
 Floor. Span : 8.0 ft
 Floor Dead Load : 25 psf
 Floor Live Load : 40 psf

Max Dead Load = $(8.0/2+2) \times 25$
 = 150 plf

Max Floor live Load = $(8.0/2+2) \times 20$
 = 120 plf

End reaction on beam,
 = 3630 lb

Using LGS post min (2) 600S162-54(50) to support the LGS webbed beam.

Capacity of LGS post = 12545 lb > 3630 lb

Beam Specification: Using 1'-4" deep webbed beam 600S162-54(50) with reinforced with L - plate 1-1/2" x 8" -54 mil at top and bottom chord both side with min (2)600S162-54(50). Refer "13.3 webbed beam (WB3) Design report" for more information.

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14.4 Webbed Beam Design at floor (WB1)-GRID 3

Beam span : 8 ft
 Floor span : 17 ft
 Cant. Span : 2 ft
 Floor Dead Load : 25 psf
 Floor Live Load : 40 psf

Max Dead Load = $(17/2+2) \times 25$
 = 263 plf

Max Floor live Load = $(17/2+2) \times 40$
 = 420 plf

Point load from above roof jamb post
 Dead Load = 1446 lb
 Roof live Load = 1157 lb

End reaction on beam,
 = 5730 lb

Using LGS post min (2) 600S162-54(50) to support the LGS webbed beam.

Capacity of LGS post = 12545 lb > 5730 lb

Beam Specification: Using 1'-9 5/8" ' deep webbed beam 600S162-54(50) with reinforced with L -plate 1-1/2" x 8" -54 mil at top and bottom chord both side with min (2)600S162-54(50). Refer "13.4 webbed beam (WB1) Design report" for more information.

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14.5 Webbed Beam Design at floor (WB1)-GRID 4

Beam span : 12 ft
 Floor span : 18 ft
 Cant. Span : 2 ft
 Floor Dead Load : 25 psf
 Floor Live Load : 40 psf

Max Dead Load = $(18/2+2) \times 25$
 = 275 plf

Max Floor live Load = $(18/2+2) \times 40$
 = 440 plf

Point load from above roof jamb post
 Dead Load = 1013 lb
 Roof live Load = 810 lb

End reaction on beam,
 = 7470 lb

Using LGS post min (2) 600S162-54(50) to support the LGS webbed beam.

Capacity of LGS post = 12545 lb > 7470 lb

Beam Specification: Using 1'-9 5/8" ' deep webbed beam 600S162-54(50) with reinforced with L -plate 1-1/2" x 8" -54 mil at top and bottom chord both side with min (2)600S162-54(50). Refer "13.5 webbed beam (WB1) Design report" for more information.

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14.6 Design of internal Opening Header/Jamb (H1)

Header Span : 3 ft
 Roof span 1 : 2 ft
 Roof span 2 : 2 ft
 Roof Dead Load : 30 psf
 Roof Live Load : 20 psf
 Wall height above opening : 3 ft
 Wall Dead load : 12 psf

Max DL Load at wall = $(2/2+2/2) \times 30 + (3 \times 12)$
 = 96 plf

Max LL Load at wall = $(2/2+2/2) \times 20$
 = 40 plf

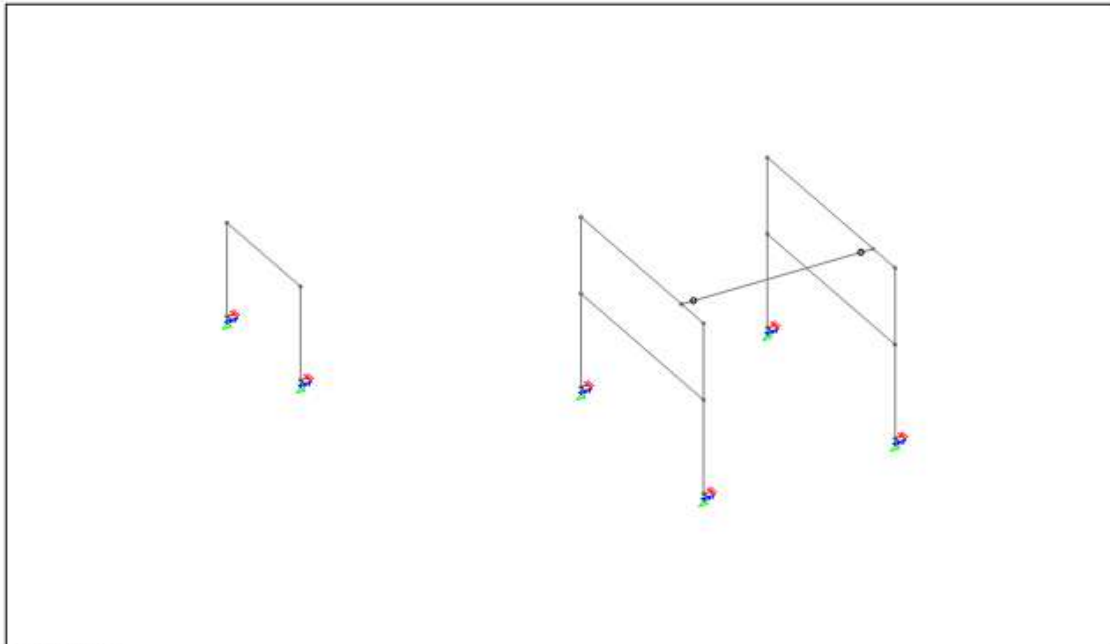
Reaction on jamb post = 348 lb

Header specification:-Use webbed header 12" deep 600S162-43 mil and for jamb post use use min 1 jack & 1 king 600S162-33 mil Studs.

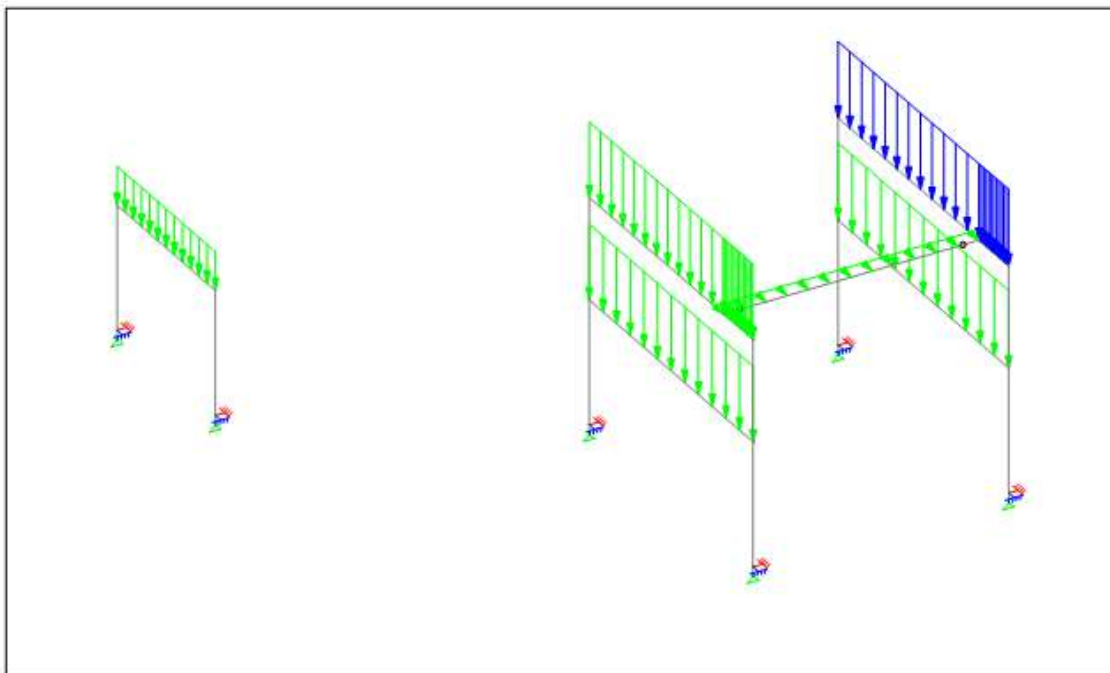
For additional information see Appendix 14.6 Design of Internal opening header and Jamb (H1).

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15. HRS FRAME DESIGN

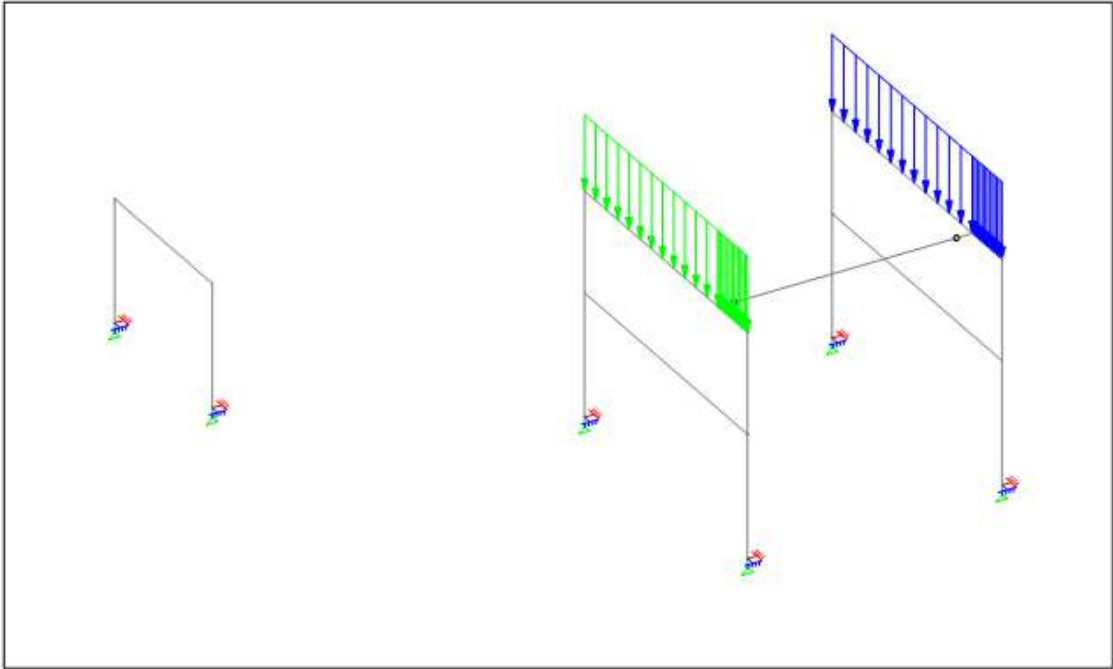


GEOMETRY

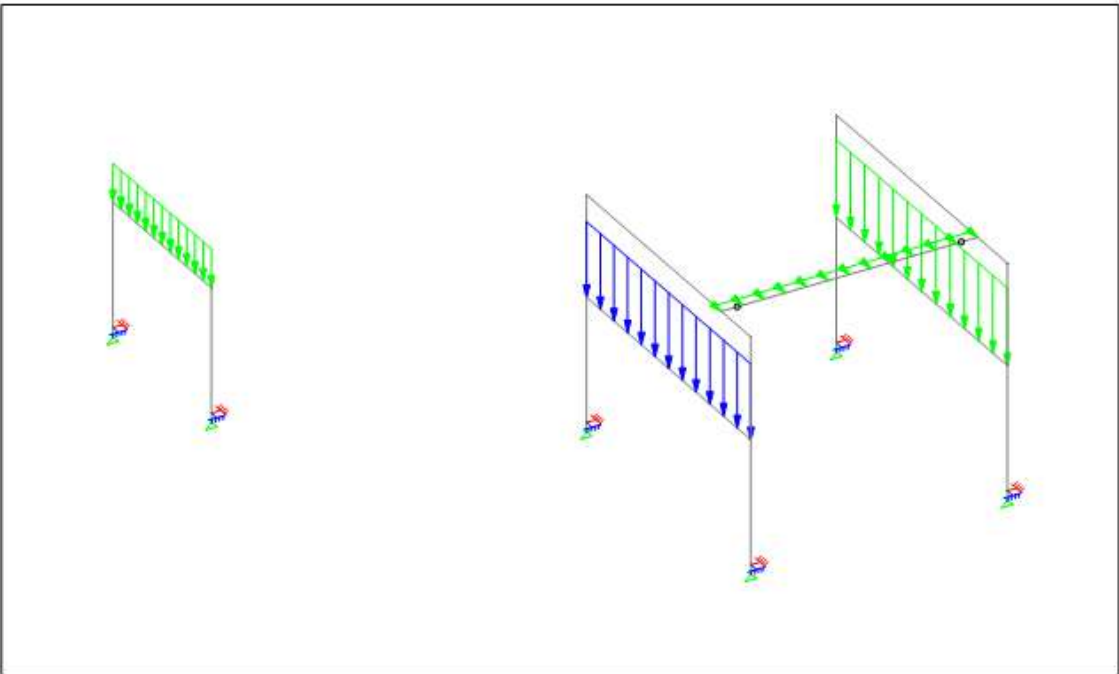


DEAD LOAD

Project: LA MARINA				JOB No:	
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Calc.By: SS	Date:22 nd SEP 25	Chk'd By:PD	Date:22 nd SEP 25	Rev: P0	Date:22 nd SEP 25

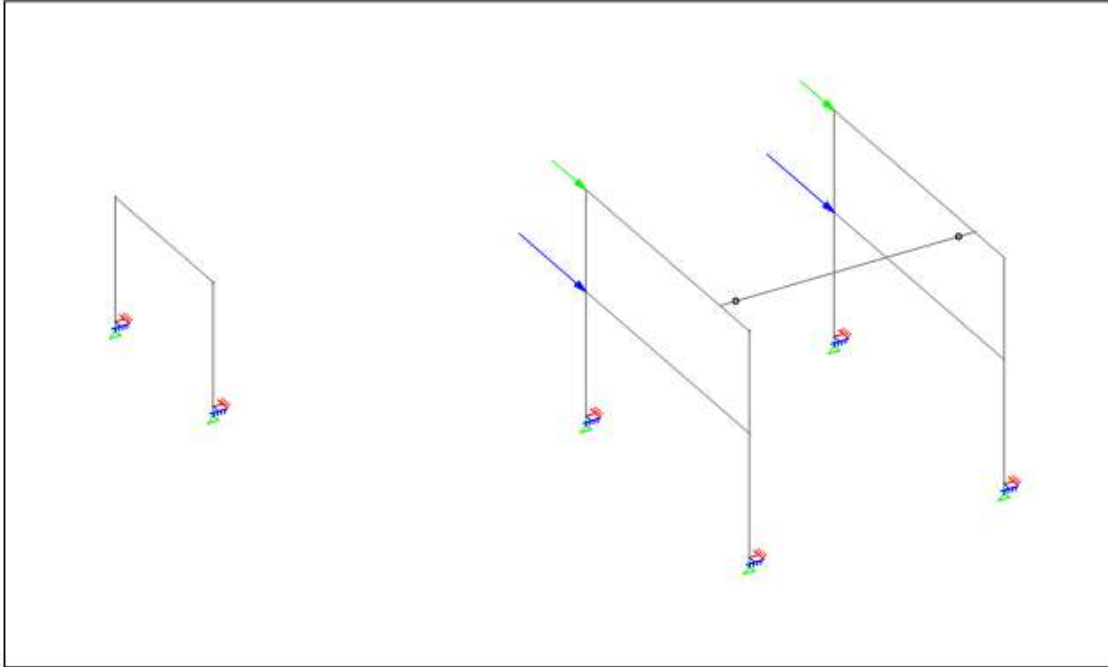


ROOF LIVE LOAD

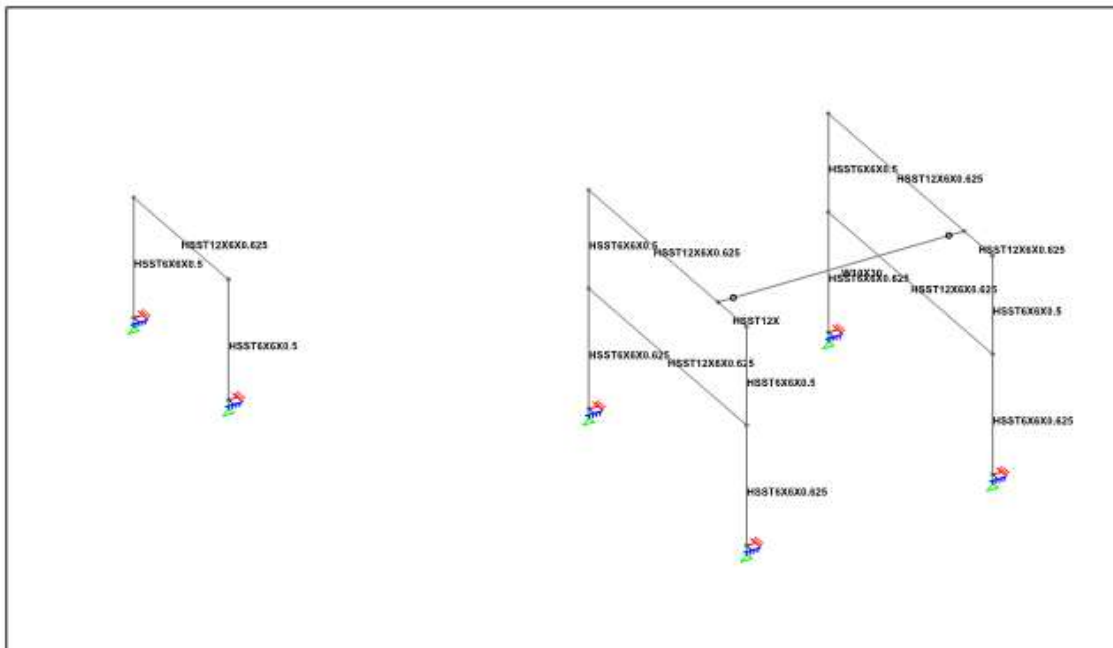


FLOOR LIVE LOAD

Project: LA MARINA				JOB No:	
Client:					
Calc.By: SS	Date:22 nd SEP 25	Chk'd By:PD	Date:22 nd SEP 25	Rev: P0	Date:22 nd SEP 25

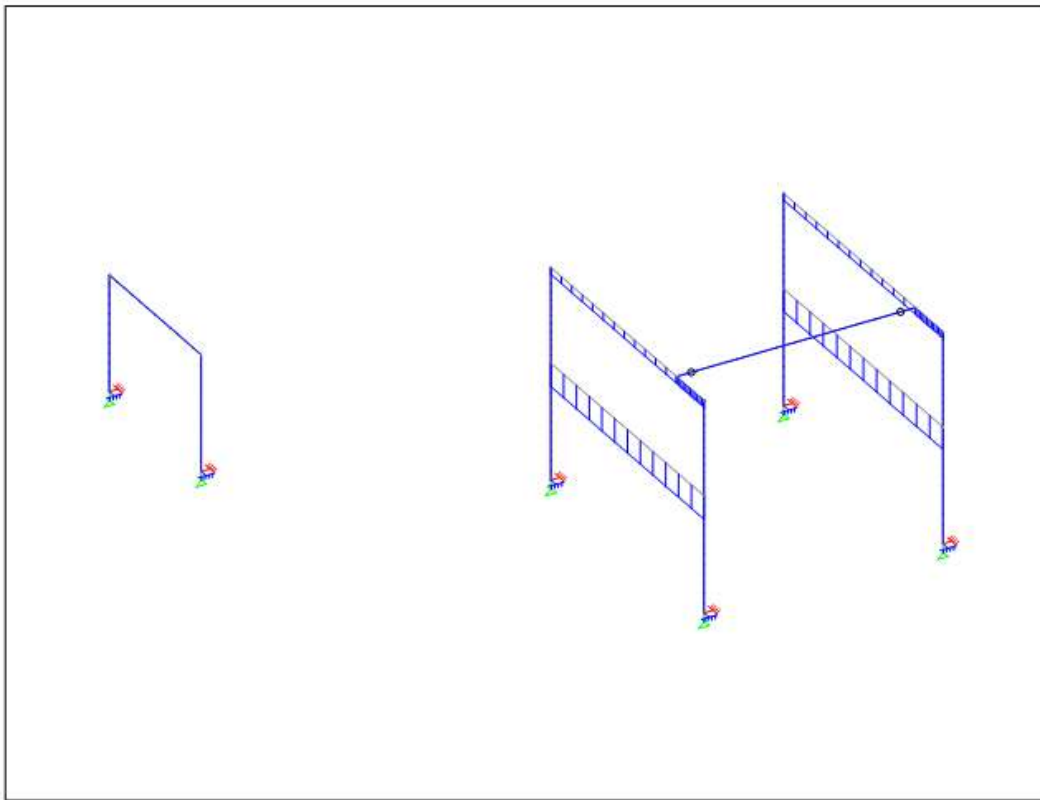


WIND LOAD

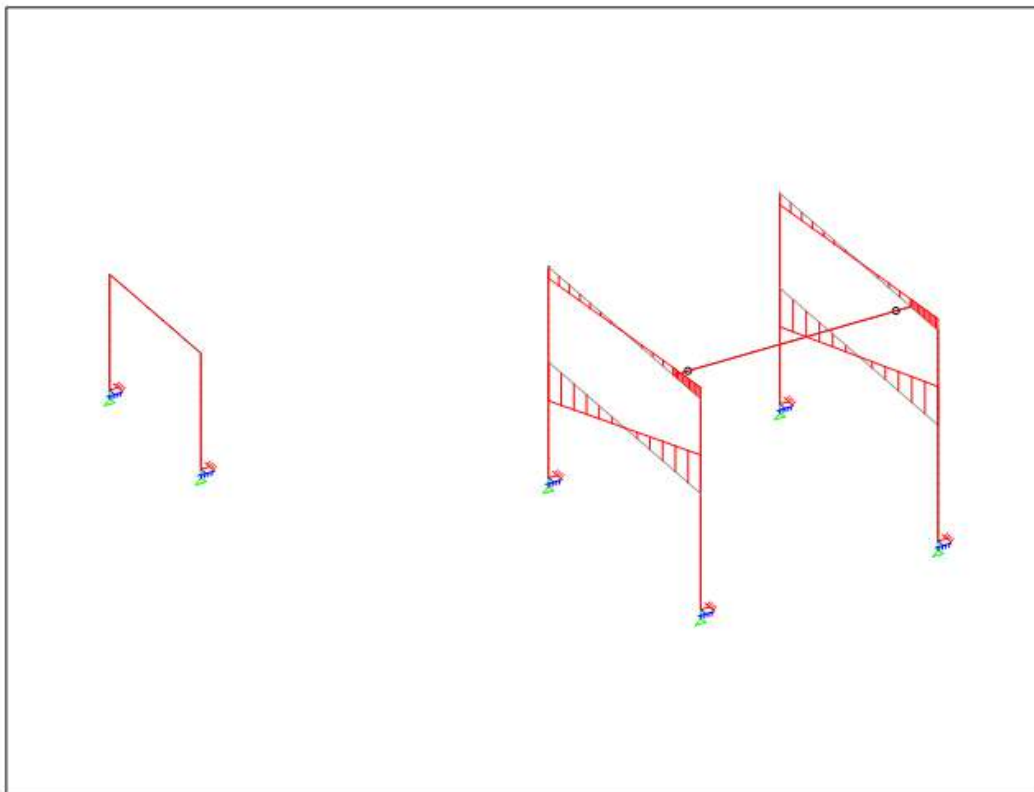


MEM - PROPERTY

Project: LA MARINA				JOB No:	
Client:					
Calc.By: SS	Date:22 nd SEP 25	Chk'd By:PD	Date:22 nd SEP 25	Rev: P0	Date:22 nd SEP 25

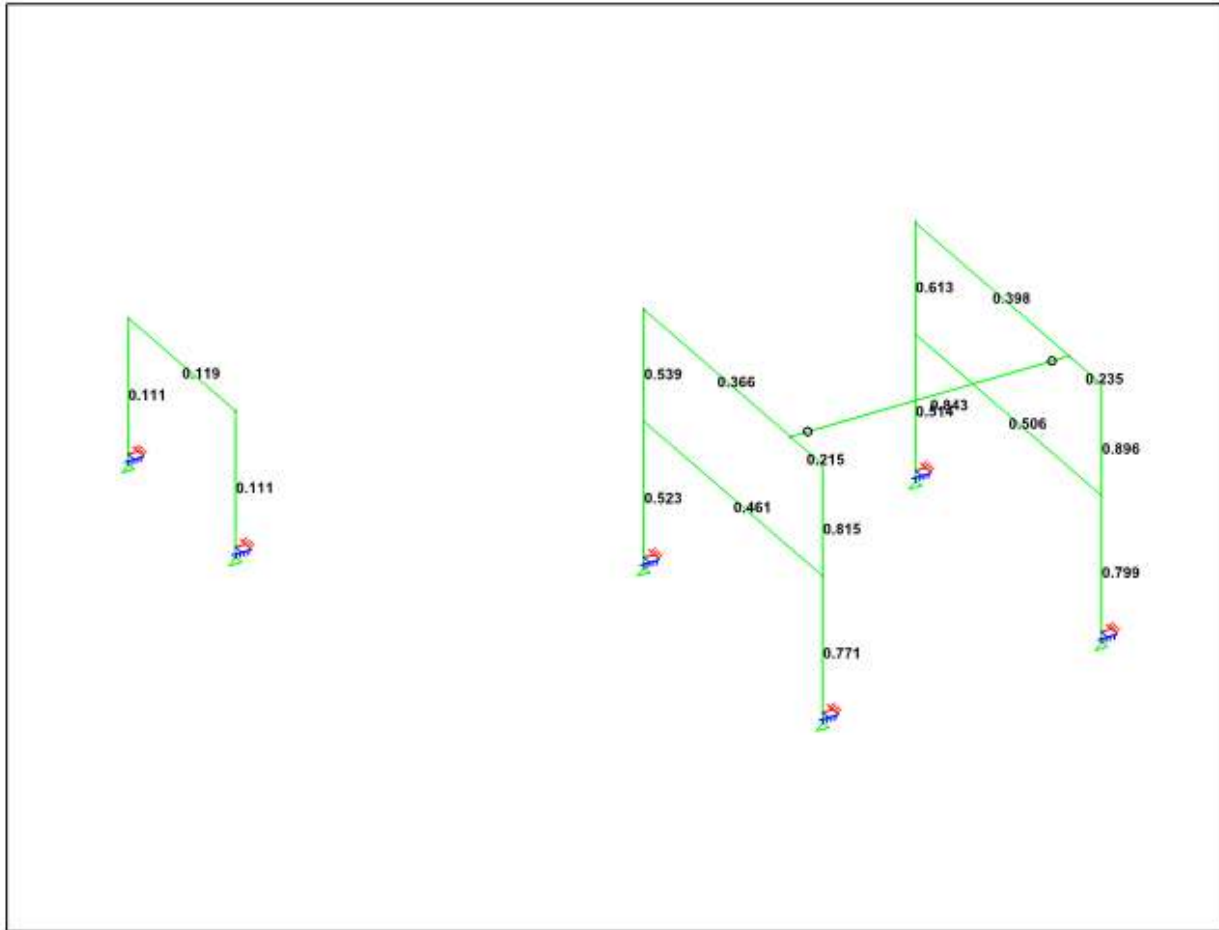


SHEAR FORCE DIAGRAM



MOMENT DIAGRAM

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For Additional information refer below attached appendix 15.1 HRS frame design.

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16. SHEAR WALL LATERAL DESIGN CALCULATION

Structures require a resisting system to transfer lateral loads imposed on them. Typically these loads are from wind or seismic forces. A commonly used type of lateral force resisting system in commercial applications is Sheathing Braced Approach. When designing this type of force resisting system there are a number of design and serviceability considerations that should be investigated. These include loads, Sheathing property, racking strength, fastener quantity, construction practices depending on type of Shear walls.

The provisions of the Code can be used to design these elements. In case of Sheathing braced design, the test report for Racking resistance of sheathing is determined and used for calculation. Connections and members must also be adequately sized to transfer loads. Where the studs are load bearing the effects of gravity loads in resisting and adding to overturning loads must be considered. Finally, it will be necessary to properly design overturning restraints (Hold downs), collectors and boundary elements. These items are included in the lateral design calculation.

The roof have design considering the dead and live load coming on to it and checking flexure, shear and deflection for the governing load using AISI code and in house spreadsheet.

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17. CONNECTION DESIGN

17.1 LGS Wall Stud Bottom Track to Concrete Base Connection

Wind pressure = 98.97 psf
 Wall stud spacing = 24" O.C
 Max Wall height = 11 ft

Maximum shear force due to wind pressure = $11/2 \times 98.97$
 = 545 plf

Using 1/2 x 4 Titen HD Anchor @48" O.C to connect wall bottom track to concrete,

Shear Capacity of anchor = 4460 lb

Shear force per anchor = 545×4
 = 2180 lb < 4460 lb Hence OK.

Connection Detail: Use 1/2 x 4 Titen HD Heavy duty screw Anchor with min 4" embedded @48" O.C to connect wall bottom track to concrete.

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17.2 Design of Webbed beam to post Connection Calculation

Wind pressure = 98.97 psf
 Depth of beam = 1'-9 5/8"
 Length of beam = 16 ft

Shear Force = (98.97x1.81)x16/2
 = 1443 lb

Provide Simpson SSC4.25 connector to resist shear force,

Allowable capacity = 1768 lb

= 1768 lb > 785 lb HENCE OK.

Model No.	No. of #10 Screws	Screw Pattern	Stud Thickness and Yield Strength								
			33 mil (20 ga.) – 33 ksi			43 mil (18 ga.) – 33 ksi			54 mil (16 ga.) – 50 ksi		
			F ₂	F ₃	F ₄	F ₂	F ₃	F ₄	F ₂	F ₃	F ₄
SSC4.25	2	Min.	350	350	355	530	485	525	660	485	890
	5	Max.	660	875	365	660	1,175	600		1,175	1,005
	4	Outer		700	235		795	330		795	625
	4	F		485	320		485	485		485	980
	8	G		970	530		970	805		970	1,625
	11	H		1,175	590		1,175	895		1,175	1,805
	14	I		1,175	730		1,175	1,105		1,175	1,860

Connection Detail: Use simpson SSC4.25 connector to connect LGS webbed beam to LGS Post w/ (14) #10 Screws.

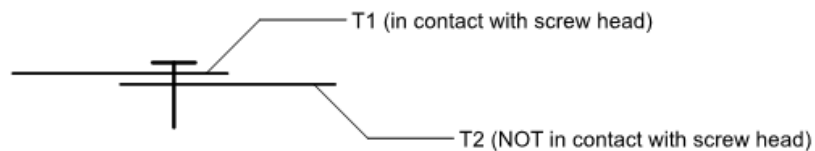
Project: LA MARINA				JOB No:	
Client:					
Calc.By: SS	Date:22 nd SEP 25	Chk'd By:PD	Date:22 nd SEP 25	Rev: P0	Date:22 nd SEP 25

17.3 LGS Wall stud to Top and bottom track connection

Wind pressure = 98.97 psf
 Wall stud spacing = 24"
 Max Wall height = 11 ft

Maximum shear force = $11/2 \times 98.97 \times 2$
 Due to wind pressure
 = 1088 lb

Using #10 screws to connect wall stud to top and bottom track,



Screw Connection Input Parameters

T1 = 0.0451 in Fu(1) = 65 ksi Edge Dist = NA
 T2 = 0.0451 in Fu(2) = 65 ksi Edge Dist = NA
 Screw Diameter = #10 (0.190 in)
 Screw Head Diameter = 0.3125

Results

	Nominal Pn (lb)	ASD Pn/Omega (lb)	LRFD phi x Pn (lb)	Min Req'd Screw Strength, Pss (lb)
Shear	1139.7	379.9	569.9	1424.7
Pullout (T2)	473.4	157.8	236.7	1717.7
Pullver (T1)	1374.1	458.0	687.1	1717.7

Notes:

1. Pullout values assume screw fully penetrates T2
2. Minimum edge distance = $1.5d = 0.285$ (in)

Pull out Capacity of #10 screws = 237 lb

No. of screws required = $1088/237$
 = 6 Nos.

Connection Detail: Use (6) #10 one each side of track to connect wall stud to top and bottom track.

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17.4 Connection of hold down and anchor :

Maximum uplift force = 30000 lb

Using (1) S/HD10 or all equivalent to connect @ hold down location.

Allowable capacity of S/HD10 = 31000 lb

Required Capacity = 30000 lb < 31000 lbHence Safe

Connection Detail: Providing (1) S/HD10 or all equivalent W/ 7/8" Dia. HIT-RE 500 V3 + ASTM A193-B7 (Grade B7) 55 Ks, min 10" embedment to concrete base.

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17.5 Connection of Roof Joist to Wall connection

Designed Wind Pressure	=	98.97 psf
Roof Dead Load	=	25 psf
Roof span	=	20 ft
Wall height	=	9 ft
Joist spacing	=	2 ft
Factored Load	=	0.9 x 25 - 98.97
	=	76.47 psf

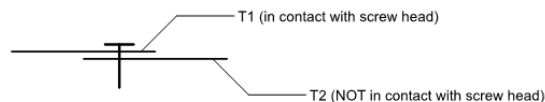
Max Roof Uplift at supports = 980 lb

Provide (1) Simpson FCB45.5 to connect roof Joist to wall,

Model No.	Connector Material Thickness mil (ga.)	L (in.)	Min./Max.	No. of #12-14 Self-Drilling Screws	Stud Thickness												Code Ref.
					33 mil (20 ga.)				43 mil (18 ga.)				54 mil (16 ga.)				
					F ^{1,4} ₁	F ₂	F ₃	F ₄	F ^{1,4} ₁	F ₂	F ₃	F ₄	F ^{1,4} ₁	F ₂	F ₃	F ₄	
FCB43.5	54 (16)	3½	Min.	4	140	755	755	755	175	1,105	905	1,055	330	1,250	905	1,235	
			Max.	6	205	1,100	1,130	1,075	260	1,105	1,105	1,350	330	1,250	2,245	1,770	
MFCB43.5	68 (14)	3½	Min.	4	140	755	755	755	220	1,105	1,105	1,055	410	1,530	2,280	1,595	
			Max.	6	205	1,130	1,130	1,075	260	1,265	1,105	1,545	410	1,530	2,630	1,770	
FCB45.5	54 (16)	5½	Min.	4	120	755	755	700	150	1,105	905	875	285	1,105	905	1,100	
			Max.	9	155	1,100	1,260	1,095	195	1,105	1,105	1,380	330	1,105	2,245	1,785	

Uplift capacity of connector = 1105 lb
= 1105 lb > 980 lb Hence OK.

Shear Force on Roof joist due to wind = 76.47 psf x (9/2 + 1) x 2 ft
= 346 lb



Screw Connection Input Parameters

T1 = 0.0451 in Fu(1) = 65 ksi Edge Dist = NA
T2 = 0.0451 in Fu(2) = 65 ksi Edge Dist = NA
Screw Diameter = #10 (0.190 in)
Screw Head Diameter = 0.3125

Results

	Nominal Pn (lb)	ASD Pn/Omega (lb)	LRFD phi x Pn (lb)	Min Req'd Screw Strength, Pss (lb)
Shear	1139.7	379.9	569.9	1424.7
Pullout (T2)	473.4	157.8	236.7	1717.7
Pullver (T1)	1374.1	458.0	687.1	1717.7

Notes:

1. Pullout values assume screw fully penetrates T2
2. Minimum edge distance = 1.5d = 0.285 (in)

Shear capacity of screw = 569.9 lb
No. of screws required = 346/569
= 2 no.

Connection Detail: Use (2) Simpson FCB45.5 mil connector each side to connect Roof joist with wall with (9) #10 screws to resist uplift & use (2) #10 screw to resist shear .

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17.6 Post to concrete base connection:

Design Wind Pressure = 98.97 psf
 Height of post = 11 ft
 Depth of floor = 1.81 ft
 Floor span = 20 ft

Using Simpson FCB45.5 cleat angle to connect post to concrete.

Shear force due to wind

Pressure on post = $98.97 \times 11/2 + 1.81 \times 98.97 \times 20/2$
 = 2335 lb

Model No.	Connector Material Thickness mil (ga.)	L (in.)	Min./Max.	No. of #12-14 Self-Drilling Screws	Stud Thickness												Code Ref.
					33 mil (20 ga.)				43 mil (18 ga.)				54 mil (16 ga.)				
					F ₁ ^{3,4}	F ₂	F ₃	F ₄	F ₁ ^{3,4}	F ₂	F ₃	F ₄	F ₁ ^{3,4}	F ₂	F ₃	F ₄	
FCB43.5	54 (16)	3½	Min.	4	140	755	755	755	175	1,105	905	1,055	330	1,250	905	1,235	
			Max.	6	205	1,100	1,130	1,075	260	1,105	1,105	1,350	330	1,250	2,245	1,770	
MFCB43.5	68 (14)	3½	Min.	4	140	755	755	755	220	1,105	1,105	1,055	410	1,530	2,280	1,595	
			Max.	6	205	1,130	1,130	1,075	260	1,265	1,105	1,545	410	1,530	2,630	1,770	
FCB45.5	54 (16)	5½	Min.	4	120	755	755	700	150	1,105	905	875	285	1,105	905	1,100	
			Max.	9	155	1,100	1,260	1,095	195	1,105	1,105	1,380	330	1,105	2,245	1,785	

Allowable shear

Capacity = 2856 lb
 = 2335 lb < 2856 lb

Hence OK.

Connection Detail: Provide FCB45.5 (9)#12 and (4) 0.157" PDPA nails to concrete to connect post to concrete.

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17.7 Floor Joist to External LGS Wall stud connection:

Design Wind Pressure = 98.97 psf
 Wall height = 11 ft
 Wall stud spacing = 2 ft

Maximum shear force
 due to wind pressure = $11 \times 98.97 \times 2$
 = 2177 lb

Using Simpson FCB45.5 to connect floor joist to wall stud,

Allowable Horizontal Shear load
 of FCB45.5 cleat = 2208 lb

Required Capacity = 2177 lb < 2208 lbHence Safe

Model No.	Connector Material Thickness mil (ga.)	L (in.)	Min./Max.	No. of #12–14 Self-Drilling Screws	Stud Thickness												Code Ref.
					33 mil (20 ga.)				43 mil (18 ga.)				54 mil (16 ga.)				
					F ₁ ^{3,4}	F ₂	F ₃	F ₄	F ₁ ^{3,4}	F ₂	F ₃	F ₄	F ₁ ^{3,4}	F ₂	F ₃	F ₄	
FCB43.5	54 (16)	3½	Min.	4	140	755	755	755	175	1,105	905	1,055	330	1,250	905	1,235	
			Max.	6	205	1,100	1,130	1,075	260	1,105	1,105	1,350	330	1,250	2,245	1,770	
MFCB43.5	68 (14)	3½	Min.	4	140	755	755	755	220	1,105	1,105	1,055	410	1,530	2,280	1,595	
			Max.	6	205	1,130	1,130	1,075	260	1,265	1,105	1,545	410	1,530	2,630	1,770	
FCB45.5	54 (16)	5½	Min.	4	120	755	755	700	150	1,105	905	875	285	1,105	905	1,100	
			Max.	9	155	1,100	1,260	1,095	195	1,105	1,105	1,380	330	1,105	2,245	1,785	

Connection Detail: Simpson FCB45.5 with (9) #10 screws directly to connect floor joist to external wall.

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Calc.By: SS	Date:22 nd SEP 25	Chk'd By:PD	Date:22 nd SEP 25	Rev: P0	Date:22 nd SEP 25

17.8 Floor joist to webbed beam connection

Floor joist span = 20 ft
 Dead load = 25 psf
 Live load = 40 psf
 Joist spacing = 24" O.C

Maximum Joist = $(1.2 \times 2 \times 25 + 1.6 \times 2 \times 40) \times 20 / 2$
 Gravity shear = 1880 lb

Using Simpson S/JCT hanger to connect floor joist to webbed beam,

Shear Capacity of hanger = 2105
 = 2105 lb > 1880 lb HENCE OK.

Model No.	Fasteners ⁷			Allowable Load ¹ (lb.)	
	Top	Face	Joist	Uplift	Down
Attached to CFS Header: 54 mil (16 ga.) ³ — Straight Hanger					
S/JCT (min.)	(1) #10	(2) #10	(4) #10	940	1,195
S/JCT (max.)	(1) #10	(4) #10	(6) #10	1,435	<u>2,105</u>
S/HJCT (min.)	(2) #10	(4) #14	(6) #14	1,510	2,920
S/HJCT (max.)	(2) #10	(8) #14	(9) #14	1,670	3,855

Connection Detail: Provide Simpson S/JCT hanger with (1) #10 screws at top, (4) #10 screws to face, (6) #10 screws to side to connect floor joist to webbed beam.

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17.9 Floor Joist to Steel beam connection

Floor joist span = 20 ft
 Dead load = 25 psf
 Live load = 40 psf
 Joist spacing = 24" O.C

Maximum joist = $(1.2 \times 2 \times 25 + 1.6 \times 2 \times 40) \times 20 / 2$
 Gravity shear = 1880 lb

Using Simpson S/JCT hanger to connect floor joist to steel beam,

Shear Capacity of hanger = 2105 lb
 = 2105 lb > 1880 lb HENCE OK.

Model No.	Fasteners ⁷			Allowable Load ¹ (lb.)	
	Top	Face	Joist	Uplift	Down
Attached to CFS Header: 54 mil (16 ga.) ³ — Straight Hanger					
S/JCT (min.)	(1) #10	(2) #10	(4) #10	940	1,195
S/JCT (max.)	(1) #10	(4) #10	(6) #10	1,435	<u>2,105</u>
S/HJCT (min.)	(2) #10	(4) #14	(6) #14	1,510	2,920
S/HJCT (max.)	(2) #10	(8) #14	(9) #14	1,670	3,855

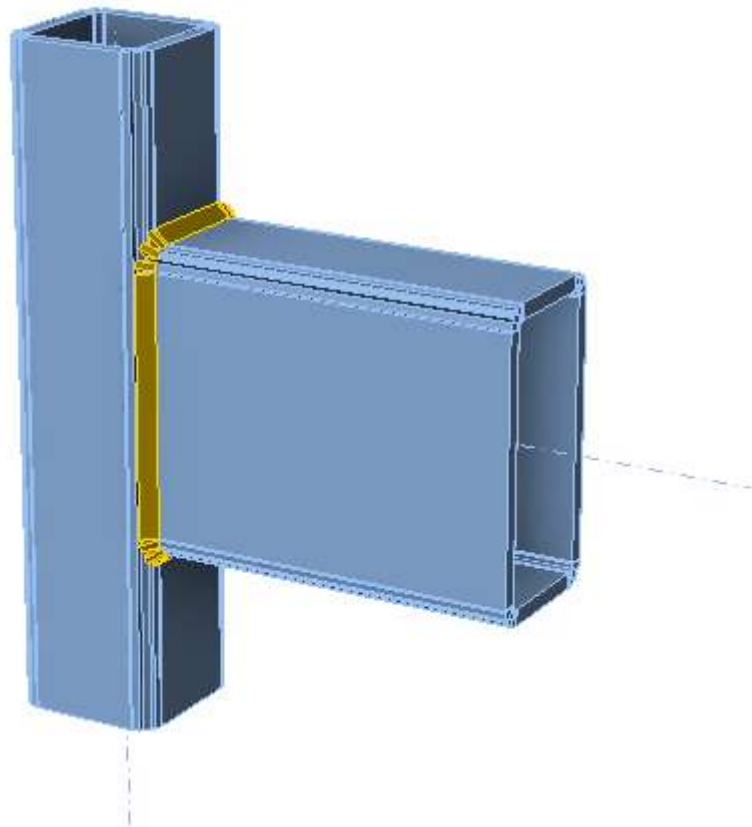
Connection Detail: Provide Simpson S/JCT hanger with (1) #10 screws at top, (4) #10 screws to face, (6) #10 screws to side to connect floor joist to steel beam.

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17.10 **HRS steel beam to HRS steel column connection :**

Maximum Moment = 62.5 kips/ft
 Maximum Shear = 23.5 kips
 Maximum Axial = 3.1 kips
 Column Size = 6x6x5/8
 Beam Size = 12x6x5/8

Connection Detail: Provide 3/8" fillet weld all around for connection of beam to column.

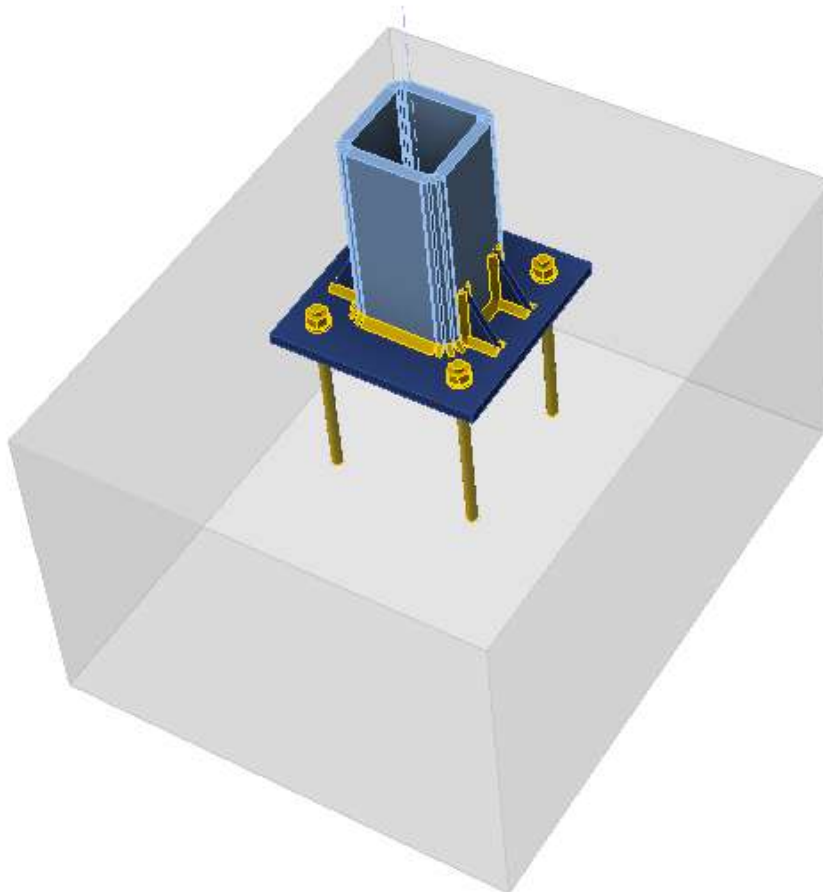


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Client:					
Calc.By: SS	Date:22 nd SEP 25	Chk'd By:PD	Date:22 nd SEP 25	Rev: P0	Date:22 nd SEP 25

17.11 Column base plate connection :

Maximum Moment	=	32 kips/ft
Maximum uplift	=	3.4 kips
Maximum shear	=	5.2 kips
Column Size	=	6x6x5/8

Connection Detail: Provide 12"x12"x5/8" cast in place base plate with 1/4" thk ribs and column welded to base plate with 3/8" fillet weld and use (4) 7/8" dia Welded stud anchor to base plate of grade B with min. 12" embedded to concrete.

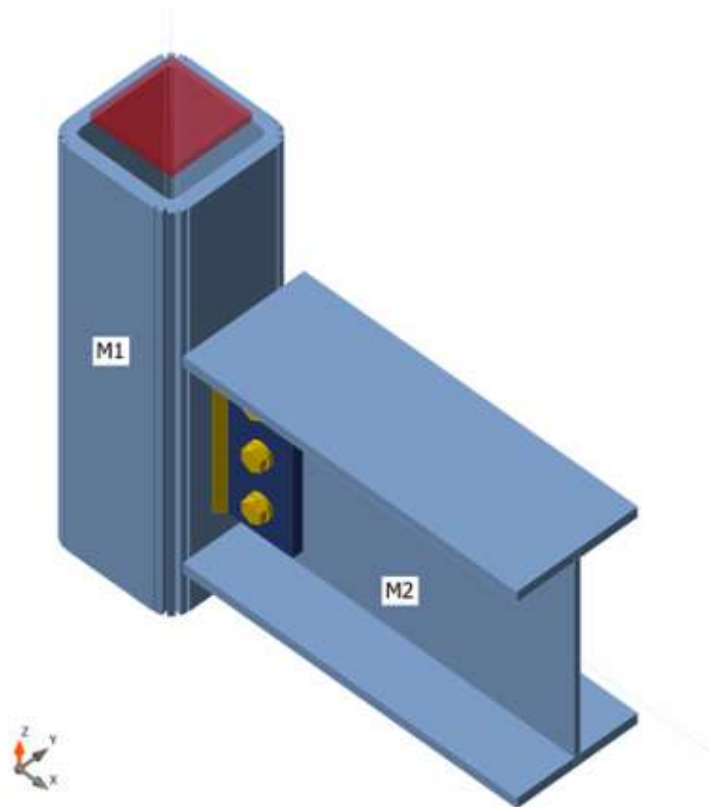


Project: LA MARINA				JOB No:	
Client:					
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17.12 Column to Beam connection :

Maximum shear = 3.5 kips

Connection Detail: Provide 1/2" thick fin plate with (3) 1/2 A325 anchor with 1/4" fillet weld to connect beam to column.

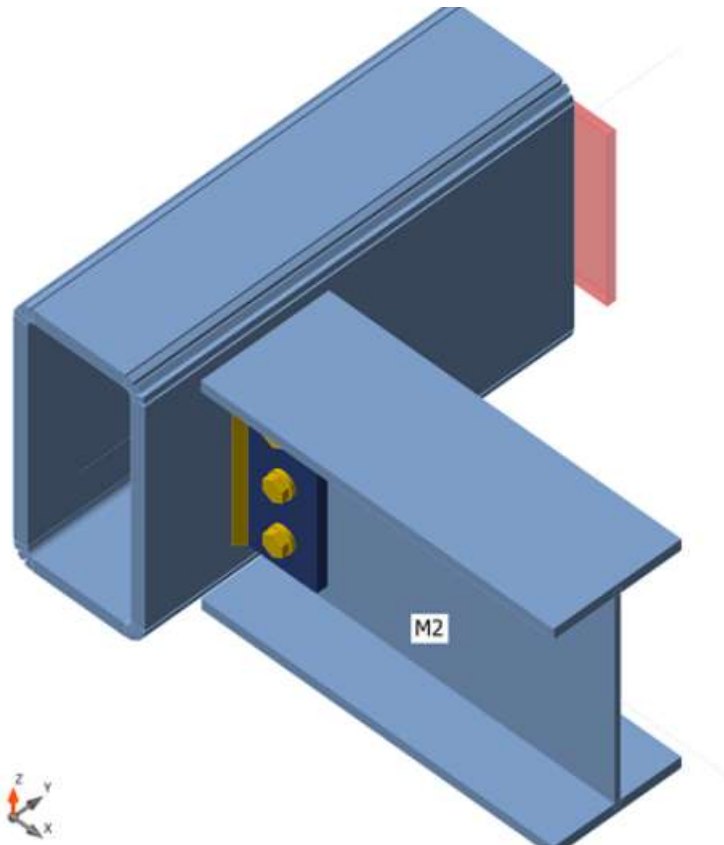


Project: LA MARINA				JOB No:	
Client:					
Calc.By: SS	Date:22 nd SEP 25	Chk'd By:PD	Date:22 nd SEP 25	Rev: P0	Date:22 nd SEP 25

17.13 Beam to Beam connection :

Maximum shear = 3.5 kips

Connection Detail: Provide 1/2" thick fin plate with (3) 1/2 A325 anchor with 1/4" fillet weld to connect beam to Beam.

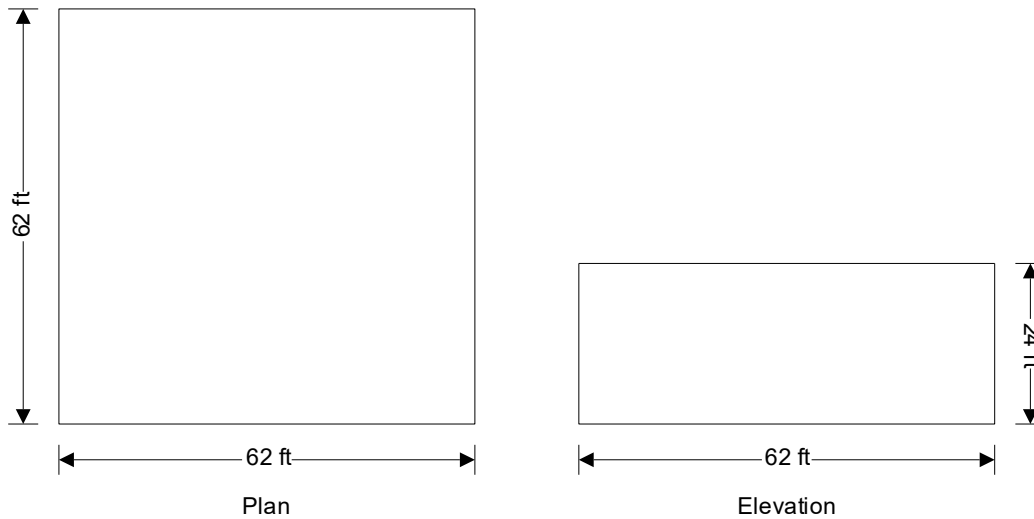


WIND LOADING

In accordance with ASCE7-16

Using the components and cladding design method

Tedds calculation version 2.1.13



Building data

Type of roof	Flat
Length of building	b = 62.00 ft
Width of building	d = 62.00 ft
Height to eaves	H = 24.00 ft
Mean height	h = 24.00 ft
End zone width	a = max(min(0.1×min(b, d), 0.4×h), 0.04×min(b, d), 3ft) = 6.20 ft

General wind load requirements

Basic wind speed	V = 175.0 mph
Risk category	II
Velocity pressure exponent coef (Table 26.6-1)	K _d = 0.85
Ground elevation above sea level	Z _{gl} = 0 ft
Ground elevation factor	K _e = exp(-0.0000362 × Z _{gl} /1ft) = 1.00
Exposure category (cl 26.7.3)	D
Enclosure classification (cl.26.12)	Partially enclosed buildings
Internal pressure coef +ve (Table 26.13-1)	GC _{pi_p} = 0.55
Internal pressure coef -ve (Table 26.13-1)	GC _{pi_n} = -0.55
Gust effect factor	G _f = 0.85

Topography

Topography factor not significant	K _{zt} = 1.0
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Velocity pressure

Velocity pressure coefficient (Table 26.10-1)	K _z = 1.11
Velocity pressure	q _h = 0.00256 × K _z × K _{zt} × K _d × K _e × V ² × 1psf/mph ² = 74.1 psf

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Peak velocity pressure for internal pressure

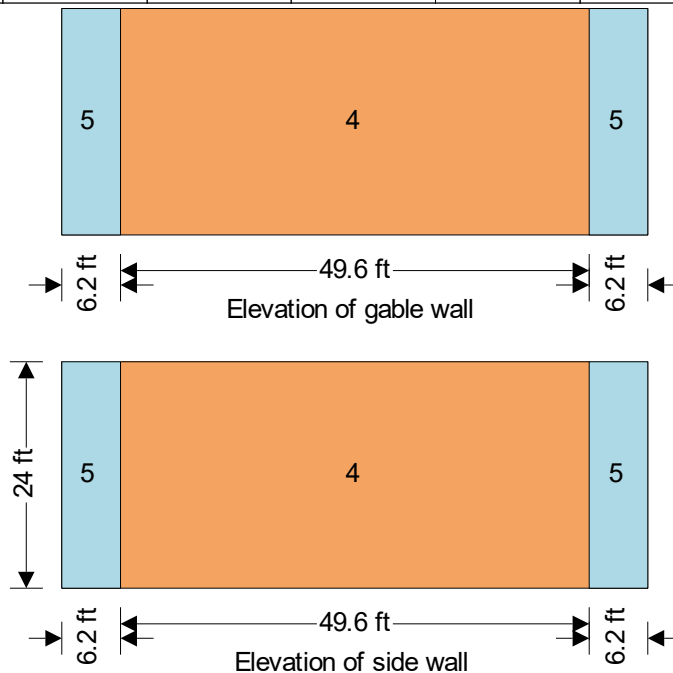
Peak velocity pressure – internal (as roof press.) $q_i = 74.10$ psf

Equations used in tables

Net pressure $p = q_i \times [GC_p - GC_{pi}]$

Components and cladding pressures - Wall (Table 30.3-1)

Component	Zone	Length (ft)	Width (ft)	Eff. area (ft ²)	+GC _p	-GC _p	Pres (+ve) (psf)	Pres (-ve) (psf)
<=10 sf	4	-	-	10.0	0.90	-0.99	107.5	-114.1
50 sf	4	-	-	50.0	0.79	-0.88	99.2	-105.9
200 sf	4	-	-	200.0	0.69	-0.78	92.1	-98.8
>500 sf	4	-	-	500.1	0.63	-0.72	87.4	-94.1
<=10 sf	5	-	-	10.0	0.90	-1.26	107.5	-134.1
50 sf	5	-	-	50.0	0.79	-1.04	99.2	-117.7
200 sf	5	-	-	200.0	0.69	-0.85	92.1	-103.5
>500 sf	5	-	-	500.1	0.63	-0.72	87.4	-94.1

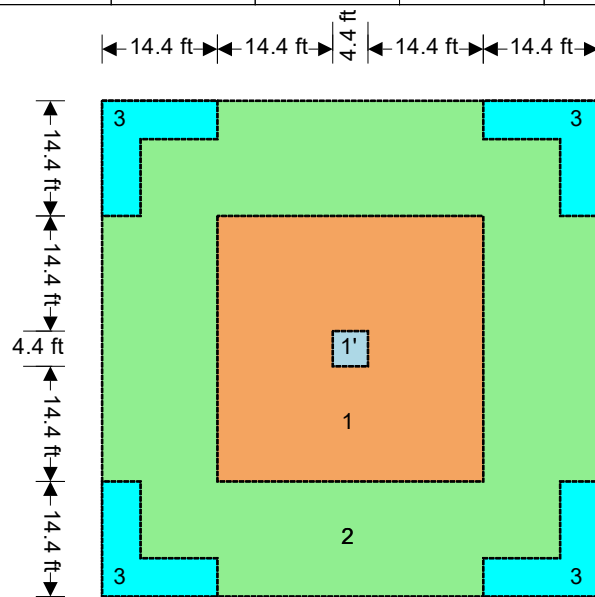


Components and cladding pressures - Roof (Figure 30.3-2A)

Component	Zone	Length (ft)	Width (ft)	Eff. area (ft ²)	+GC _p	-GC _p	Pres (+ve) (psf)	Pres (-ve) (psf)
<=10 sf	1	-	-	10.0	0.30	-1.70	63.0	-166.7
100 sf	1	-	-	100.0	0.20	-1.29	55.6	-136.2
200 sf	1	-	-	200.0	0.20	-1.16	55.6	-127.0
>500 sf	1	-	-	500.1	0.20	-1.00	55.6	-114.9

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Component	Zone	Length (ft)	Width (ft)	Eff. area (ft ²)	+GC _p	-GC _p	Pres (+ve) (psf)	Pres (-ve) (psf)
<=10 sf	1'	-	-	10.0	0.30	-0.90	63.0	-107.5
100 sf	1'	-	-	100.0	0.20	-0.90	55.6	-107.5
500 sf	1'	-	-	500.0	0.20	-0.55	55.6	-81.6
>1000 sf	1'	-	-	1000.1	0.20	-0.40	55.6	-70.4
<=10 sf	2	-	-	10.0	0.30	-2.30	63.0	-211.2
100 sf	2	-	-	100.0	0.20	-1.77	55.6	-171.9
200 sf	2	-	-	200.0	0.20	-1.61	55.6	-160.1
>500 sf	2	-	-	500.1	0.20	-1.40	55.6	-144.5
<=10 sf	3	-	-	10.0	0.30	-3.20	63.0	-277.9
100 sf	3	-	-	100.0	0.20	-2.14	55.6	-199.4
200 sf	3	-	-	200.0	0.20	-1.82	55.6	-175.7
>500 sf	3	-	-	500.1	0.20	-1.40	55.6	-144.5



Plan on roof

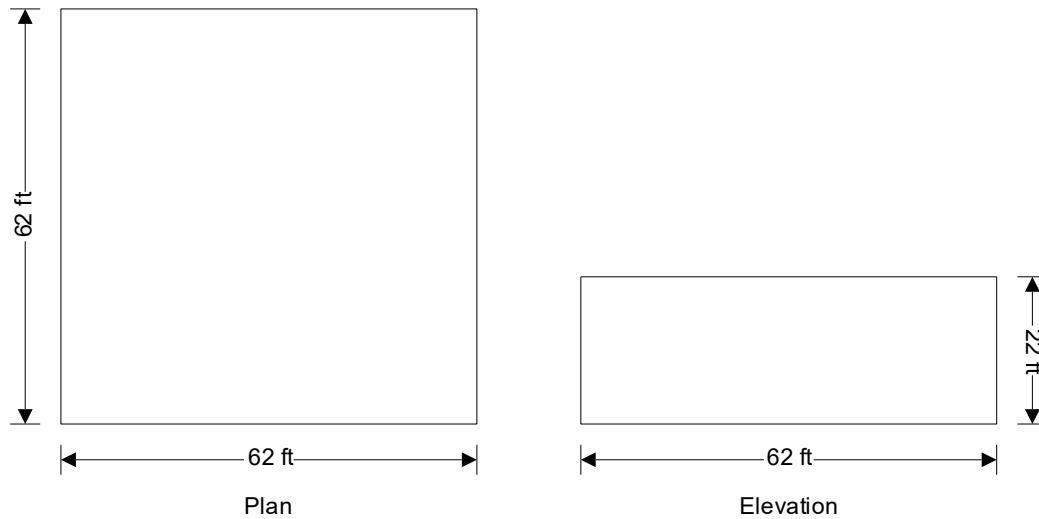
Project				Job Ref.	
Section Wind Analysis mwfrs				Sheet no./rev. 1	
Calc. by P	Date 16-07-2025	Chk'd by	Date	App'd by	Date

WIND LOADING

In accordance with ASCE7-16

Using the directional design method

Tedds calculation version 2.1.13



Building data

Type of roof	Flat
Length of building	b = 62.00 ft
Width of building	d = 62.00 ft
Height to eaves	H = 22.00 ft
Mean height	h = 22.00 ft

General wind load requirements

Basic wind speed	V = 175.0 mph
Risk category	II
Velocity pressure exponent coef (Table 26.6-1)	$K_d = \mathbf{0.85}$
Ground elevation above sea level	$z_{gl} = \mathbf{0}$ ft
Ground elevation factor	$K_e = \exp(-0.0000362 \times z_{gl}/1\text{ft}) = \mathbf{1.00}$
Exposure category (cl 26.7.3)	D
Enclosure classification (cl.26.12)	Partially enclosed buildings
Internal pressure coef +ve (Table 26.13-1)	$GC_{pi_p} = \mathbf{0.55}$
Internal pressure coef -ve (Table 26.13-1)	$GC_{pi_n} = \mathbf{-0.55}$
Gust effect factor	$G_f = \mathbf{0.85}$
Minimum design wind loading (cl.27.1.5)	$p_{min_r} = \mathbf{8}$ lb/ft ²

Topography

Topography factor not significant	$K_{zt} = 1.0$
Velocity pressure equation	$q = 0.00256 \times K_z \times K_{zt} \times K_d \times V^2 \times 1\text{psf}/\text{mph}^2$

Velocity pressures table

z (ft)	K_z (Table 26.10-1)	q_z (psf)
15.00	1.03	68.64

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Section				Sheet no./rev. 2	
Calc. by P	Date 16-07-2025	Chk'd by	Date	App'd by	Date

z (ft)	K _z (Table 26.10-1)	q _z (psf)
15.00	1.03	68.64
22.00	1.10	73.04

Peak velocity pressure for internal pressure

Peak velocity pressure – internal (as roof press.) $q_i = 73.04$ psf

Pressures and forces

Net pressure

$$p = q \times G_f \times C_{pe} - q_i \times GC_{pi}$$

Net force

$$F_w = p \times A_{ref}$$

Roof load case 1 - Wind 0, $GC_{pi} 0.55$, $-C_{pe}$

Zone	Ref. height (ft)	Ext pressure coefficient C_{pe}	Peak velocity pressure q_p (psf)	Net pressure p (psf)	Area A_{ref} (ft ²)	Net force F_w (kips)
A (-ve)	22.00	-0.90	73.04	-96.04	682.00	-65.50
B (-ve)	22.00	-0.90	73.04	-96.04	682.00	-65.50
C (-ve)	22.00	-0.50	73.04	-71.21	1364.00	-97.13
D (-ve)	22.00	-0.30	73.04	-58.80	1116.00	-65.62

Total vertical net force $F_{w,v} = -293.75$ kips

Total horizontal net force $F_{w,h} = 0.00$ kips

Walls load case 1 - Wind 0, $GC_{pi} 0.55$, $-C_{pe}$

Zone	Ref. height (ft)	Ext pressure coefficient C_{pe}	Peak velocity pressure q_p (psf)	Net pressure p (psf)	Area A_{ref} (ft ²)	Net force F_w (kips)
A ₁	15.00	0.80	68.64	6.50	930.00	6.05
A ₂	15.00	0.80	68.64	6.50	0.00	0.00
A ₃	22.00	0.80	73.04	9.49	434.00	4.12
B	22.00	-0.50	73.04	-71.21	1364.00	-97.13
C	22.00	-0.70	73.04	-83.63	1364.00	-114.07
D	22.00	-0.70	73.04	-83.63	1364.00	-114.07

Overall loading

Projected vertical plan area of wall

$$A_{vert_w_0} = b \times H = 1364.00 \text{ ft}^2$$

Projected vertical area of roof

$$A_{vert_r_0} = 0.00 \text{ ft}^2$$

Minimum overall horizontal loading

$$F_{w,total_min} = p_{min_w} \times A_{vert_w_0} + p_{min_r} \times A_{vert_r_0} = 21.82 \text{ kips}$$

Leeward net force

$$F_l = F_{w,wB} = -97.1 \text{ kips}$$

Windward net force

$$F_w = F_{w,wA_1} + F_{w,wA_2} + F_{w,wA_3} = 10.2 \text{ kips}$$

Overall horizontal loading

$$F_{w,total} = \max(F_w - F_l + F_{w,h}, F_{w,total_min}) = 107.3 \text{ kips}$$

Roof load case 2 - Wind 0, $GC_{pi} -0.55$, $-1C_{pe}$

Zone	Ref. height (ft)	Ext pressure coefficient C_{pe}	Peak velocity pressure q_p (psf)	Net pressure p (psf)	Area A_{ref} (ft ²)	Net force F_w (kips)
A (+ve)	22.00	-0.18	73.04	29.00	682.00	19.78
B (+ve)	22.00	-0.18	73.04	29.00	682.00	19.78

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Zone	Ref. height (ft)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (psf)	Net pressure p (psf)	Area A_{ref} (ft ²)	Net force F_w (kips)
C (+ve)	22.00	-0.18	73.04	29.00	1364.00	39.55
D (+ve)	22.00	-0.18	73.04	29.00	1116.00	32.36

Total vertical net force $F_{w,v} = 111.46$ kips

Total horizontal net force $F_{w,h} = 0.00$ kips

Walls load case 2 - Wind 0, $GC_{pi} -0.55$, $-1c_{pe}$

Zone	Ref. height (ft)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (psf)	Net pressure p (psf)	Area A_{ref} (ft ²)	Net force F_w (kips)
A ₁	15.00	0.80	68.64	86.85	930.00	80.77
A ₂	15.00	0.80	68.64	86.85	0.00	0.00
A ₃	22.00	0.80	73.04	89.84	434.00	38.99
B	22.00	-0.50	73.04	9.13	1364.00	12.45
C	22.00	-0.70	73.04	-3.29	1364.00	-4.48
D	22.00	-0.70	73.04	-3.29	1364.00	-4.48

Overall loading

Projected vertical plan area of wall

$$A_{vert_w_0} = b \times H = 1364.00 \text{ ft}^2$$

Projected vertical area of roof

$$A_{vert_r_0} = 0.00 \text{ ft}^2$$

Minimum overall horizontal loading

$$F_{w,total_min} = p_{min_w} \times A_{vert_w_0} + p_{min_r} \times A_{vert_r_0} = 21.82 \text{ kips}$$

Leeward net force

$$F_l = F_{w,wB} = 12.5 \text{ kips}$$

Windward net force

$$F_w = F_{w,WA_1} + F_{w,WA_2} + F_{w,WA_3} = 119.8 \text{ kips}$$

Overall horizontal loading

$$F_{w,total} = \max(F_w - F_l + F_{w,h}, F_{w,total_min}) = 107.3 \text{ kips}$$

Roof load case 3 - Wind 90, $GC_{pi} 0.55$, $-c_{pe}$

Zone	Ref. height (ft)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (psf)	Net pressure p (psf)	Area A_{ref} (ft ²)	Net force F_w (kips)
A (-ve)	22.00	-0.90	73.04	-96.04	682.00	-65.50
B (-ve)	22.00	-0.90	73.04	-96.04	682.00	-65.50
C (-ve)	22.00	-0.50	73.04	-71.21	1364.00	-97.13
D (-ve)	22.00	-0.30	73.04	-58.80	1116.00	-65.62

Total vertical net force $F_{w,v} = -293.75$ kips

Total horizontal net force $F_{w,h} = 0.00$ kips

Walls load case 3 - Wind 90, $GC_{pi} 0.55$, $-c_{pe}$

Zone	Ref. height (ft)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (psf)	Net pressure p (psf)	Area A_{ref} (ft ²)	Net force F_w (kips)
A ₁	15.00	0.80	68.64	6.50	930.00	6.05
A ₂	15.00	0.80	68.64	6.50	0.00	0.00
A ₃	22.00	0.80	73.04	9.49	434.00	4.12

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Zone	Ref. height (ft)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (psf)	Net pressure p (psf)	Area A_{ref} (ft ²)	Net force F_w (kips)
B	22.00	-0.50	73.04	-71.21	1364.00	-97.13
C	22.00	-0.70	73.04	-83.63	1364.00	-114.07
D	22.00	-0.70	73.04	-83.63	1364.00	-114.07

Overall loading

Projected vertical plan area of wall

$$A_{vert_w_90} = d \times H = \mathbf{1364.00 \text{ ft}^2}$$

Projected vertical area of roof

$$A_{vert_r_90} = \mathbf{0.00 \text{ ft}^2}$$

Minimum overall horizontal loading

$$F_{w,total_min} = p_{min_w} \times A_{vert_w_90} + p_{min_r} \times A_{vert_r_90} = \mathbf{21.82 \text{ kips}}$$

Leeward net force

$$F_l = F_{w,wB} = \mathbf{-97.1 \text{ kips}}$$

Windward net force

$$F_w = F_{w,wA_1} + F_{w,wA_2} + F_{w,wA_3} = \mathbf{10.2 \text{ kips}}$$

Overall horizontal loading

$$F_{w,total} = \max(F_w - F_l + F_{w,h}, F_{w,total_min}) = \mathbf{107.3 \text{ kips}}$$

Roof load case 4 - Wind 90, $GC_{pi} -0.55, +c_{pe}$

Zone	Ref. height (ft)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (psf)	Net pressure p (psf)	Area A_{ref} (ft ²)	Net force F_w (kips)
A (+ve)	22.00	-0.18	73.04	29.00	682.00	19.78
B (+ve)	22.00	-0.18	73.04	29.00	682.00	19.78
C (+ve)	22.00	-0.18	73.04	29.00	1364.00	39.55
D (+ve)	22.00	-0.18	73.04	29.00	1116.00	32.36

Total vertical net force

$$F_{w,v} = \mathbf{111.46 \text{ kips}}$$

Total horizontal net force

$$F_{w,h} = \mathbf{0.00 \text{ kips}}$$

Walls load case 4 - Wind 90, $GC_{pi} -0.55, +c_{pe}$

Zone	Ref. height (ft)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (psf)	Net pressure p (psf)	Area A_{ref} (ft ²)	Net force F_w (kips)
A ₁	15.00	0.80	68.64	86.85	930.00	80.77
A ₂	15.00	0.80	68.64	86.85	0.00	0.00
A ₃	22.00	0.80	73.04	89.84	434.00	38.99
B	22.00	-0.50	73.04	9.13	1364.00	12.45
C	22.00	-0.70	73.04	-3.29	1364.00	-4.48
D	22.00	-0.70	73.04	-3.29	1364.00	-4.48

Overall loading

Projected vertical plan area of wall

$$A_{vert_w_90} = d \times H = \mathbf{1364.00 \text{ ft}^2}$$

Projected vertical area of roof

$$A_{vert_r_90} = \mathbf{0.00 \text{ ft}^2}$$

Minimum overall horizontal loading

$$F_{w,total_min} = p_{min_w} \times A_{vert_w_90} + p_{min_r} \times A_{vert_r_90} = \mathbf{21.82 \text{ kips}}$$

Leeward net force

$$F_l = F_{w,wB} = \mathbf{12.5 \text{ kips}}$$

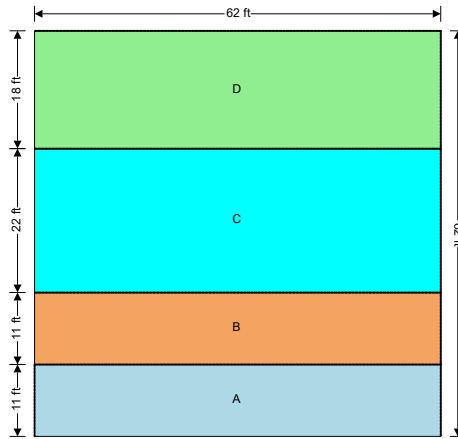
Windward net force

$$F_w = F_{w,wA_1} + F_{w,wA_2} + F_{w,wA_3} = \mathbf{119.8 \text{ kips}}$$

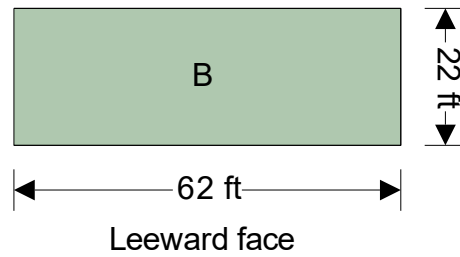
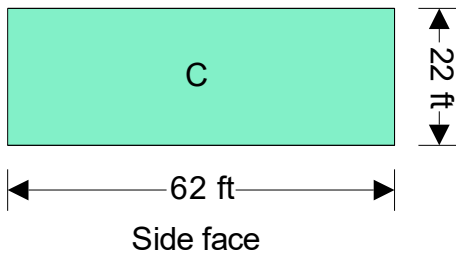
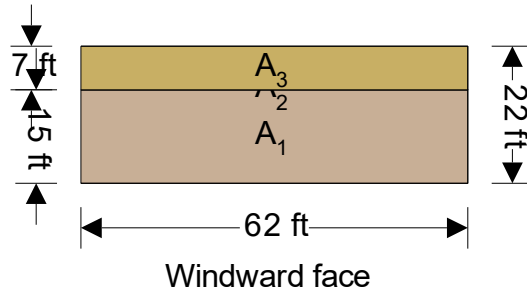
Overall horizontal loading

$$F_{w,total} = \max(F_w - F_l + F_{w,h}, F_{w,total_min}) = \mathbf{107.3 \text{ kips}}$$

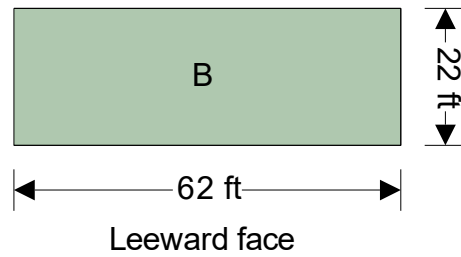
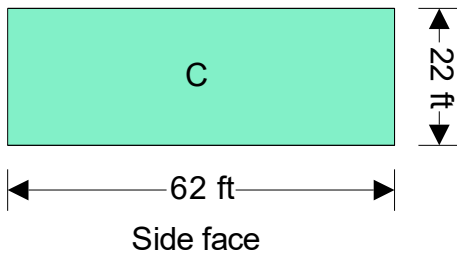
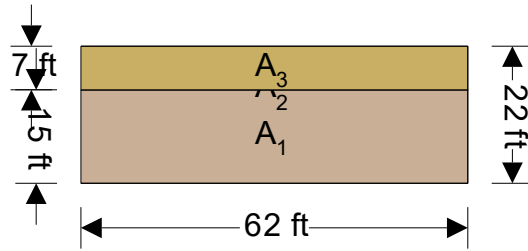
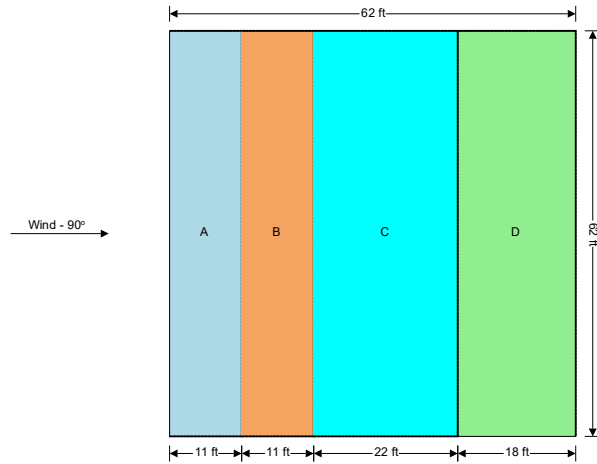
Project				Job Ref.	
Section				Sheet no./rev. 5	
Calc. by P	Date 16-07-2025	Chk'd by	Date	App'd by	Date



Wind - 0°
↑
Plan view - Flat roof



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Section				Sheet no./rev. 6	
Calc. by P	Date 16-07-2025	Chk'd by	Date	App'd by	Date



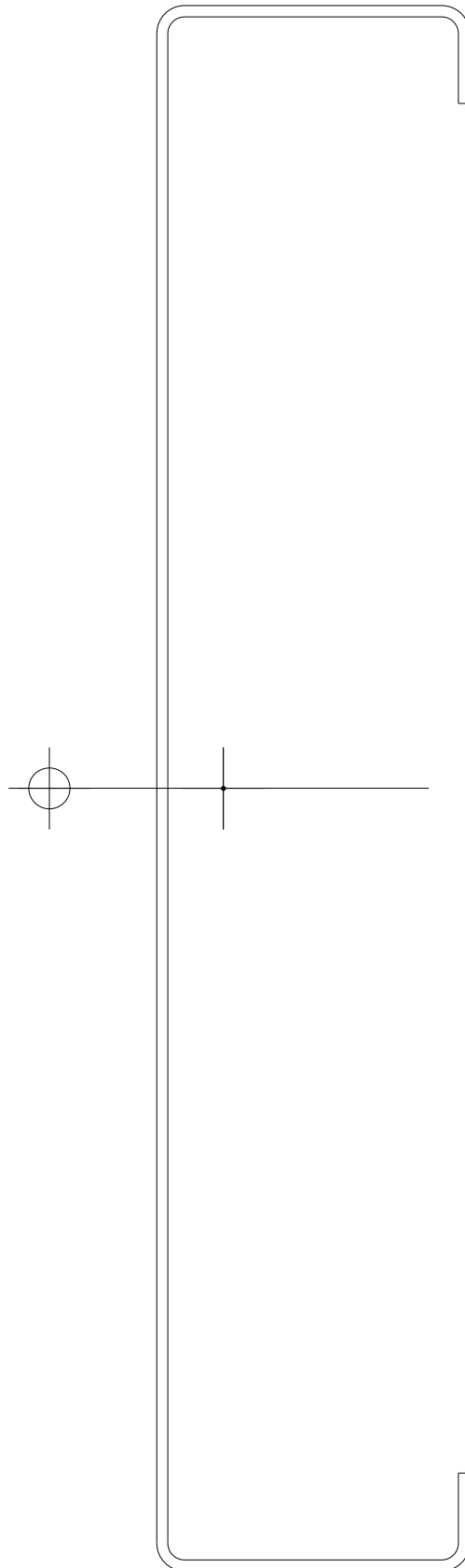
10.1 Roof joist Design

CFS Version 14.0.1
Section: Section 1.cfss
Channel 10x2x0.625-68 mil

PD
FCOM

Page 1

Rev. Date: 24-07-2025 16:26:28
By: PD
Printed: 24-07-2025 16:27:14



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By: PD
Printed: 24-07-2025 16:27:14

Section Inputs

Material: A653 SS Grade 50/1
No cold work of forming strength increase.
No inelastic reserve strength increase.
Modulus of Elasticity, E 29500 ksi
Yield Strength, Fy 50 ksi
Tensile Strength, Fu 65 ksi
Min Elongation in 2 inches 12 %
Torsion Constant Override, J 0 in⁴
Warping Constant Override, Cw 0 in⁶

Stiffened Channel, Thickness 0.0713 in (68 mil)

Placement of Part from Origin:

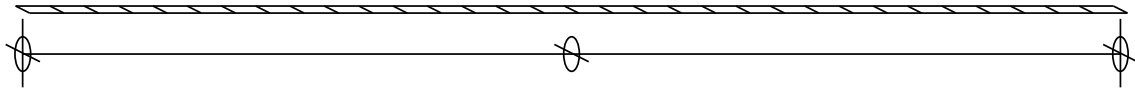
X to center of gravity 0 in

Y to center of gravity 0 in

Outside dimensions, Open shape

	Length (in)	Angle (deg)	Radius (in)	Web	k Coef.	Hole Size (in)	Distance (in)
1	0.625	270.000	0.10690	None	0.000	0.000	0.313
2	2.000	180.000	0.10690	Single	0.000	0.000	1.000
3	10.000	90.000	0.10690	Cee	0.000	0.000	5.000
4	2.000	0.000	0.10690	Single	0.000	0.000	1.000
5	0.625	-90.000	0.10690	None	0.000	0.000	0.313

Rev. Date: 24-07-2025 16:26:57
By: PD
Printed: 24-07-2025 16:27:14



Analysis Inputs

General

Member Orientation: Horizontal
Calculate global buckling using specification equations
Do not include torsion in member checks

Members

Section File				Revision Date and Time				
1 Section 1.cfss				24-07-2025 16:26:28				
Material				Area	Length	Weight		
				(in²)	(ft)	(lb)		
1 A653 SS Grade 50/1				1.0495	19.500	69.585		
Total					19.500	69.585		
Start Loc.		End Loc.	Braced	R	kφ	Lm	ex	ey
(ft)		(ft)	Flange		(lb)	(ft)	(in)	(in)
1 0.000		19.500	Top	0.0000	0.0000	19.500	0.000	0.000

Supports

Type	Location (ft)	Bearing (in)	Fastened	K
1 XYT	0.000	3.25	No	1.0000
2 XT	9.750	1.00	No	1.0000
3 XYT	19.500	3.25	No	1.0000

Loading: Dead Load

Type	Angle (deg)	Start Loc. (ft)	End Loc. (ft)	Start Magnitude	End Magnitude
1 Distributed	90.000	0.000	19.500	-60.000	-60.000 lb/ft

Loading: Roof Live Load

Type	Angle (deg)	Start Loc. (ft)	End Loc. (ft)	Start Magnitude	End Magnitude
1 Distributed	90.000	0.000	19.500	-40.000	-40.000 lb/ft

Load Combination: 1.4D

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.400

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Load Combination: 1.2D+1.6L+0.5Lr
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Live Load	1.600
3 Product Load	1.600
4 Roof Live Load	0.500

Load Combination: 1.2D+1.6L+0.3S
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Live Load	1.600
3 Product Load	1.600
4 Snow Load	0.300

Load Combination: 1.2D+1.6Lr+L
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Roof Live Load	1.600
3 Live Load	1.000
4 Product Load	1.000

Load Combination: 0.9D+W
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	0.900
2 Wind Load	1.000

Load Combination: 1.2D+E+L+0.15S
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Earthquake Load	1.000
3 Live Load	1.000
4 Product Load	1.000
5 Snow Load	0.150

Load Combination: 0.9D+E
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	0.900
2 Earthquake Load	1.000

Member Check - AISI S100-16/S3-22, US, LRFD

Load Combination: 1.2D+1.6Lr+L
Design Parameters at 9.750 ft, Left side:

Lx	19.500 ft	Ly	9.750 ft	Lt	9.750 ft
Kx	1.0000	Ky	1.0000	Kt	1.0000

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Section: Section 1.cfss

Material Type: A653 SS Grade 50/1, Fy=50 ksi

Cbx	1.2987	Cby	1.0000	ex	0.0000 in
Cmx	1.0000	Cmy	1.0000	ey	0.0000 in
Braced Flange: Top		kφ	0 lb		
Red. Factor, R: 0		Lm	19.500 ft		

Loads:	P (lb)	Mx (lb-ft)	Vy (lb)	My (lb-ft)	Vx (lb)
Total	0.0	6464.3	0.0	0.0	0.0
Applied	0.0	6464.3	0.0	0.0	0.0
Strength	9721.2	8260.4	6920.1	925.4	6328.2

Interaction Equations

Eq. H1.2-1 (P, Mx, My) $0.000 + 0.783 + 0.000 = 0.783 \leq 1.0$
Eq. H2-1 (Mx, Vy) $\text{Sqrt}(0.506 + 0.000) = 0.711 \leq 1.0$
Eq. H2-1 (My, Vx) $\text{Sqrt}(0.000 + 0.000) = 0.000 \leq 1.0$

Web Crippling Check - AISI S100-16/S3-22, US, LRFD

Load Combination: 1.2D+1.6Lr+L

Parameters at 19.500 ft:

Total Load: 1326.0 lb on bottom flange

Total Moment: 0.000 lb-ft

Bearing: 3.2500 in

Flange fastened to bearing surface: No

Distance from edge of bearing to edge of opposite load: 19.500 ft

Section: Section 1.cfss

Material Type: A653 SS Grade 50/1, Fy=50 ksi

Applied Load: 1326.0 lb on bottom flange

Applied Moment: 0.0 lb-ft

Distance from edge of bearing to end of member: 0.0000 ft

Part	Elem	Calculation Type	φPn (lb)	Pny (lb)	Notes
1	3	Cee, US-EOF	1739.3	1739.3	
		Web Crippling Strength		1739.3	

Web Crippling Check: 1326.0 lb $\leq$ 1739.3 lb

Moment Check: 0.0 lb-ft $\leq$ 9089.6 lb-ft

Interaction Equations

Eq. H3-1b (P, M) $0.555 + 0.000 = 0.555 \leq 1.197$

10.2 Roof joist Design

CFS Version 14.0.1

Section: Section 1.cfss

Double Channel 10x2x0.625-54 mil

PD

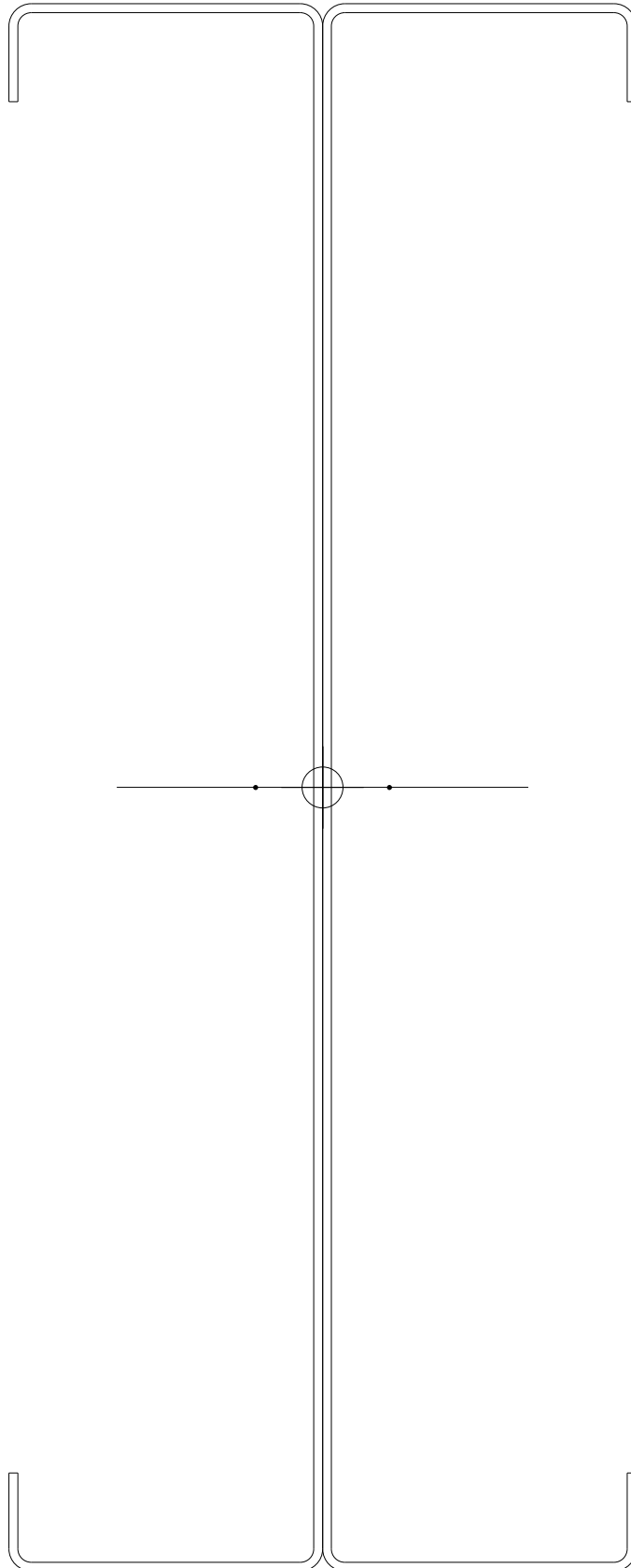
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By: PD

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Section: Section 1.cfss
Double Channel 10x2x0.625-54 mil

PD
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By: PD
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Section Inputs

Material: A653 SS Grade 50/1

No cold work of forming strength increase.

No inelastic reserve strength increase.

Modulus of Elasticity, E 29500 ksi

Yield Strength, Fy 50 ksi

Tensile Strength, Fu 65 ksi

Min Elongation in 2 inches 12 %

Torsion Constant Override, J 0 in⁴

Warping Constant Override, Cw 26.239 in⁶

Connector Spacing 0 in

Right Channel, Thickness 0.0566 in (54 mil)

Placement of Part from Origin:

X to left edge 0 in

Y to center of gravity 0 in

Outside dimensions, Open shape

	Length (in)	Angle (deg)	Radius (in)	Web	k Coef.	Hole Size (in)	Distance (in)
1	0.625	270.000	0.084900	None	0.000	0.000	0.313
2	2.000	180.000	0.084900	Single	0.000	0.000	1.000
3	10.000	90.000	0.084900	Double	0.000	0.000	5.000
4	2.000	0.000	0.084900	Single	0.000	0.000	1.000
5	0.625	-90.000	0.084900	None	0.000	0.000	0.313

Left Channel, Thickness 0.0566 in (54 mil)

Placement of Part from Origin:

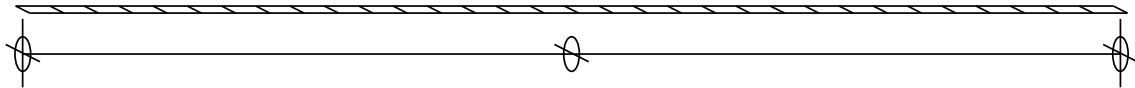
X to right edge 0 in

Y to center of gravity 0 in

Outside dimensions, Open shape

	Length (in)	Angle (deg)	Radius (in)	Web	k Coef.	Hole Size (in)	Distance (in)
1	0.625	-90.000	0.084900	None	0.000	0.000	0.313
2	2.000	0.000	0.084900	Single	0.000	0.000	1.000
3	10.000	90.000	0.084900	Double	0.000	0.000	5.000
4	2.000	180.000	0.084900	Single	0.000	0.000	1.000
5	0.625	270.000	0.084900	None	0.000	0.000	0.313

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Analysis Inputs

General

Member Orientation: Horizontal
 Calculate global buckling using specification equations
 Do not include torsion in member checks

Members

Section File				Revision Date and Time			
1 Section 1.cfss				24-07-2025 16:30:34			
Material		Area	Length	Weight			
		(in²)	(ft)	(lb)			
1 A653 SS Grade 50/1		1.6787	22.670	129.39			
Total			22.670	129.39			
Start Loc.	End Loc.	Braced	R	kφ	Lm	ex	ey
(ft)	(ft)	Flange		(lb)	(ft)	(in)	(in)
1 0.000	22.670	Top	0.0000	0.0000	22.670	0.000	0.000

Supports

Type	Location (ft)	Bearing (in)	Fastened	K
1 XYT	0.000	3.25	No	1.0000
2 XT	11.335	1.00	No	1.0000
3 XYT	22.670	3.25	No	1.0000

Loading: Dead Load

Type	Angle (deg)	Start Loc. (ft)	End Loc. (ft)	Start Magnitude	End Magnitude
1 Distributed	90.000	0.000	22.670	-60.000	-60.000 lb/ft

Loading: Roof Live Load

Type	Angle (deg)	Start Loc. (ft)	End Loc. (ft)	Start Magnitude	End Magnitude
1 Distributed	90.000	0.000	22.670	-40.000	-40.000 lb/ft

Load Combination: 1.4D

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.400

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Load Combination: 1.2D+1.6L+0.5Lr
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Live Load	1.600
3 Product Load	1.600
4 Roof Live Load	0.500

Load Combination: 1.2D+1.6L+0.3S
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Live Load	1.600
3 Product Load	1.600
4 Snow Load	0.300

Load Combination: 1.2D+1.6Lr+L
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Roof Live Load	1.600
3 Live Load	1.000
4 Product Load	1.000

Load Combination: 0.9D+W
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	0.900
2 Wind Load	1.000

Load Combination: 1.2D+E+L+0.15S
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Earthquake Load	1.000
3 Live Load	1.000
4 Product Load	1.000
5 Snow Load	0.150

Load Combination: 0.9D+E
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	0.900
2 Earthquake Load	1.000

Member Check - AISI S100-16/S3-22, US, LRFD

Load Combination: 1.2D+1.6Lr+L
Design Parameters at 11.335 ft, Left side:

Lx	22.670 ft	Ly	11.335 ft	Lt	11.335 ft
Kx	1.0000	Ky	1.0000	Kt	1.0000

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 By: PD
 Printed: 24-07-2025 16:31:32

Section: Section 1.cfss

Material Type: A653 SS Grade 50/1, Fy=50 ksi

Cbx	1.2987	Cby	1.0000	ex	0.0000 in
Cmx	1.0000	Cmy	1.0000	ey	0.0000 in
Braced Flange: Top		kφ	0 lb		
Red. Factor, R: 0		Lm	22.670 ft		

Loads:	P (lb)	Mx (lb-ft)	Vy (lb)	My (lb-ft)	Vx (lb)
Total	0	8737	0	0	0
Applied	0	8737	0	0	0
Strength	8803	11926	8371	1639	10496

Interaction Equations

Eq. H1.2-1	(P, Mx, My)	$0.000 + 0.733 + 0.000 = 0.733 \leq 1.0$
Eq. H2-1	(Mx, Vy)	$\text{Sqrt}(0.465 + 0.000) = 0.682 \leq 1.0$
Eq. H2-1	(My, Vx)	$\text{Sqrt}(0.000 + 0.000) = 0.000 \leq 1.0$

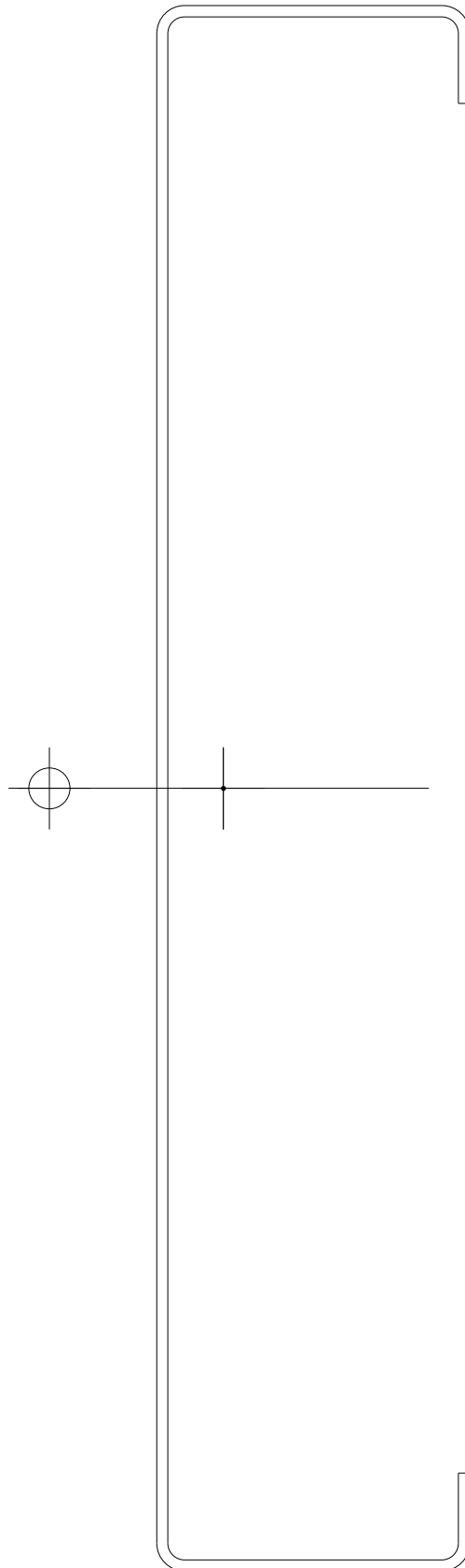
11.1 Floor joist Design

CFS Version 14.0.1
Section: Section 1.cfss
Channel 10x2x0.625-68 mil

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Section Inputs

Material: A653 SS Grade 50/1
No cold work of forming strength increase.
No inelastic reserve strength increase.
Modulus of Elasticity, E 29500 ksi
Yield Strength, Fy 50 ksi
Tensile Strength, Fu 65 ksi
Min Elongation in 2 inches 12 %
Torsion Constant Override, J 0 in⁴
Warping Constant Override, Cw 0 in⁶

Stiffened Channel, Thickness 0.0713 in (68 mil)

Placement of Part from Origin:

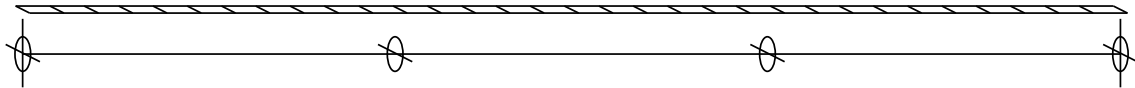
X to center of gravity 0 in

Y to center of gravity 0 in

Outside dimensions, Open shape

	Length (in)	Angle (deg)	Radius (in)	Web	k Coef.	Hole Size (in)	Distance (in)
1	0.625	270.000	0.10690	None	0.000	0.000	0.313
2	2.000	180.000	0.10690	Single	0.000	0.000	1.000
3	10.000	90.000	0.10690	Cee	0.000	0.000	5.000
4	2.000	0.000	0.10690	Single	0.000	0.000	1.000
5	0.625	-90.000	0.10690	None	0.000	0.000	0.313

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Analysis Inputs

General

Member Orientation: Horizontal
 Calculate global buckling using specification equations
 Do not include torsion in member checks

Members

Section File				Revision Date and Time			
1 Section 1.cfss				22-07-2025 15:29:32			
Material			Area	Length	Weight		
			(in²)	(ft)	(lb)		
1	A653 SS Grade 50/1		1.0495	18.670	66.623		
Total				18.670	66.623		
Start Loc.	End Loc.	Braced	R	kφ	Lm	ex	ey
(ft)	(ft)	Flange		(lb)	(ft)	(in)	(in)
1	0.000	18.670 Top	0.0000	0.0000	19.000	0.000	0.000

Supports

Type	Location (ft)	Bearing (in)	Fastened	K
1 XYT	0.000	3.25	No	1.0000
2 XT	6.333	1.00	No	1.0000
3 XT	12.667	1.00	No	1.0000
4 XYT	18.670	3.25	No	1.0000

Loading: Dead Load

Type	Angle (deg)	Start Loc. (ft)	End Loc. (ft)	Start Magnitude	End Magnitude
1 Distributed	90.000	0.000	18.670	-50.000	-50.000 lb/ft

Loading: Live Load

Type	Angle (deg)	Start Loc. (ft)	End Loc. (ft)	Start Magnitude	End Magnitude
1 Distributed	90.000	0.000	18.670	-80.000	-80.000 lb/ft

Load Combination: 1.4D

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.400

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Load Combination: 1.2D+1.6L+0.5Lr
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Live Load	1.600
3 Product Load	1.600
4 Roof Live Load	0.500

Load Combination: 1.2D+1.6Lr+L
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Roof Live Load	1.600
3 Live Load	1.000
4 Product Load	1.000

Load Combination: 1.2D+1.6Lr+0.5W
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Roof Live Load	1.600
3 Wind Load	0.500

Load Combination: 0.9D+W
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	0.900
2 Wind Load	1.000

Load Combination: 1.2D+E+L+0.15S
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Earthquake Load	1.000
3 Live Load	1.000
4 Product Load	1.000
5 Snow Load	0.150

Load Combination: 0.9D+E
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	0.900
2 Earthquake Load	1.000

Member Check - AISI S100-16/S3-22, US, LRFD

Load Combination: 1.2D+1.6L+0.5Lr
Design Parameters at 9.335 ft:

Lx	18.670 ft	Ly	6.333 ft	Lt	6.333 ft
Kx	1.0000	Ky	1.0000	Kt	1.0000

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 Printed: 22-07-2025 15:51:41

Section: Section 1.cfss

Material Type: A653 SS Grade 50/1, Fy=50 ksi

Cbx	1.0143	Cby	1.0000	ex	0.0000 in
Cmx	1.0000	Cmy	1.0000	ey	0.0000 in
Braced Flange: Top		kφ	0 lb		
Red. Factor, R: 0		Lm	19.000 ft		

Loads:	P (lb)	Mx (lb-ft)	Vy (lb)	My (lb-ft)	Vx (lb)
Total	0	8191.4	0	0.0	0
Applied	0	8191.4	0	0.0	0
Strength	11432	8260.4	6920	1085.7	6328

Interaction Equations

Eq. H1.2-1	(P, Mx, My)	$0.000 + 0.992 + 0.000 = 0.992 \leq 1.0$
Eq. H2-1	(Mx, Vy)	$\text{Sqrt}(0.812 + 0.000) = 0.901 \leq 1.0$
Eq. H2-1	(My, Vx)	$\text{Sqrt}(0.000 + 0.000) = 0.000 \leq 1.0$

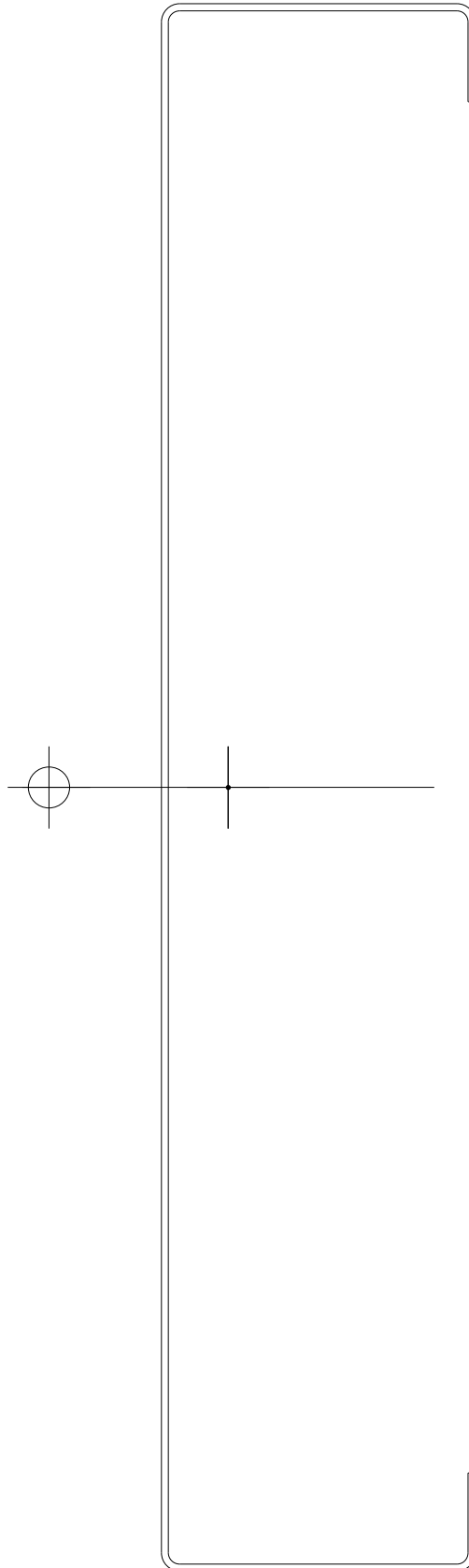
11.2 Floor joist Design

CFS Version 14.0.1
Section: Section 1.cfss
Channel 10x2x0.625-43 mil

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Section Inputs

Material: A653 SS Grade 50/1
No cold work of forming strength increase.
No inelastic reserve strength increase.
Modulus of Elasticity, E 29500 ksi
Yield Strength, Fy 50 ksi
Tensile Strength, Fu 65 ksi
Min Elongation in 2 inches 12 %
Torsion Constant Override, J 0 in⁴
Warping Constant Override, Cw 0 in⁶

Stiffened Channel, Thickness 0.0451 in (43 mil)

Placement of Part from Origin:

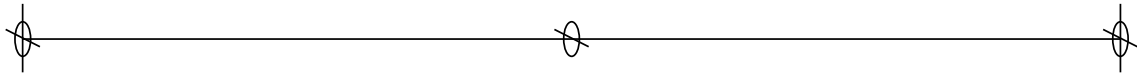
X to center of gravity 0 in

Y to center of gravity 0 in

Outside dimensions, Open shape

	Length (in)	Angle (deg)	Radius (in)	Web	k Coef.	Hole Size (in)	Distance (in)
1	0.625	270.000	0.071200	None	0.000	0.000	0.313
2	2.000	180.000	0.071200	Single	0.000	0.000	1.000
3	10.000	90.000	0.071200	Cee	0.000	0.000	5.000
4	2.000	0.000	0.071200	Single	0.000	0.000	1.000
5	0.625	-90.000	0.071200	None	0.000	0.000	0.313

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Analysis Inputs

General

Member Orientation: Horizontal
Calculate global buckling using specification equations
Do not include torsion in member checks

Members

Section File				Revision Date and Time				
1 Section 1.cfss				22-07-2025 15:53:09				
Material				Area	Length	Weight		
				(in²)	(ft)	(lb)		
1 A653 SS Grade 50/1				0.67238	8.0000	18.289		
Total					8.0000	18.289		
Start Loc.		End Loc.	Braced	R	kφ	Lm	ex	ey
(ft)		(ft)	Flange		(lb)	(ft)	(in)	(in)
1 0.0000		8.0000	None	0.0000	0.0000	8.0000	0.000	0.000

Supports

Type	Location (ft)	Bearing (in)	Fastened	K
1 XYT	0.0000	3.25	No	1.0000
2 XT	4.0000	1.00	No	1.0000
3 XYT	8.0000	3.25	No	1.0000

Loading: Dead Load

Type	Angle (deg)	Start Loc. (ft)	End Loc. (ft)	Start Magnitude	End Magnitude
1 Distributed	90.000	0.0000	8.0000	-50.000	-50.000 lb/ft

Loading: Live Load

Type	Angle (deg)	Start Loc. (ft)	End Loc. (ft)	Start Magnitude	End Magnitude
1 Distributed	90.000	0.0000	8.0000	-80.000	-80.000 lb/ft

Load Combination: 1.4D

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.400

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Load Combination: 1.2D+1.6L+0.5Lr
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Live Load	1.600
3 Product Load	1.600
4 Roof Live Load	0.500

Load Combination: 1.2D+1.6Lr+L
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Roof Live Load	1.600
3 Live Load	1.000
4 Product Load	1.000

Load Combination: 1.2D+1.6Lr+0.5W
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Roof Live Load	1.600
3 Wind Load	0.500

Load Combination: 0.9D+W
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	0.900
2 Wind Load	1.000

Load Combination: 1.2D+E+L+0.15S
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Earthquake Load	1.000
3 Live Load	1.000
4 Product Load	1.000
5 Snow Load	0.150

Load Combination: 0.9D+E
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	0.900
2 Earthquake Load	1.000

Member Check - AISI S100-16/S3-22, US, LRFD

Load Combination: 1.2D+1.6L+0.5Lr

Design Parameters at 4.0000 ft, Left side:

Lx	8.0000 ft	Ly	4.0000 ft	Lt	4.0000 ft
Kx	1.0000	Ky	1.0000	Kt	1.0000

Rev. Date: 22-07-2025 15:53:29
 By: PD
 Printed: 22-07-2025 15:53:42

Section: Section 1.cfss

Material Type: A653 SS Grade 50/1, Fy=50 ksi

Cbx	1.2987	Cby	1.0000	ex	0.0000 in
Cmx	1.0000	Cmy	1.0000	ey	0.0000 in
Braced Flange:	None	kφ	0 lb		
Red. Factor, R:	0	Lm	8.0000 ft		

Loads:	P	Mx	Vy	My	Vx
	(lb)	(lb-ft)	(lb)	(lb-ft)	(lb)
Total	0.0	1504.0	0.0	0.0	0.0
Applied	0.0	1504.0	0.0	0.0	0.0
Strength	8430.2	4304.6	2250.5	623.3	4304.3

Interaction Equations

Eq. H1.2-1 (P, Mx, My) $0.000 + 0.349 + 0.000 = 0.349 \leq 1.0$
 Eq. H2-1 (Mx, Vy) $\text{Sqrt}(0.099 + 0.000) = 0.315 \leq 1.0$
 Eq. H2-1 (My, Vx) $\text{Sqrt}(0.000 + 0.000) = 0.000 \leq 1.0$

Stiffened Channel element 3 h/t exceeds 200.

11.3 Floor joist Design

CFS Version 14.0.1

Section: Section 1.cfss

Double Channel 10x2x0.625-54 mil

PD

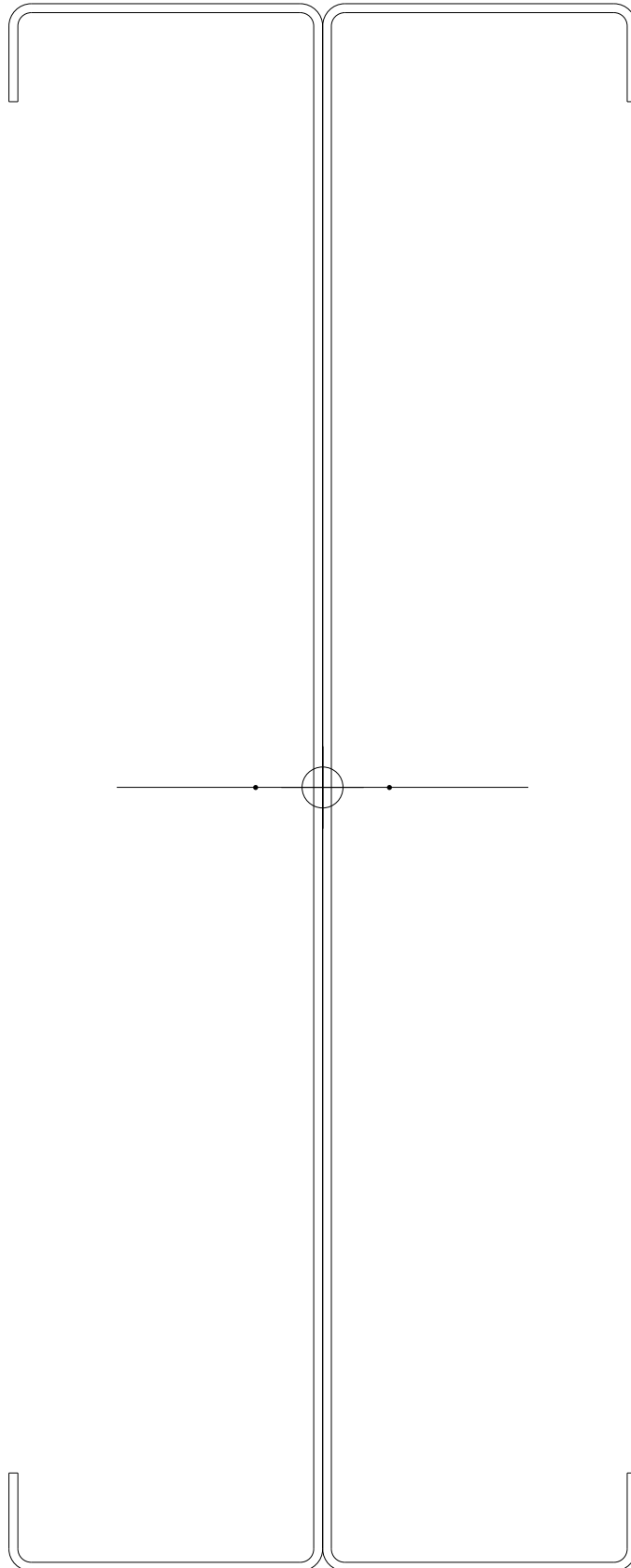
FCOM

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Rev. Date: 22-07-2025 14:53:07

By: PD

Printed: 22-07-2025 14:54:10



Section: Section 1.cfss
Double Channel 10x2x0.625-54 mil

PD
FCOM

Rev. Date: 22-07-2025 14:53:07

By: PD

Printed: 22-07-2025 14:54:10

Section Inputs

Material: A653 SS Grade 50/1

No cold work of forming strength increase.

No inelastic reserve strength increase.

Modulus of Elasticity, E 29500 ksi

Yield Strength, Fy 50 ksi

Tensile Strength, Fu 65 ksi

Min Elongation in 2 inches 12 %

Torsion Constant Override, J 0 in⁴

Warping Constant Override, Cw 26.239 in⁶

Connector Spacing 0 in

Right Channel, Thickness 0.0566 in (54 mil)

Placement of Part from Origin:

X to left edge 0 in

Y to center of gravity 0 in

Outside dimensions, Open shape

	Length (in)	Angle (deg)	Radius (in)	Web	k Coef.	Hole Size (in)	Distance (in)
1	0.625	270.000	0.084900	None	0.000	0.000	0.313
2	2.000	180.000	0.084900	Single	0.000	0.000	1.000
3	10.000	90.000	0.084900	Double	0.000	0.000	5.000
4	2.000	0.000	0.084900	Single	0.000	0.000	1.000
5	0.625	-90.000	0.084900	None	0.000	0.000	0.313

Left Channel, Thickness 0.0566 in (54 mil)

Placement of Part from Origin:

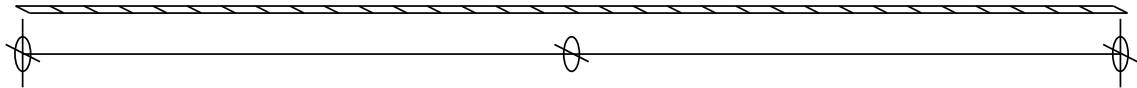
X to right edge 0 in

Y to center of gravity 0 in

Outside dimensions, Open shape

	Length (in)	Angle (deg)	Radius (in)	Web	k Coef.	Hole Size (in)	Distance (in)
1	0.625	-90.000	0.084900	None	0.000	0.000	0.313
2	2.000	0.000	0.084900	Single	0.000	0.000	1.000
3	10.000	90.000	0.084900	Double	0.000	0.000	5.000
4	2.000	180.000	0.084900	Single	0.000	0.000	1.000
5	0.625	270.000	0.084900	None	0.000	0.000	0.313

Rev. Date: 22-07-2025 14:53:50
By: PD
Printed: 22-07-2025 14:54:10



Analysis Inputs

General

Member Orientation: Horizontal
Calculate global buckling using specification equations
Do not include torsion in member checks

Members

Section File				Revision Date and Time			
1 Section 1.cfss				22-07-2025 14:53:07			
Material		Area	Length	Weight			
		(in²)	(ft)	(lb)			
1	A653 SS Grade 50/1	1.6787	22.500	128.42			
Total			22.500	128.42			
Start Loc.	End Loc.	Braced	R	kφ	Lm	ex	ey
(ft)	(ft)	Flange		(lb)	(ft)	(in)	(in)
1 0.000	22.500	Top	0.0000	0.0000	22.500	0.000	0.000

Supports

Type	Location (ft)	Bearing (in)	Fastened	K
1 XYT	0.000	3.25	No	1.0000
2 XT	11.250	1.00	No	1.0000
3 XYT	22.500	3.25	No	1.0000

Loading: Dead Load

Type	Angle (deg)	Start Loc. (ft)	End Loc. (ft)	Start Magnitude	End Magnitude
1 Distributed	90.000	0.000	22.500	-50.000	-50.000 lb/ft

Loading: Live Load

Type	Angle (deg)	Start Loc. (ft)	End Loc. (ft)	Start Magnitude	End Magnitude
1 Distributed	90.000	0.000	22.500	-80.000	-80.000 lb/ft

Load Combination: 1.4D

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.400

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Load Combination: 1.2D+1.6L+0.5Lr
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Live Load	1.600
3 Product Load	1.600
4 Roof Live Load	0.500

Load Combination: 1.2D+1.6Lr+L
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Roof Live Load	1.600
3 Live Load	1.000
4 Product Load	1.000

Load Combination: 1.2D+1.6Lr+0.5W
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Roof Live Load	1.600
3 Wind Load	0.500

Load Combination: 0.9D+W
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	0.900
2 Wind Load	1.000

Load Combination: 1.2D+E+L+0.15S
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Earthquake Load	1.000
3 Live Load	1.000
4 Product Load	1.000
5 Snow Load	0.150

Load Combination: 0.9D+E
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	0.900
2 Earthquake Load	1.000

Member Check - AISI S100-16/S3-22, US, LRFD

Load Combination: 1.2D+1.6L+0.5Lr

Design Parameters at 11.250 ft, Left side:

Lx	22.500 ft	Ly	11.250 ft	Lt	11.250 ft
Kx	1.0000	Ky	1.0000	Kt	1.0000

Rev. Date: 22-07-2025 14:53:50
 By: PD
 Printed: 22-07-2025 14:54:10

Section: Section 1.cfss

Material Type: A653 SS Grade 50/1, Fy=50 ksi

Cbx	1.2987	Cby	1.0000	ex	0.0000 in
Cmx	1.0000	Cmy	1.0000	ey	0.0000 in
Braced Flange: Top		kφ	0 lb		
Red. Factor, R: 0		Lm	22.500 ft		

Loads:	P	Mx	Vy	My	Vx
	(lb)	(lb-ft)	(lb)	(lb-ft)	(lb)
Total	0	11897	0	0	0
Applied	0	11897	0	0	0
Strength	8908	11926	8371	1639	10496

Interaction Equations

Eq. H1.2-1 (P, Mx, My) $0.000 + 0.998 + 0.000 = 0.998 \leq 1.0$
 Eq. H2-1 (Mx, Vy) $\text{Sqrt}(0.862 + 0.000) = 0.928 \leq 1.0$
 Eq. H2-1 (My, Vx) $\text{Sqrt}(0.000 + 0.000) = 0.000 \leq 1.0$

Web Crippling Check - AISI S100-16/S3-22, US, LRFD

Load Combination: 1.2D+1.6L+0.5Lr

Parameters at 22.500 ft:

Total Load: 2115.0 lb on bottom flange

Total Moment: -0.002 lb-ft

Bearing: 3.2500 in

Flange fastened to bearing surface: No

Distance from edge of bearing to edge of opposite load: 22.500 ft

Section: Section 1.cfss

Material Type: A653 SS Grade 50/1, Fy=50 ksi

Applied Load: 2115.0 lb on bottom flange

Applied Moment: 0 lb-ft

Distance from edge of bearing to end of member: 0.0000 ft

Part	Elem	Calculation Type	φPn (lb)	Pny (lb)	Notes
1	3	Double, US-EOF	3066.5	3066.5	
2	3	Double, US-EOF	3066.5	3066.5	
		Web Crippling Strength		6133.0	

Web Crippling Check: 2115.0 lb ≤ 6133.0 lb

Moment Check: 0 lb-ft ≤ 12814 lb-ft

Interaction Equations

Eq. H3-2b (P, M) $0.228 + 0.000 = 0.228 \leq 1.314$

12.1 External Wall Design

CFS Version 14.0.1

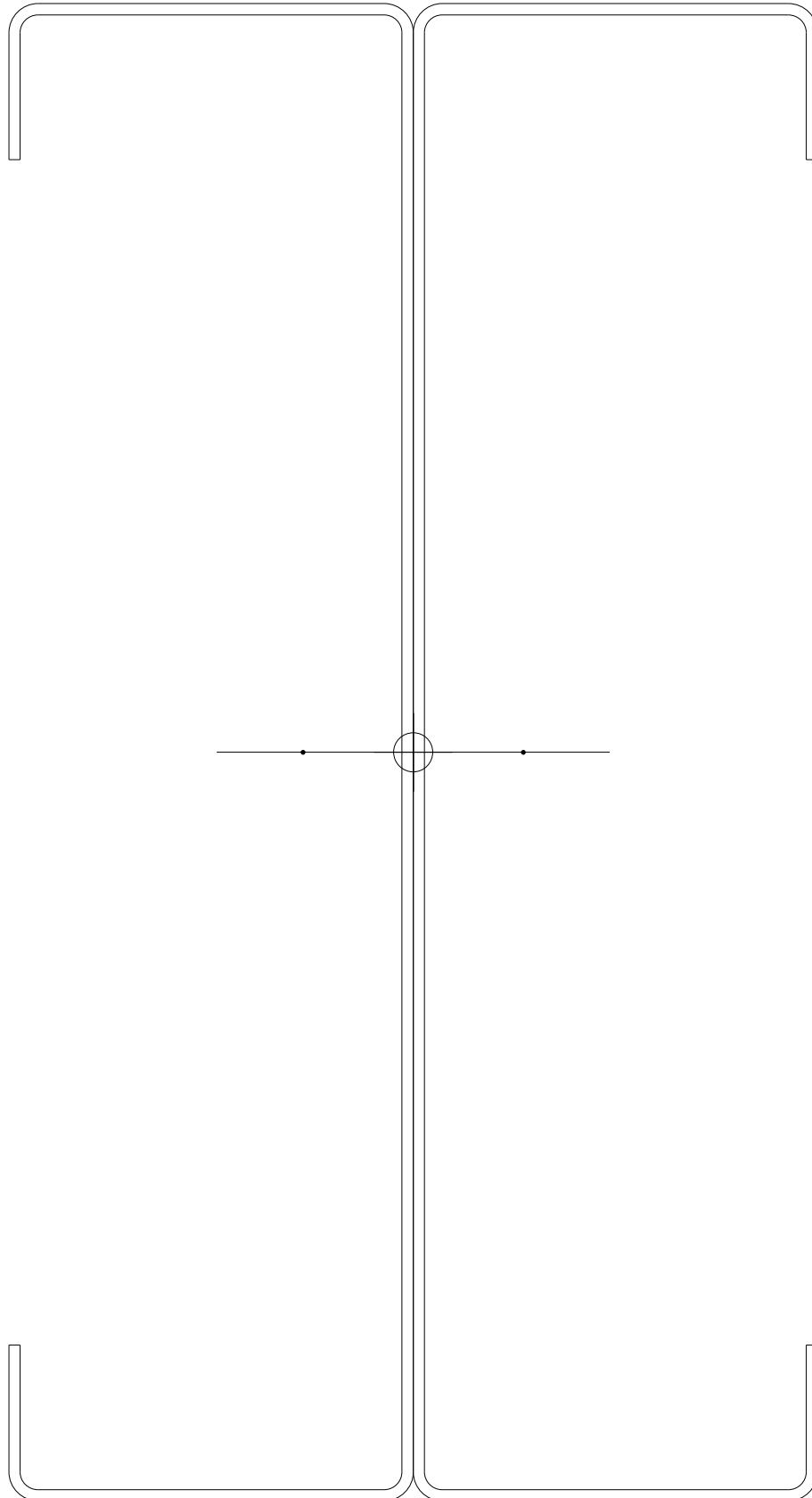
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Section: Double Channel 6x1.62x0.625-43 mil.cfss PD
Double Channel 6x1.62x0.625-43 mil FCOM

Rev. Date: 16-07-2025 16:16:22

By: PD

Printed: 24-07-2025 16:47:21



Section: Double Channel 6x1.62x0.625-43 mil.cfss PD
 Double Channel 6x1.62x0.625-43 mil FCOM

Rev. Date: 16-07-2025 16:16:22

By: PD

Printed: 24-07-2025 16:47:21

Section Inputs

Material: A653 SS Grade 50/1

No cold work of forming strength increase.

No inelastic reserve strength increase.

Modulus of Elasticity, E 29500 ksi

Yield Strength, Fy 50 ksi

Tensile Strength, Fu 65 ksi

Min Elongation in 2 inches 12 %

Torsion Constant Override, J 0 in⁴

Warping Constant Override, Cw 4.4687 in⁶

Connector Spacing 0 in

Right Channel, Thickness 0.0451 in (43 mil)

Placement of Part from Origin:

X to left edge 0 in

Y to center of gravity 0 in

Outside dimensions, Open shape

	Length (in)	Angle (deg)	Radius (in)	Web	k Coef.	Hole Size (in)	Distance (in)
1	0.6250	270.000	0.071200	None	0.000	0.0000	0.3125
2	1.6200	180.000	0.071200	Single	0.000	0.0000	0.8100
3	6.0000	90.000	0.071200	Double	0.000	0.0000	3.0000
4	1.6200	0.000	0.071200	Single	0.000	0.0000	0.8100
5	0.6250	-90.000	0.071200	None	0.000	0.0000	0.3125

Left Channel, Thickness 0.0451 in (43 mil)

Placement of Part from Origin:

X to right edge 0 in

Y to center of gravity 0 in

Outside dimensions, Open shape

	Length (in)	Angle (deg)	Radius (in)	Web	k Coef.	Hole Size (in)	Distance (in)
1	0.6250	-90.000	0.071200	None	0.000	0.0000	0.3125
2	1.6200	0.000	0.071200	Single	0.000	0.0000	0.8100
3	6.0000	90.000	0.071200	Double	0.000	0.0000	3.0000
4	1.6200	180.000	0.071200	Single	0.000	0.0000	0.8100
5	0.6250	270.000	0.071200	None	0.000	0.0000	0.3125

Rev. Date: 24-07-2025 16:46:47
 By: PD
 Printed: 24-07-2025 16:47:21



Analysis Inputs

General

Member Orientation: Vertical
 Calculate global buckling using specification equations
 Do not include torsion in member checks

Members

Section File	Revision Date and Time
1 Double Channel 6x1.62x0.625-43 mil.cfss	16-07-2025 16:16:22

Material	Area (in ²)	Length (ft)	Weight (lb)
1 A653 SS Grade 50/1	0.91541	9.0000	28.011
Total		9.0000	28.011

	Start Loc. (ft)	End Loc. (ft)	Braced Flange	R	kφ (lb)	Lm (ft)	ex (in)	ey (in)
1	0.0000	9.0000	None	0.0000	0.0000	9.0000	0.000	0.000

Supports

Type	Location (ft)	Bearing (in)	Fastened	K
1 XYT	0.0000	2.00	No	1.0000
2 XT	4.5000	1.00	No	1.0000
3 XYT	9.0000	2.00	No	1.0000

Loading: Dead Load

Type	Angle (deg)	Start Loc. (ft)	End Loc. (ft)	Start Magnitude	End Magnitude
1 Axial		0.0000	9.0000	1260.0	1260.0 lb

Analysis: 9 ft Tall Beam-Column.cfsa
9 ft Tall Beam-Column

PD
FCOM

Rev. Date: 24-07-2025 16:46:47

By: PD

Printed: 24-07-2025 16:47:21

Loading: Roof Live Load

Type	Angle (deg)	Start Loc. (ft)	End Loc. (ft)	Start Magnitude	End Magnitude	
1 Axial		0.0000	9.0000	480.0	480.0	lb

Loading: Wind Load

Type	Angle (deg)	Start Loc. (ft)	End Loc. (ft)	Start Magnitude	End Magnitude	
1 Distributed	90.000	0.0000	9.0000	-230.00	-230.00	lb/ft

Load Combination: 1.4D

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.400

Load Combination: 1.2D+1.6L+0.5Lr

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Live Load	1.600
3 Product Load	1.600
4 Roof Live Load	0.500

Load Combination: 1.2D+1.6L+0.3S

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Live Load	1.600
3 Product Load	1.600
4 Snow Load	0.300

Load Combination: 1.2D+1.6Lr+L

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Roof Live Load	1.600
3 Live Load	1.000
4 Product Load	1.000

Load Combination: 1.2D+1.6Lr+0.5W

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Roof Live Load	1.600
3 Wind Load	0.500

Load Combination: 1.2D+S+0.5W

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Snow Load	1.000
3 Wind Load	0.500

Analysis: 9 ft Tall Beam-Column.cfss
9 ft Tall Beam-Column

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Rev. Date: 24-07-2025 16:46:47

By: PD

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Load Combination: 1.2D+W+L+0.5Lr

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Wind Load	1.000
3 Live Load	1.000
4 Product Load	1.000
5 Roof Live Load	0.500

Load Combination: 1.2D+W+L+0.3S

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Wind Load	1.000
3 Live Load	1.000
4 Product Load	1.000
5 Snow Load	0.300

Load Combination: 0.9D+W

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	0.900
2 Wind Load	1.000

Load Combination: 1.2D+E+L+0.15S

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Earthquake Load	1.000
3 Live Load	1.000
4 Product Load	1.000
5 Snow Load	0.150

Load Combination: 0.9D+E

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	0.900
2 Earthquake Load	1.000

Member Check - AISI S100-16/S3-22, US, LRFD

Load Combination: 1.2D+W+L+0.5Lr

Design Parameters at 4.5000 ft:

Lx	9.0000 ft	Ly	4.5000 ft	Lt	4.5000 ft
Kx	1.0000	Ky	1.0000	Kt	1.0000

Section: Double Channel 6x1.62x0.625-43 mil.cfss

Material Type: A653 SS Grade 50/1, Fy=50 ksi

Cbx	1.2987	Cby	2.0833	ex	0.0000 in
Cmx	1.0000	Cmy	1.0000	ey	0.0000 in

Braced Flange: None kφ 0 lb

Red. Factor, R: 0 Lm 9.0000 ft

Analysis: 9 ft Tall Beam-Column.cfsa
9 ft Tall Beam-Column

PD
FCOM

Rev. Date: 24-07-2025 16:46:47

By: PD

Printed: 24-07-2025 16:47:21

Loads:	P (lb)	Mx (lb-ft)	Vy (lb)	My (lb-ft)	Vx (lb)
Total	1752	2328.8	0	0.0	0
Applied	1752	2363.6	0	0.0	0
Strength	17445	4923.5	5582	1023.5	6758

Interaction Equations

Eq. H1.2-1 (P, Mx, My) $0.100 + 0.480 + 0.000 = 0.580 \leq 1.0$

Eq. H2-1 (Mx, Vy) $\text{Sqrt}(0.167 + 0.000) = 0.408 \leq 1.0$

Eq. H2-1 (My, Vx) $\text{Sqrt}(0.000 + 0.000) = 0.000 \leq 1.0$

12.2 External Wall Design

CFS Version 14.0.1

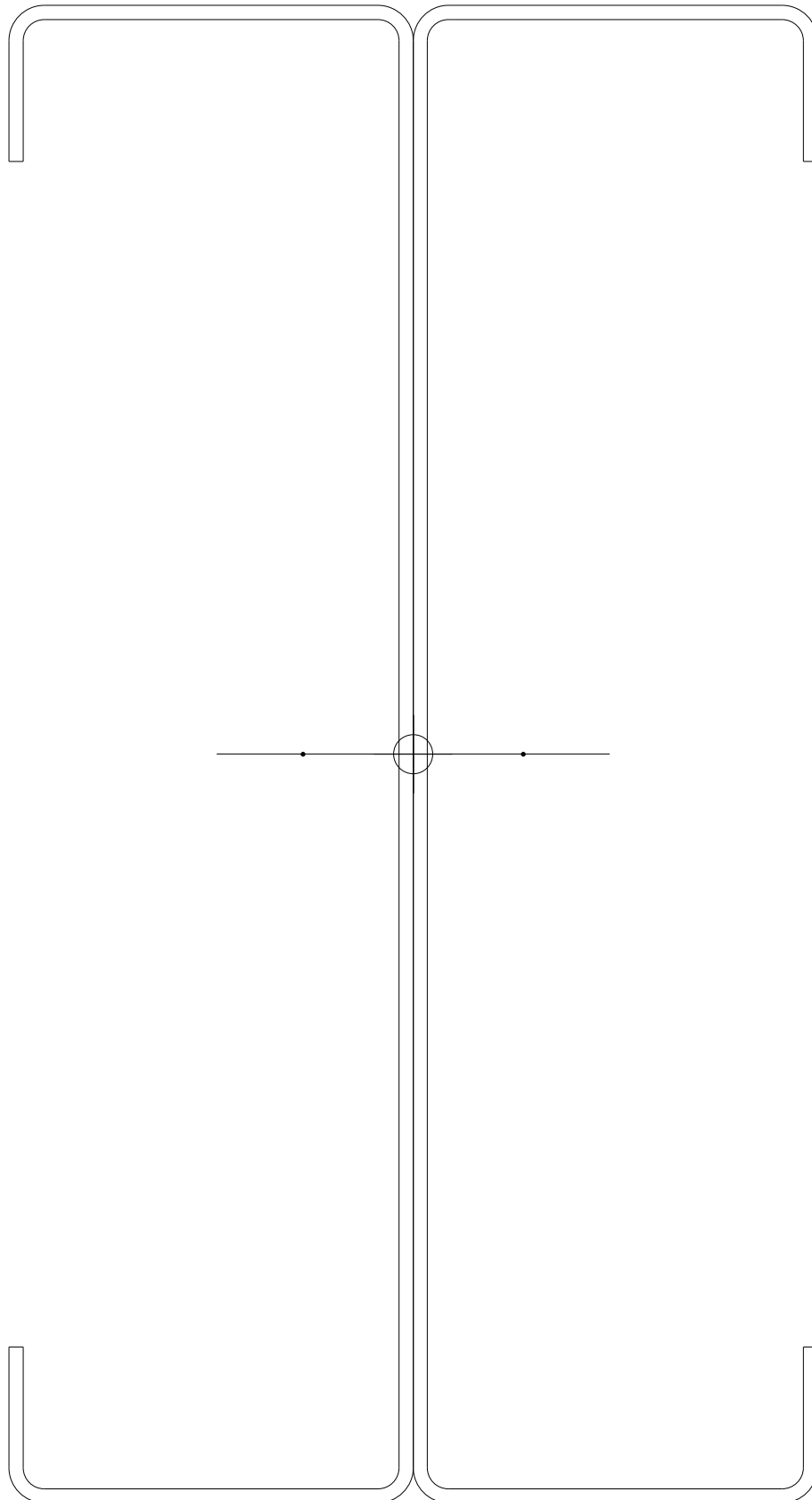
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Section: Double Channel 6x1.62x0.625-54 mil.cfss PD
Double Channel 6x1.62x0.625-54 mil FCOM

Rev. Date: 16-07-2025 16:40:32

By: PD

Printed: 24-07-2025 16:58:42



Section: Double Channel 6x1.62x0.625-54 mil.cfss PD
 Double Channel 6x1.62x0.625-54 mil FCOM

Rev. Date: 16-07-2025 16:40:32

By: PD

Printed: 24-07-2025 16:58:42

Section Inputs

Material: A653 SS Grade 50/1

No cold work of forming strength increase.

No inelastic reserve strength increase.

Modulus of Elasticity, E 29500 ksi

Yield Strength, Fy 50 ksi

Tensile Strength, Fu 65 ksi

Min Elongation in 2 inches 12 %

Torsion Constant Override, J 0 in⁴

Warping Constant Override, Cw 5.4665 in⁶

Connector Spacing 0 in

Right Channel, Thickness 0.0566 in (54 mil)

Placement of Part from Origin:

X to left edge 0 in

Y to center of gravity 0 in

Outside dimensions, Open shape

	Length (in)	Angle (deg)	Radius (in)	Web	k Coef.	Hole Size (in)	Distance (in)
1	0.6250	270.000	0.084900	None	0.000	0.0000	0.3125
2	1.6200	180.000	0.084900	Single	0.000	0.0000	0.8100
3	6.0000	90.000	0.084900	Double	0.000	0.0000	3.0000
4	1.6200	0.000	0.084900	Single	0.000	0.0000	0.8100
5	0.6250	-90.000	0.084900	None	0.000	0.0000	0.3125

Left Channel, Thickness 0.0566 in (54 mil)

Placement of Part from Origin:

X to right edge 0 in

Y to center of gravity 0 in

Outside dimensions, Open shape

	Length (in)	Angle (deg)	Radius (in)	Web	k Coef.	Hole Size (in)	Distance (in)
1	0.6250	-90.000	0.084900	None	0.000	0.0000	0.3125
2	1.6200	0.000	0.084900	Single	0.000	0.0000	0.8100
3	6.0000	90.000	0.084900	Double	0.000	0.0000	3.0000
4	1.6200	180.000	0.084900	Single	0.000	0.0000	0.8100
5	0.6250	270.000	0.084900	None	0.000	0.0000	0.3125

Rev. Date: 24-07-2025 16:58:03
By: PD
Printed: 24-07-2025 16:58:42



Analysis Inputs

General

Member Orientation: Vertical
Calculate global buckling using specification equations
Do not include torsion in member checks

Members

Section File	Revision Date and Time
1 Double Channel 6x1.62x0.625-54 mil.cfss	16-07-2025 16:40:32

Material	Area (in ²)	Length (ft)	Weight (lb)
1 A653 SS Grade 50/1	1.1398	11.000	42.630
Total		11.000	42.630

Start Loc. (ft)	End Loc. (ft)	Braced Flange	R	kφ (lb)	Lm (ft)	ex (in)	ey (in)
1 0.000	11.000	None	0.0000	0.0000	11.000	0.000	0.000

Supports

Type	Location (ft)	Bearing (in)	Fastened	K
1 XYT	0.000	2.00	No	1.0000
2 XT	3.660	1.00	No	1.0000
3 XT	7.320	1.00	No	1.0000
4 XYT	11.000	2.00	No	1.0000

Analysis: 11 ft Tall Beam-Column.cfsa
11 ft Tall Beam-Column

PD
FCOM

Rev. Date: 24-07-2025 16:58:03

By: PD

Printed: 24-07-2025 16:58:42

Loading: Dead Load

Type	Angle (deg)	Start Loc. (ft)	End Loc. (ft)	Start Magnitude	End Magnitude	
1 Axial		0.000	11.000	2520.0	2520.0	lb

Loading: Live Load

Type	Angle (deg)	Start Loc. (ft)	End Loc. (ft)	Start Magnitude	End Magnitude	
1 Axial		0.000	11.000	960.0	960.0	lb

Loading: Roof Live Load

Type	Angle (deg)	Start Loc. (ft)	End Loc. (ft)	Start Magnitude	End Magnitude	
1 Axial		0.000	11.000	480.0	480.0	lb

Loading: Wind Load

Type	Angle (deg)	Start Loc. (ft)	End Loc. (ft)	Start Magnitude	End Magnitude	
1 Distributed	90.000	0.000	11.000	-230.00	-230.00	lb/ft

Load Combination: 1.4D

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.400

Load Combination: 1.2D+1.6L+0.5Lr

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Live Load	1.600
3 Product Load	1.600
4 Roof Live Load	0.500

Load Combination: 1.2D+1.6L+0.3S

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Live Load	1.600
3 Product Load	1.600
4 Snow Load	0.300

Load Combination: 1.2D+1.6Lr+L

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Roof Live Load	1.600
3 Live Load	1.000
4 Product Load	1.000

Analysis: 11 ft Tall Beam-Column.cfsa
11 ft Tall Beam-Column

PD
FCOM

Rev. Date: 24-07-2025 16:58:03

By: PD

Printed: 24-07-2025 16:58:42

Load Combination: 1.2D+1.6Lr+0.5W

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Roof Live Load	1.600
3 Wind Load	0.500

Load Combination: 1.2D+S+L

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Snow Load	1.000
3 Live Load	1.000
4 Product Load	1.000

Load Combination: 1.2D+S+0.5W

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Snow Load	1.000
3 Wind Load	0.500

Load Combination: 1.2D+W+L+0.5Lr

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Wind Load	1.000
3 Live Load	1.000
4 Product Load	1.000
5 Roof Live Load	0.500

Load Combination: 1.2D+W+L+0.3S

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Wind Load	1.000
3 Live Load	1.000
4 Product Load	1.000
5 Snow Load	0.300

Load Combination: 0.9D+W

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	0.900
2 Wind Load	1.000

Analysis: 11 ft Tall Beam-Column.cfsa
11 ft Tall Beam-Column

PD
FCOM

Rev. Date: 24-07-2025 16:58:03

By: PD

Printed: 24-07-2025 16:58:42

Load Combination: 1.2D+E+L+0.15S

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Earthquake Load	1.000
3 Live Load	1.000
4 Product Load	1.000
5 Snow Load	0.150

Load Combination: 0.9D+E

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	0.900
2 Earthquake Load	1.000

Member Check - AISI S100-16/S3-22, US, LRFD

Load Combination: 1.2D+W+L+0.5Lr

Design Parameters at 5.500 ft:

Lx	11.000 ft	Ly	3.660 ft	Lt	3.660 ft
Kx	1.0000	Ky	1.0000	Kt	1.0000

Section: Double Channel 6x1.62x0.625-54 mil.cfs

Material Type: A653 SS Grade 50/1, Fy=50 ksi

Cbx	1.0135	Cby	3.2013	ex	0.0000 in
Cmx	1.0000	Cmy	1.0000	ey	0.0000 in
Braced Flange: None		kφ	0 lb		
Red. Factor, R: 0		Lm	11.000 ft		

Loads:	P (lb)	Mx (lb-ft)	Vy (lb)	My (lb-ft)	Vx (lb)
Total	4224	3478.8	0	0.0	0
Applied	4224	3635.0	0	0.0	0
Strength	26025	6647.6	9010	1367.0	8173

Interaction Equations

Eq. H1.2-1 (P, Mx, My) $0.162 + 0.547 + 0.000 = 0.709 \leq 1.0$

Eq. H2-1 (Mx, Vy) $\text{Sqrt}(0.245 + 0.000) = 0.495 \leq 1.0$

Eq. H2-1 (My, Vx) $\text{Sqrt}(0.000 + 0.000) = 0.000 \leq 1.0$

12.3 External Wall Design

CFS Version 14.0.1

Page 1

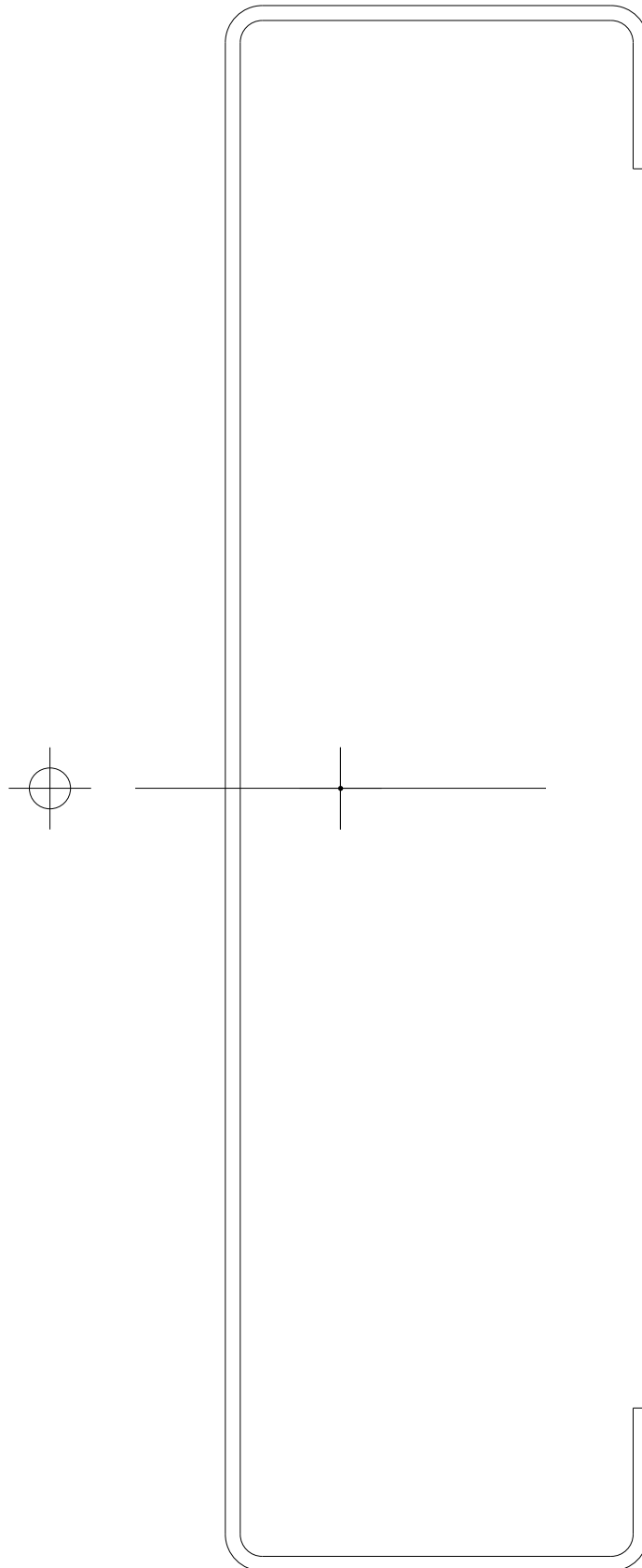
Section: Channel 6x1.62x0.625-54 mil.cfss
Channel 6x1.62x0.625-54 mil

PD
FCOM

Rev. Date: 16-07-2025 15:57:00

By: PD

Printed: 24-07-2025 17:01:39



Section: Channel 6x1.62x0.625-54 mil.cfss
 Channel 6x1.62x0.625-54 mil

PD
 FCOM

Rev. Date: 16-07-2025 15:57:00

By: PD

Printed: 24-07-2025 17:01:39

Section Inputs

Material: A653 SS Grade 50/1

No cold work of forming strength increase.

No inelastic reserve strength increase.

Modulus of Elasticity, E 29500 ksi

Yield Strength, Fy 50 ksi

Tensile Strength, Fu 65 ksi

Min Elongation in 2 inches 12 %

Torsion Constant Override, J 0 in⁴

Warping Constant Override, Cw 0 in⁶

Stiffened Channel, Thickness 0.0566 in (54 mil)

Placement of Part from Origin:

X to center of gravity 0 in

Y to center of gravity 0 in

Outside dimensions, Open shape

	Length (in)	Angle (deg)	Radius (in)	Web	k Coef.	Hole Size (in)	Distance (in)
1	0.6250	270.000	0.084900	None	0.000	0.0000	0.3125
2	1.6200	180.000	0.084900	Single	0.000	0.0000	0.8100
3	6.0000	90.000	0.084900	Cee	0.000	0.0000	3.0000
4	1.6200	0.000	0.084900	Single	0.000	0.0000	0.8100
5	0.6250	-90.000	0.084900	None	0.000	0.0000	0.3125

Rev. Date: 24-07-2025 17:01:22

By: PD

Printed: 24-07-2025 17:01:39



Analysis Inputs

General

Member Orientation: Vertical

Calculate global buckling using specification equations

Do not include torsion in member checks

Members

Section File	Revision Date and Time
1 Channel 6x1.62x0.625-54 mil.cfss	16-07-2025 15:57:00

Material	Area (in ²)	Length (ft)	Weight (lb)
1 A653 SS Grade 50/1	0.56992	9.0000	17.440
Total		9.0000	17.440

	Start Loc. (ft)	End Loc. (ft)	Braced Flange	R	kφ (lb)	Lm (ft)	ex (in)	ey (in)
1	0.0000	9.0000	None	0.0000	0.0000	20.0000	0.000	0.000

Supports

Type	Location (ft)	Bearing (in)	Fastened	K
1 XYT	0.0000	2.00	No	1.0000
2 XT	3.0000	1.00	No	1.0000
3 XT	6.0000	1.00	No	1.0000
4 XYT	9.0000	2.00	No	1.0000

Analysis: 9 ft Tall Beam-Column Parallel.cfsa
9 ft Tall Beam-Column

PD
FCOM

Rev. Date: 24-07-2025 17:01:22

By: PD

Printed: 24-07-2025 17:01:39

Loading: Dead Load

Type	Angle (deg)	Start Loc. (ft)	End Loc. (ft)	Start Magnitude	End Magnitude	
1 Axial		0.0000	9.0000	720.00	720.00	lb

Loading: Roof Live Load

Type	Angle (deg)	Start Loc. (ft)	End Loc. (ft)	Start Magnitude	End Magnitude	
1 Axial		0.0000	9.0000	120.00	120.00	lb

Loading: Wind Load

Type	Angle (deg)	Start Loc. (ft)	End Loc. (ft)	Start Magnitude	End Magnitude	
1 Distributed	90.000	0.0000	9.0000	-230.00	-230.00	lb/ft

Load Combination: 1.4D

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.400

Load Combination: 1.2D+1.6L+0.5Lr

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Live Load	1.600
3 Product Load	1.600
4 Roof Live Load	0.500

Load Combination: 1.2D+1.6L+0.3S

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Live Load	1.600
3 Product Load	1.600
4 Snow Load	0.300

Load Combination: 1.2D+1.6Lr+L

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Roof Live Load	1.600
3 Live Load	1.000
4 Product Load	1.000

Load Combination: 1.2D+1.6Lr+0.5W

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Roof Live Load	1.600
3 Wind Load	0.500

Analysis: 9 ft Tall Beam-Column Parallel.cfsa
9 ft Tall Beam-Column

PD
FCOM

Rev. Date: 24-07-2025 17:01:22

By: PD

Printed: 24-07-2025 17:01:39

Load Combination: 1.2D+S+0.5W

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Snow Load	1.000
3 Wind Load	0.500

Load Combination: 1.2D+W+L+0.5Lr

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Wind Load	1.000
3 Live Load	1.000
4 Product Load	1.000
5 Roof Live Load	0.500

Load Combination: 1.2D+W+L+0.3S

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Wind Load	1.000
3 Live Load	1.000
4 Product Load	1.000
5 Snow Load	0.300

Load Combination: 0.9D+W

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	0.900
2 Wind Load	1.000

Load Combination: 1.2D+E+L+0.15S

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Earthquake Load	1.000
3 Live Load	1.000
4 Product Load	1.000
5 Snow Load	0.150

Load Combination: 0.9D+E

Specification: AISI S100-16/S3-22, US, LRFD

Inflection Point Bracing: No

Loading	Factor
1 Dead Load	0.900
2 Earthquake Load	1.000

Member Check - AISI S100-16/S3-22, US, LRFD

Load Combination: 1.2D+W+L+0.5Lr

Design Parameters at 4.5000 ft:

Lx	9.0000 ft	Ly	3.0000 ft	Lt	3.0000 ft
Kx	1.0000	Ky	1.0000	Kt	1.0000

Analysis: 9 ft Tall Beam-Column Parallel.cfss
 9 ft Tall Beam-Column

PD
 FCOM

Rev. Date: 24-07-2025 17:01:22

By: PD

Printed: 24-07-2025 17:01:39

Section: Channel 6x1.62x0.625-54 mil.cfss

Material Type: A653 SS Grade 50/1, Fy=50 ksi

Cbx 1.0135 Cby 1.0000 ex 0.0000 in

Cmx 1.0000 Cmy 1.0000 ey 0.0000 in

Braced Flange: None kφ 0 lb

Red. Factor, R: 0 Lm 20.0000 ft

Loads:	P (lb)	Mx (lb-ft)	Vy (lb)	My (lb-ft)	Vx (lb)
Total	924	2328.8	0	0.0	0
Applied	924	2358.4	0	0.0	0
Strength	12882	3323.8	4505	631.6	4086

Interaction Equations

Eq. H1.2-1 (P, Mx, My) $0.072 + 0.710 + 0.000 = 0.781 \leq 1.0$

Eq. H2-1 (Mx, Vy) $\text{Sqrt}(0.412 + 0.000) = 0.642 \leq 1.0$

Eq. H2-1 (My, Vx) $\text{Sqrt}(0.000 + 0.000) = 0.000 \leq 1.0$

12.4 External Wall Design

CFS Version 14.0.1

Page 1

Section: Section 1.cfss

PD

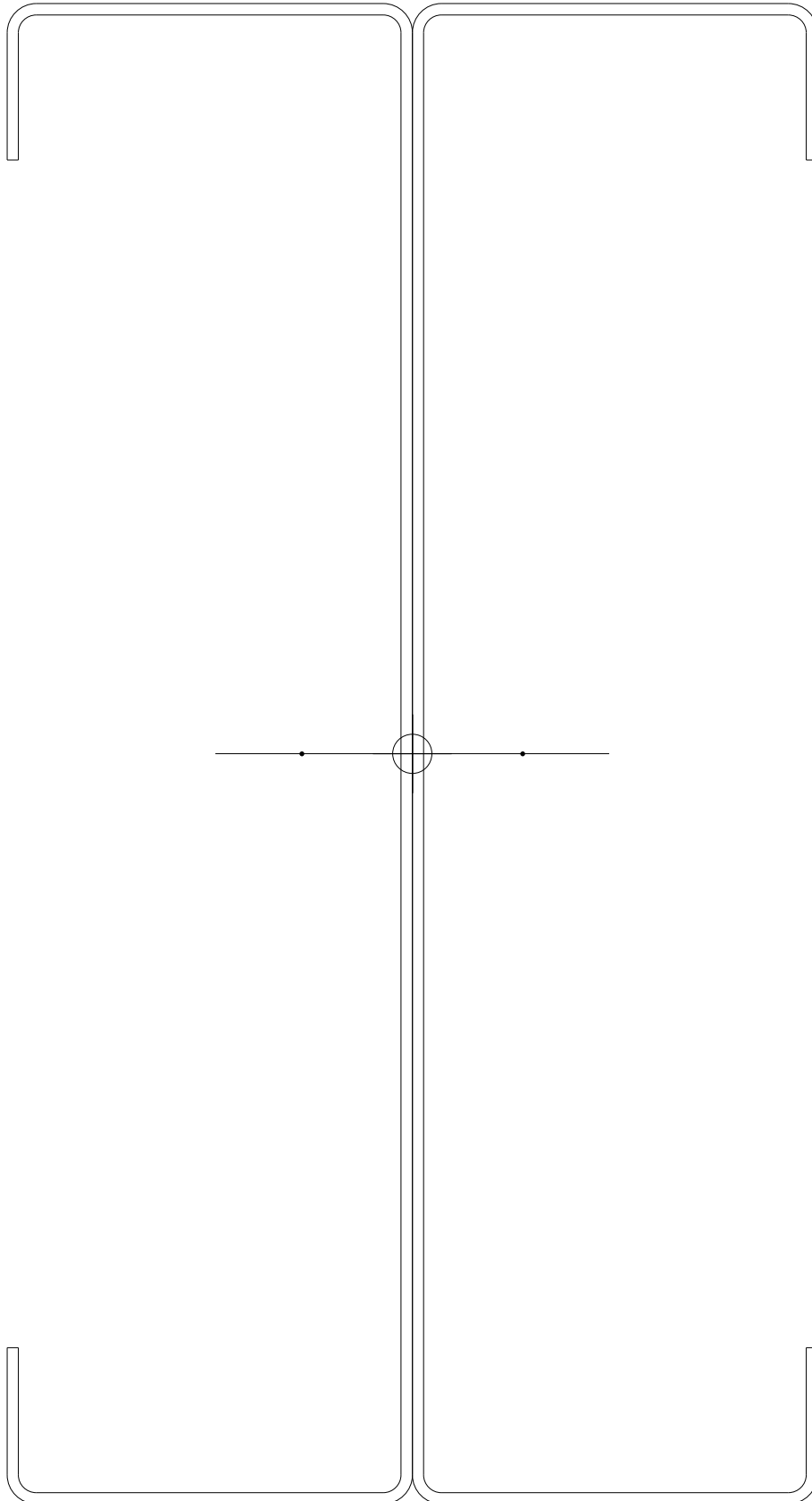
Double Channel 6x1.62x0.625-43 mil

FCOM

Rev. Date: 24-07-2025 17:04:45

By: PD

Printed: 24-07-2025 17:05:51



Section: Section 1.cfss
Double Channel 6x1.62x0.625-43 mil

PD
FCOM

Rev. Date: 24-07-2025 17:04:45

By: PD

Printed: 24-07-2025 17:05:51

Section Inputs

Material: A653 SS Grade 50/1

No cold work of forming strength increase.

No inelastic reserve strength increase.

Modulus of Elasticity, E 29500 ksi

Yield Strength, Fy 50 ksi

Tensile Strength, Fu 65 ksi

Min Elongation in 2 inches 12 %

Torsion Constant Override, J 0 in⁴

Warping Constant Override, Cw 4.4687 in⁶

Connector Spacing 0 in

Right Channel, Thickness 0.0451 in (43 mil)

Placement of Part from Origin:

X to left edge 0 in

Y to center of gravity 0 in

Outside dimensions, Open shape

	Length (in)	Angle (deg)	Radius (in)	Web	k Coef.	Hole Size (in)	Distance (in)
1	0.6250	270.000	0.071200	None	0.000	0.0000	0.3125
2	1.6200	180.000	0.071200	Single	0.000	0.0000	0.8100
3	6.0000	90.000	0.071200	Double	0.000	0.0000	3.0000
4	1.6200	0.000	0.071200	Single	0.000	0.0000	0.8100
5	0.6250	-90.000	0.071200	None	0.000	0.0000	0.3125

Left Channel, Thickness 0.0451 in (43 mil)

Placement of Part from Origin:

X to right edge 0 in

Y to center of gravity 0 in

Outside dimensions, Open shape

	Length (in)	Angle (deg)	Radius (in)	Web	k Coef.	Hole Size (in)	Distance (in)
1	0.6250	-90.000	0.071200	None	0.000	0.0000	0.3125
2	1.6200	0.000	0.071200	Single	0.000	0.0000	0.8100
3	6.0000	90.000	0.071200	Double	0.000	0.0000	3.0000
4	1.6200	180.000	0.071200	Single	0.000	0.0000	0.8100
5	0.6250	270.000	0.071200	None	0.000	0.0000	0.3125

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By: PD
Printed: 24-07-2025 17:05:51



Analysis Inputs

General

Member Orientation: Vertical
Calculate global buckling using specification equations
Do not include torsion in member checks

Members

Section File	Revision Date and Time
1 Section 1.cfss	24-07-2025 17:04:45

Material	Area (in ²)	Length (ft)	Weight (lb)
1 A653 SS Grade 50/1	0.91541	11.000	34.236
Total		11.000	34.236

Start Loc. (ft)	End Loc. (ft)	Braced Flange	R	kφ (lb)	Lm (ft)	ex (in)	ey (in)
1 0.000	11.000	None	0.0000	0.0000	11.000	0.000	0.000

Supports

Type	Location (ft)	Bearing (in)	Fastened	K
1 XYT	0.000	3.25	No	1.0000
2 XT	5.500	1.00	No	1.0000
3 XYT	11.000	3.25	No	1.0000

Loading: Dead Load

Type	Angle (deg)	Start Loc. (ft)	End Loc. (ft)	Start Magnitude	End Magnitude
1 Axial		0.000	11.000	1530.0	1530.0 lb

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By: PD
Printed: 24-07-2025 17:05:51

Loading: Live Load

Type	Angle (deg)	Start Loc. (ft)	End Loc. (ft)	Start Magnitude	End Magnitude	
1 Axial		0.000	11.000	240.0	240.0	lb

Loading: Roof Live Load

Type	Angle (deg)	Start Loc. (ft)	End Loc. (ft)	Start Magnitude	End Magnitude	
1 Axial		0.000	11.000	120.0	120.0	lb

Loading: Wind Load

Type	Angle (deg)	Start Loc. (ft)	End Loc. (ft)	Start Magnitude	End Magnitude	
1 Distributed	90.000	0.000	11.000	-228.00	-228.00	lb/ft

Load Combination: 1.4D

Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.400

Load Combination: 1.2D+1.6L+0.5Lr

Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Live Load	1.600
3 Product Load	1.600
4 Roof Live Load	0.500

Load Combination: 1.2D+1.6L+0.3S

Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Live Load	1.600
3 Product Load	1.600
4 Snow Load	0.300

Load Combination: 1.2D+1.6Lr+L

Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Roof Live Load	1.600
3 Live Load	1.000
4 Product Load	1.000

Load Combination: 1.2D+1.6Lr+0.5W

Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Roof Live Load	1.600
3 Wind Load	0.500

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Load Combination: 1.2D+S+L
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Snow Load	1.000
3 Live Load	1.000
4 Product Load	1.000

Load Combination: 1.2D+S+0.5W
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Snow Load	1.000
3 Wind Load	0.500

Load Combination: 1.2D+W+L+0.5Lr
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Wind Load	1.000
3 Live Load	1.000
4 Product Load	1.000
5 Roof Live Load	0.500

Load Combination: 1.2D+W+L+0.3S
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Wind Load	1.000
3 Live Load	1.000
4 Product Load	1.000
5 Snow Load	0.300

Load Combination: 0.9D+W
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	0.900
2 Wind Load	1.000

Load Combination: 1.2D+E+L+0.15S
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	1.200
2 Earthquake Load	1.000
3 Live Load	1.000
4 Product Load	1.000
5 Snow Load	0.150

Rev. Date: 24-07-2025 17:05:39
By: PD
Printed: 24-07-2025 17:05:51

Load Combination: 0.9D+E
Specification: AISI S100-16/S3-22, US, LRFD
Inflection Point Bracing: No

Loading	Factor
1 Dead Load	0.900
2 Earthquake Load	1.000

Member Check - AISI S100-16/S3-22, US, LRFD

Load Combination: 1.2D+W+L+0.5Lr

Design Parameters at 5.500 ft, Upper side:

Lx	11.000 ft	Ly	5.500 ft	Lt	5.500 ft
Kx	1.0000	Ky	1.0000	Kt	1.0000

Section: Section 1.cfss

Material Type: A653 SS Grade 50/1, Fy=50 ksi

Cbx	1.2987	Cby	2.0833	ex	0.0000 in
Cmx	1.0000	Cmy	1.0000	ey	0.0000 in

Braced Flange: None kφ 0 lb

Red. Factor, R: 0 Lm 11.000 ft

Loads:	P (lb)	Mx (lb-ft)	Vy (lb)	My (lb-ft)	Vx (lb)
Total	2136	3448.5	0	0.0	0
Applied	2136	3543.7	0	0.0	0
Strength	14886	4923.5	5582	1023.5	6758

Interaction Equations

Eq. H1.2-1 (P, Mx, My) $0.143 + 0.720 + 0.000 = 0.863 \leq 1.0$

Eq. H2-1 (Mx, Vy) $\text{Sqrt}(0.374 + 0.000) = 0.612 \leq 1.0$

Eq. H2-1 (My, Vx) $\text{Sqrt}(0.000 + 0.000) = 0.000 \leq 1.0$

12.5 Internal wall design



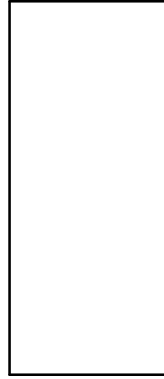
ClarkWestern Building Systems
CW Tech Support: (888) 437-3244
clarkwestern.com

2007 North American Specification LRFD
DATE: 7/3/2025

SECTION DESIGNATION: 362S162-43 [33] Single

Section Dimensions:

Web Height = 3.625 in
Top Flange = 1.625 in
Bottom Flange = 1.625 in
Stiffening Lip = 0.500 in
Inside Corner Radius = 0.0712 in
Punchout Width = 1.500 in
Punchout Length = 4.000 in
Design Thickness = 0.0451 in



Steel Properties:

Fy = 33.000 ksi
Fu = 45.000 ksi
Fya = 33.000 ksi

COMBINED AXIAL AND BENDING LOADS

INPUT PARAMETERS

Overall Wall Height = 9.0
Lateral Load = 5.0 psf
Load Factor for Lateral Load = 1.00
Lateral Load multiplied by 0.70 for deflection calculations
Studs Considered Fully Braced for Bending

K-phi (flexure) for Distortional Buckling = 0.00 lb*in/in
K-phi (axial) for Distortional Buckling = 0.00 lb*in/in

MAXIMUM FACTORED AXIAL LOADS (lb)

<u>BRACING</u>	<u>SPACING</u>			Maximum <u>KL/r</u>
	<u>12 in</u>	<u>16 in</u>	<u>24 in</u>	
NONE	1516	1486	1424	177
MID Pt	3671	3585	3417	88
THIRD Pt	4492	4381	4165	75
SHEATH 2 SIDES	N/A	N/A	N/A	75
DEFLECTION	L/4377	L/3283	L/2188	

Note: Axial loads for sheathing braced design are based on the North American Standard for Cold-Formed Steel Framing - Wall Stud Design, 2007 Edition with 1/2 inch gypsum sheathing and No. 6 fasteners max 12 inches on center



12.6 Internal Wall Design

ClarkWestern Building Systems
CW Tech Support: (888) 437-3244
clarkwestern.com

2007 North American Specification LRFD
DATE: 7/3/2025

SECTION DESIGNATION: 362S162-43 [33] Single

Section Dimensions:

Web Height = 3.625 in
Top Flange = 1.625 in
Bottom Flange = 1.625 in
Stiffening Lip = 0.500 in
Inside Corner Radius = 0.0712 in
Punchout Width = 1.500 in
Punchout Length = 4.000 in
Design Thickness = 0.0451 in



Steel Properties:

Fy = 33.000 ksi
Fu = 45.000 ksi
Fya = 33.000 ksi

COMBINED AXIAL AND BENDING LOADS

INPUT PARAMETERS

Overall Wall Height = 10.0
Lateral Load = 5.0 psf
Load Factor for Lateral Load = 1.00
Lateral Load multiplied by 0.70 for deflection calculations
Studs Considered Fully Braced for Bending

K-phi (flexure) for Distortional Buckling = 0.00 lb*in/in
K-phi (axial) for Distortional Buckling = 0.00 lb*in/in

MAXIMUM FACTORED AXIAL LOADS (lb)

<u>BRACING</u>	<u>SPACING</u>			Maximum <u>KL/r</u>
	<u>12 in</u>	<u>16 in</u>	<u>24 in</u>	
NONE	1267	1235	1171	196
MID Pt	3175	3082	2898	98
THIRD Pt	4041	3913	3664	83
SHEATH 2 SIDES	N/A	N/A	N/A	83
DEFLECTION	L/3191	L/2393	L/1595	

Note: Axial loads for sheathing braced design are based on the North American Standard for Cold-Formed Steel Framing - Wall Stud Design, 2007 Edition with 1/2 inch gypsum sheathing and No. 6 fasteners max 12 inches on center

Project				Job Ref.	
Section 13.1 CONCRETE WALL DESIGN				Sheet no./rev. 1	
Calc. by V	Date 7/4/2025	Chk'd by	Date	App'd by	Date

RC WALL DESIGN

In accordance with ACI 318-19 (22)

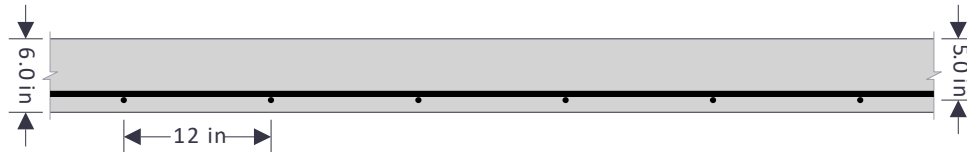
Tedds calculation version 1.2.07

Design summary

Overall design status

PASS

Description	Unit	Required	Provided	Utilization	Result
Axial	kips/ft	0.2	3.4	0.058	PASS
Moment	kip_ft/ft	0.3	5.1	0.059	PASS
Shear	kips/ft	0.6	3.4	0.176	PASS



Geometry of wall

Depth of wall	$h = 6.00$ in
Clear cover to reinforcement (both sides)	$c_c = 0.75$ in
Unsupported height of wall	$l_u = 120.0$ in
Effective height factor	$k = 1.00$

Reinforcement of wall

Numbers of reinforcement layers	$N_l = 1$
Vertical steel bar diameter number	$D_{ver_num} = 4$
Spacing of vertical steel	$s_v = 12.00$ in
Diameter of vertical steel bar	$D_{ver} = 0.500$ in
Horizontal steel bar diameter number	$D_{hor_num} = 4$
Spacing of horizontal steel	$s_h = 12.00$ in
Diameter of horizontal bar	$D_{hor} = 0.500$ in
Specified yield strength of reinforcement	$f_y = 60000$ psi
Specified compressive strength of concrete	$f'_c = 4000$ psi
Modulus of elasticity of bar reinforcement	$E_s = 29 \times 10^6$ psi
Modulus of elasticity of concrete	$E_c = 57000 \times f'_c^{1/2} \times (1\text{psi})^{1/2} = 3604997$ psi
Ultimate design strain	$\epsilon_c = 0.003$ in/in
Compression-controlled strain limit	$\epsilon_{ty} = 0.002$

Check for minimum area of vertical steel of single layer reinforcement wall to cl. 11.6.1

Gross area of wall per running foot length	$A_g = h \times 12\text{in} = 72.000$ in ²
Numbers of vertical bars per running foot length	$N_v = 12\text{in}/s_v = 1.000$
Area of vertical steel per running foot length	$A_{st_v} = N_v \times (\pi \times D_{ver}^2) / 4 = 0.196$ in ²
Minimum area of vertical steel required	$A_{st_v_min} = 0.086$ in ²

PASS- Minimum vertical steel check

Check for minimum area of horizontal steel of single layer reinforcement wall to cl. 11.6.1

Gross area of wall per running foot height	$A_g = h \times 12\text{in} = 72.000$ in ²
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Project				Job Ref.	
Section				Sheet no./rev. 2	
Calc. by V	Date 7/4/2025	Chk'd by	Date	App'd by	Date

Numbers of horizontal bar per running foot height $N_h = 12\text{in} / s_h = \mathbf{1.000}$
Area of horizontal steel per running foot height $A_{st_h} = N_h \times (\pi \times D_{hor}^2) / 4 = \mathbf{0.196\text{ in}^2}$
Minimum area of horizontal steel required $A_{st_h_min} = \mathbf{0.144\text{ in}^2}$

PASS- Minimum horizontal steel check

Braced wall slenderness check to 6.2.5

Permissible slenderness ratio $Sr_{perm} = \min(34 + 12 \times (M_{1_act} / M_{2_act}), 40) = \mathbf{34.0}$
Radius of gyration $r_{min} = 0.3 \times h = \mathbf{1.80\text{ in}}$
Actual slenderness ratio $Sr_{act} = k \times l_u / r_{min} = \mathbf{66.67}$

Wall is braced slender wall

Design loads and moments for wall subjected to shear, axial load and bending

Ultimate axial force per running foot $P_{u_act} = \mathbf{0.20\text{ kips/ft}}$
Maximum transverse moment per running foot $M_{2_act} = \mathbf{0.30\text{ kips_ft/ft}}$
Ultimate shear force per running foot $V_{u_act} = \mathbf{0.60\text{ kips/ft}}$
Ratio of DL moment to total moment $\beta_d = \mathbf{0.650}$

Magnified moment for braced slender wall to 6.6.4

Moment of inertia of section $I_g = (12\text{in} \times h^3) / 12 = \mathbf{216.000\text{ in}^4}$
Euler's buckling load $P_c = (\pi^2 \times 0.4 \times E_c \times I_g) / ((1 + \beta_d) \times (k \times l_u)^2 \times 1\text{ft}) = \mathbf{129.381\text{ kips/ft}}$
Wall is loaded in transverse direction between lateral supports
Correction factor for actual to equiv. mmt. diagram $C_m = \mathbf{1.0}$
Moment magnifier $\delta_{ns} = \max(1.0, C_m / (1 - (P_{u_act} / (0.75 \times P_c)))) = \mathbf{1.002}$
Minimum uniaxial moment for slender section $M_{2_min} = P_{u_act} \times (0.6\text{ in} + 0.03 \times h) = \mathbf{0.013\text{ kip_ft/ft}}$
Magnified uniaxial moment $M_c = \delta_{ns} \times \max(M_{2_min}, M_{2_act}) = \mathbf{0.301\text{ kip_ft/ft}}$ Library item: Mc braced slender trans

Axial load capacity of single layer reinforcement wall subjected to bending

c/d_t ratio $r = \mathbf{0.090}$
Effective cover to reinforcement $d' = c_c + (D_{ver}/2) = \mathbf{1.000\text{ in}}$
Depth of tension steel $d_t = h - d' = \mathbf{5.000\text{ in}}$
Depth of NA from extreme compression face $c = r \times d_t = \mathbf{0.450\text{ in}}$
Factor of depth of compressive stress block $\beta_1 = \mathbf{0.850}$
Depth of equivalent rectangular stress block $a = \min((\beta_1 \times c), h) = \mathbf{0.382\text{ in}}$
Strain in 'tension' reinforcement $\epsilon_s = \epsilon_c \times (1 - d_t / c) = \mathbf{-0.030333}$
f_s Stress in 'tension' reinforcement $f_s = \max(E_s \times \epsilon_s, -f_y) = \mathbf{-60000.0\text{ psi}}$
Compression force in concrete $C_c = 0.85 \times f'_c \times a \times 12\text{in}/1\text{ft} = \mathbf{15.606\text{ kips/ft}}$
Area of vertical tension steel per running foot $A_s = A_{st_v} = \mathbf{0.196\text{ in}^2}$
Force in 'tension' steel $T_s = A_s \times f_s / 1\text{ft} = \mathbf{-11.781\text{ kips/ft}}$
Nominal axial load strength $P_n = C_c + T_s = \mathbf{3.825\text{ kips/ft}}$
Strength reduction factor $\phi = 0.90 = \mathbf{0.9}$
Ultimate axial load carrying capacity of wall $P_u = \phi \times P_n = \mathbf{3.443\text{ kips/ft}}$

Check for axial load capacity of wall

Wall is safe in axial loading

Bending capacity of single layer reinforcement wall

Centroid of wall $y = h \times 0.5 = \mathbf{3.000\text{ in}}$
Nominal moment strength $M_n = C_c \times (y - 0.5 \times a) - T_s \times (d_t - y) = \mathbf{5.616\text{ kip_ft/ft}}$

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Calc. by V	Date 7/4/2025	Chk'd by	Date	App'd by	Date

Ultimate moment strength capacity of wall

$$M_u = \phi \times M_n = 5.055 \text{ kip_ft/ft}$$

Check for uniaxial bending capacity of wall

Wall is safe for bending

Check for shear capacity of wall cl. 22.5

Required shear strength

$$V_{u_act} = 0.600 \text{ kips/ft}$$

Strength reduction factor

$$\phi_v = 0.75$$

Effective cover to reinforcement

$$d' = c_c + (D_{ver}/2) = 1.000 \text{ in}$$

Depth of tension steel

$$d_t = h - d' = 5.000 \text{ in}$$

Size effect factor – cl. 22.5.5.1.3

$$\lambda_s = \min(\sqrt{2 / (1 + (d_t / 1 \text{ in}) / 10)}), 1.0) = 1.00$$

Ratio of longitudinal reinforcement

$$\rho_w = A_s / (d_t \times 12 \text{ in}) = 0.003$$

Maximum shear force resisting capacity of wall

$$V_{max} = 5 \times \lambda \times \sqrt{f'_c \times 1 \text{ psi}} \times d_t = 18.974 \text{ kips/ft}$$

Shear strength provided by concrete

$$V_{c1} = (8 \times \lambda_s \times \lambda \times (\rho_w)^{1/3} \times \sqrt{f'_c \times 1 \text{ psi}}) + P_{u_act} / (6 \times A_g \times 12 \text{ in}) \times d_t = 4.535 \text{ kips/ft}$$

Design shear capacity of section

$$\phi V_c = \phi_v \times \min(V_{c1}, V_{max}) = 3.401 \text{ kips/ft}$$

PASS- Wall is safe in shear force



SECTION DESIGNATION: 600S162-43 [33] (2) Back-to-Back

Section Dimensions:

Web Height =	6.000 in
Top Flange =	1.625 in
Bottom Flange =	1.625 in
Stiffening Lip =	0.500 in
Inside Corner Radius =	0.0712 in
Punchout Width =	1.500 in
Punchout Length =	4.000 in
Design Thickness =	0.0451 in



Steel Properties:

Fy =	33.000 ksi
Fu =	45.000 ksi
Fya =	36.303 ksi

MAXIMUM FACTORED AXIAL LOADS, Pu

INPUT PARAMETERS

Overall Stud Length = 9 ft

Member Configuration: (2) BACK-TO-BACK MEMBERS

K-phi (axial) for Distortional Buckling = 0.00 lb*in/in

TOTAL FACTORED AXIAL LOADS, Pu (lb)

<u>WEAK AXIS BRACING</u>	<u>MAXIMUM KL/r</u>	<u>CONCENTRIC LOADING</u>
NONE	154	5501
MID Pt	79	11098
THIRD Pt	55	12676

Note: For (2) Back-to-Back Members, Individual Members
Must be Adequately Interconnected

Powered by FRAMECAD Structure E

Minimum number of fasteners required is 2 per joint

14.2 WB2

22'-6"

1'-9 5/8"

W5

W6

T3

T4

B1

B2

22'-6"

1'-9 5/8"

-3.28kip

0.32kip

-3.28kip

0.32kip

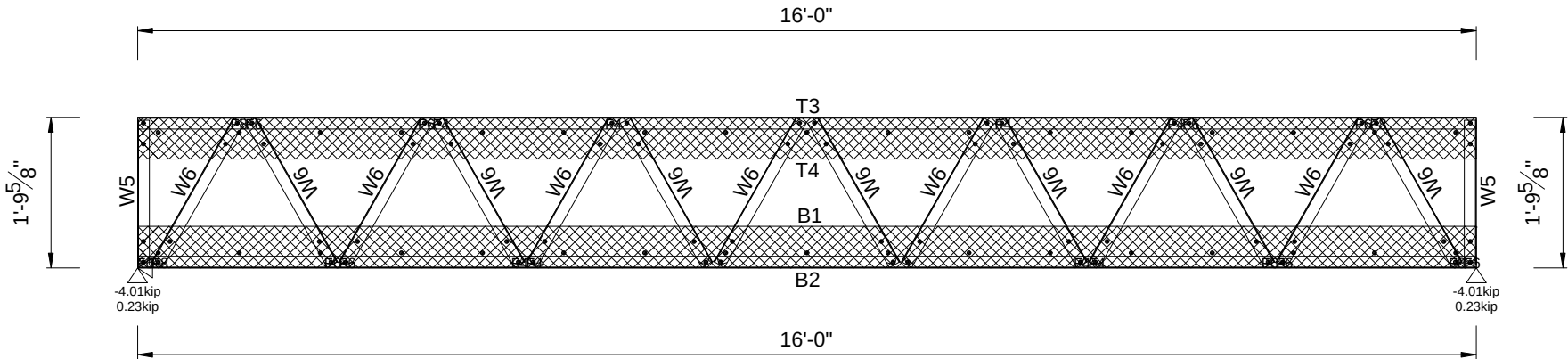
System Name:	FRAMECAD_FT_i	Roof Type:	SHEET	Truss Model:	M11 362 LE System	Wind Speed:	W170
Truss Type:	In-plane Truss	Truss Pitch:	0.000	Truss Spacing:	1'-0"	Design Code:	AISI S100-20 LRFD
Loading Code:	IBC 2024 LRFD	Material Grade Reduction:	Not Applied	Top Chord Live Load (psf):	120.0	Top Chord Dead Load (psf):	70.0
Bottom Chord Dead Load (psf):	5.0	Number of on-coming Trusses:	0	Envelope:	1'-9 5/8"h x 22'-6"w		

B1 600S162-43-50	1	16'-0"	B2 600S162-43-50	1	16'-0"	T3 600S162-43-50	1	16'-0"	T4 600S162-43-50	1	16'-0"
W5 600S162-43-50	2	1'-8 13/16"	W6 600S162-43-50	14	1'-11 3/4"	FRAMECAD 10g-16mm Flathead	80		FRAMECAD 10g-19mm XDrive	196	
Working Sheet: WB3			Assembly Weight			149.1lbs					

Powered by FRAMECAD Structure E

Minimum number of fasteners required is 2 per joint

14.3 WB3



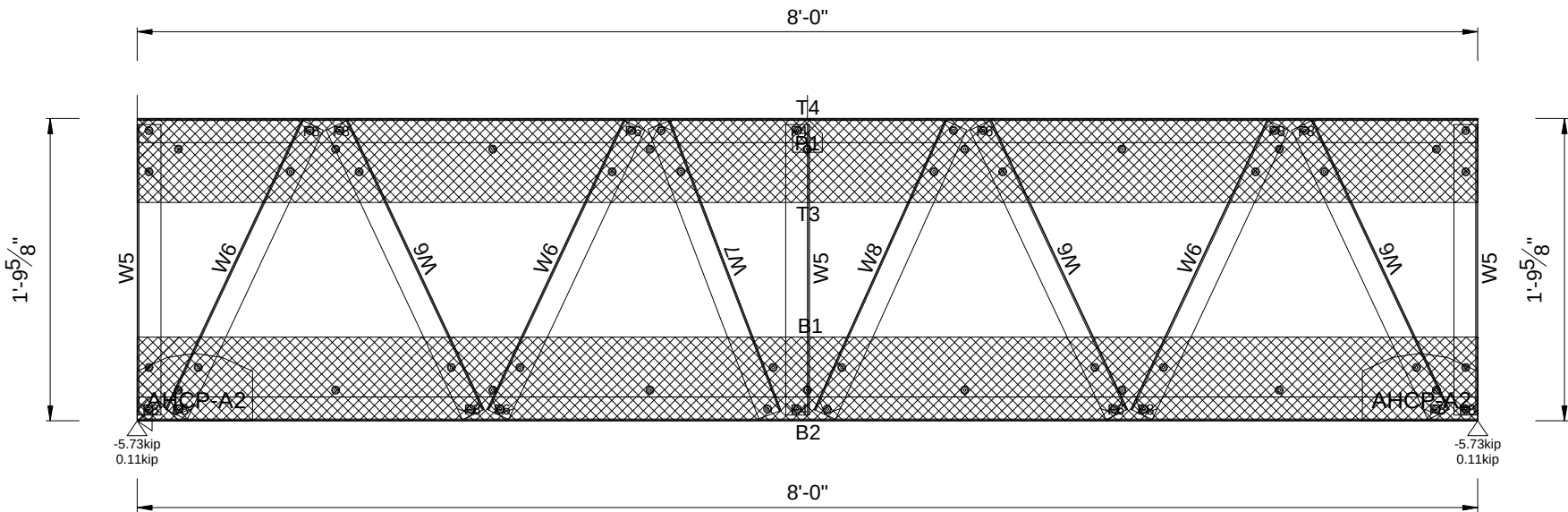
Quantity Required = 1 Mark as WB3 Engineering Status = 75%

System Name:	FRAMECAD_FT_i	Roof Type:	SHEET	Truss Model:	M11 362 LE System	Wind Speed:	W170
Truss Type:	In-plane Truss	Truss Pitch:	0.000	Truss Spacing:	1'-0"	Design Code:	AISI S100-20 LRFD
Loading Code:	IBC 2024 LRFD	Material Grade Reduction:	Not Applied	Top Chord Live Load (psf):	210.0	Top Chord Dead Load (psf):	125.0
Bottom Chord Dead Load (psf):	7.0	Number of on-coming Trusses:	0	Envelope:	1'-9 5/8"h x 16'-0"w		

B1 600S162-54-50	1	8'-0"	B2 600S162-54-50	1	8'-0"	T3 600S162-54-50	1	8'-0"	T4 600S162-54-50	1	8'-0"
W5 600S162-54-50	3	1'-8 13/16"	W6 600S162-54-50	6	1'-10 7/8"	W7 600S162-54-50	1	1'-10 3/16"	W8 600S162-54-50	1	1'-10 3/4"
FRAMECAD 1.15mm Apex/Heel Plate4			FRAMECAD 10g-16mm Flathead 128			FRAMECAD 10g-19mm XDrive 120			Working Sheet: WB4		
Assembly Weight 101.7lbs											

Powered by FRAMECAD Structure E Minimum number of fasteners required is 2 per joint

14.4 WB4



Quantity Required = 1 Mark as WB4 Engineering Status = 70%

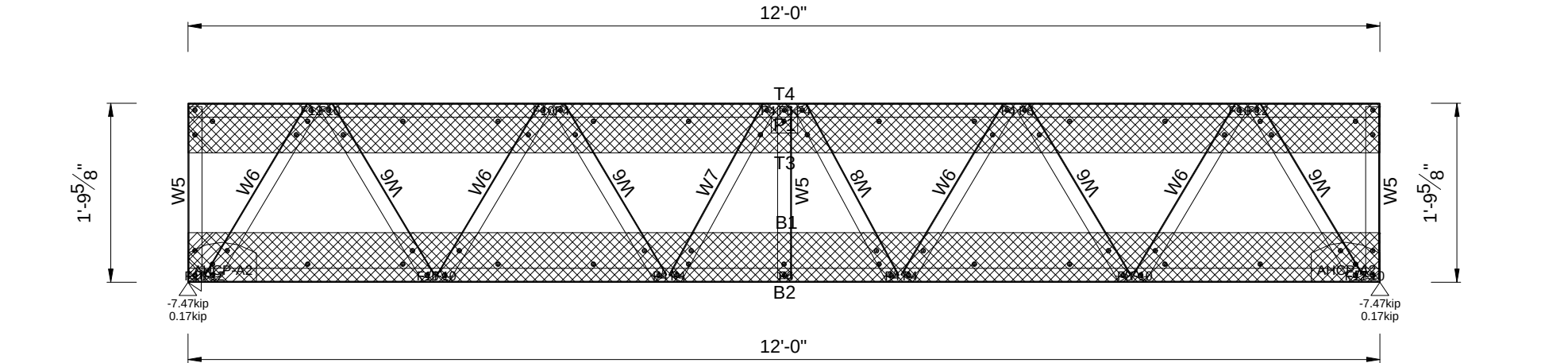
System Name:	FRAMECAD_FT_i	Roof Type:	SHEET	Truss Model:	M11 362 LE System	Wind Speed:	W170
Truss Type:	In-plane Truss	Truss Pitch:	0.000	Truss Spacing:	1'-0"	Design Code:	AISI S100-20 LRFD
Loading Code:	IBC 2024 LRFD	Material Grade Reduction:	Not Applied	Top Chord Live Load (psf):	420.0	Top Chord Dead Load (psf):	250.0
Bottom Chord Dead Load (psf):	13.0	Number of on-coming Trusses:	0	Envelope:	1'-9 5/8"h x 8'-0"w		

B1 600S162-54-50	1	12'-0"	B2 600S162-54-50	1	12'-0"	T3 600S162-54-50	1	12'-0"	T4 600S162-54-50	1	12'-0"
W5 600S162-54-50	3	1'-8 13/16"	W6 600S162-54-50	8	2'-0 1/16"	W7 600S162-54-50	1	1'-11 9/16"	W8 600S162-54-50	1	1'-11 1/2"
FRAMECAD 1.15mm Apex/Heel Plate 4			FRAMECAD 10g-16mm Flathead 188			FRAMECAD 10g-19mm XDrive 152			Working Sheet: WB5		
Assembly Weight 142.1lbs											

Powered by FRAMECAD Structure E

Minimum number of fasteners required is 2 per joint
Warning: Some joints contain excessive fasteners.
Please consider an alternative solution.

14.5 WB5



Quantity Required = 1 Mark as WB5 Engineering Status = 92%

System Name:	FRAMECAD_FT_i	Roof Type:	SHEET	Truss Model:	M11 362 LE System	Wind Speed:	W170
Truss Type:	In-plane Truss	Truss Pitch:	0.000	Truss Spacing:	1'-0"	Design Code:	AISI S100-20 LRFD
Loading Code:	IBC 2024 LRFD	Material Grade Reduction:	Not Applied	Top Chord Live Load (psf):	440.0	Top Chord Dead Load (psf):	250.0
Bottom Chord Dead Load (psf):	25.0	Number of on-coming Trusses:	0	Envelope:	1'-9 5/8"h x 12'-0"w		

Webbed Beam Post design



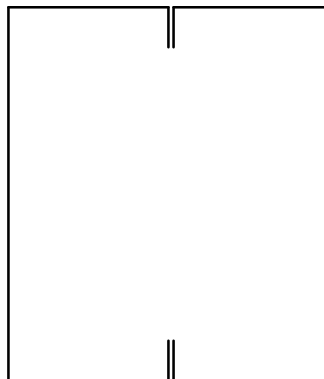
ClarkWestern Building Systems
CW Tech Support: (888) 437-3244
clarkwestern.com

2007 North American Specification LRFD
DATE: 7/2/2025

SECTION DESIGNATION: 600S162-54 [50] (2) Boxed

Section Dimensions:

Web Height =	6.000 in
Top Flange =	1.625 in
Bottom Flange =	1.625 in
Stiffening Lip =	0.500 in
Inside Corner Radius =	0.0849 in
Punchout Width =	1.500 in
Punchout Length =	4.000 in
Design Thickness =	0.0566 in



Steel Properties:

Fy =	50.000 ksi
Fu =	65.000 ksi
Fya =	55.318 ksi

COMBINED AXIAL AND BENDING LOADS

INPUT PARAMETERS

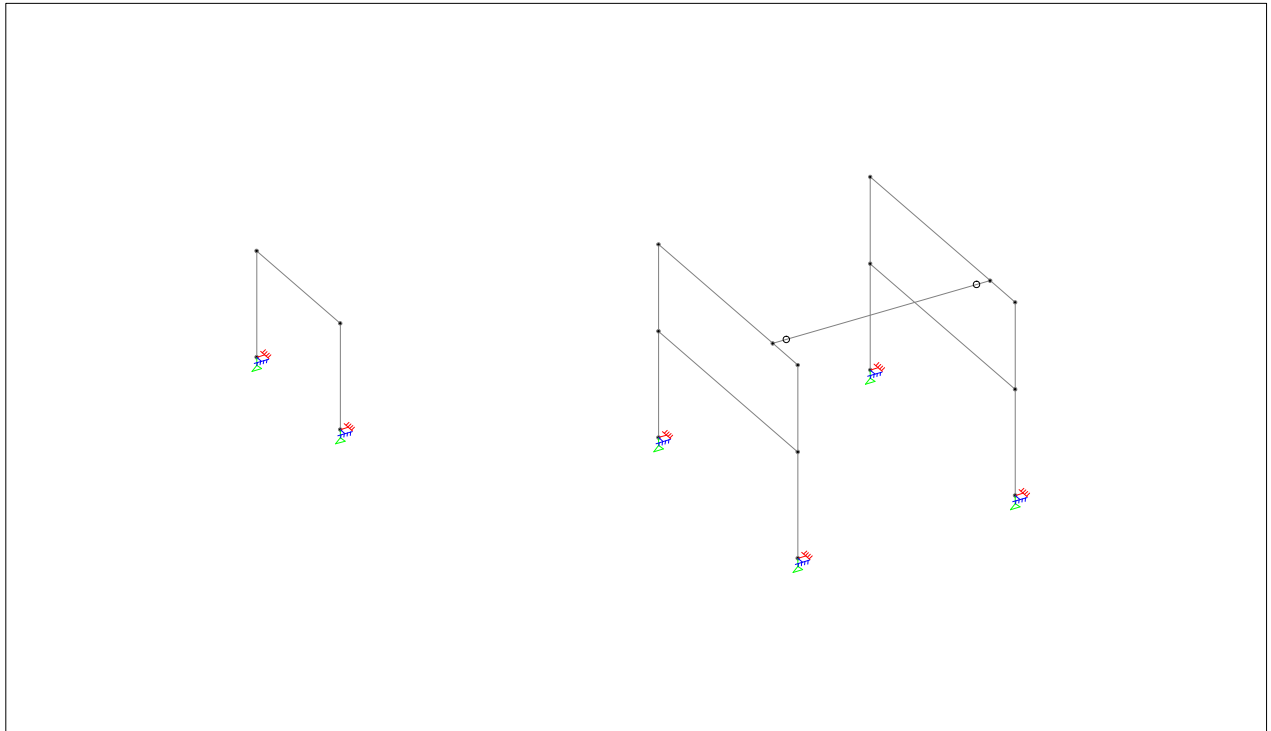
Overall Wall Height = 10.0
Lateral Load = 63.1 psf
Load Factor for Lateral Load = 1.00
Lateral Load multiplied by 0.70 for deflection calculations
Studs Considered Fully Braced for Bending

MAXIMUM FACTORED AXIAL LOADS (lb)

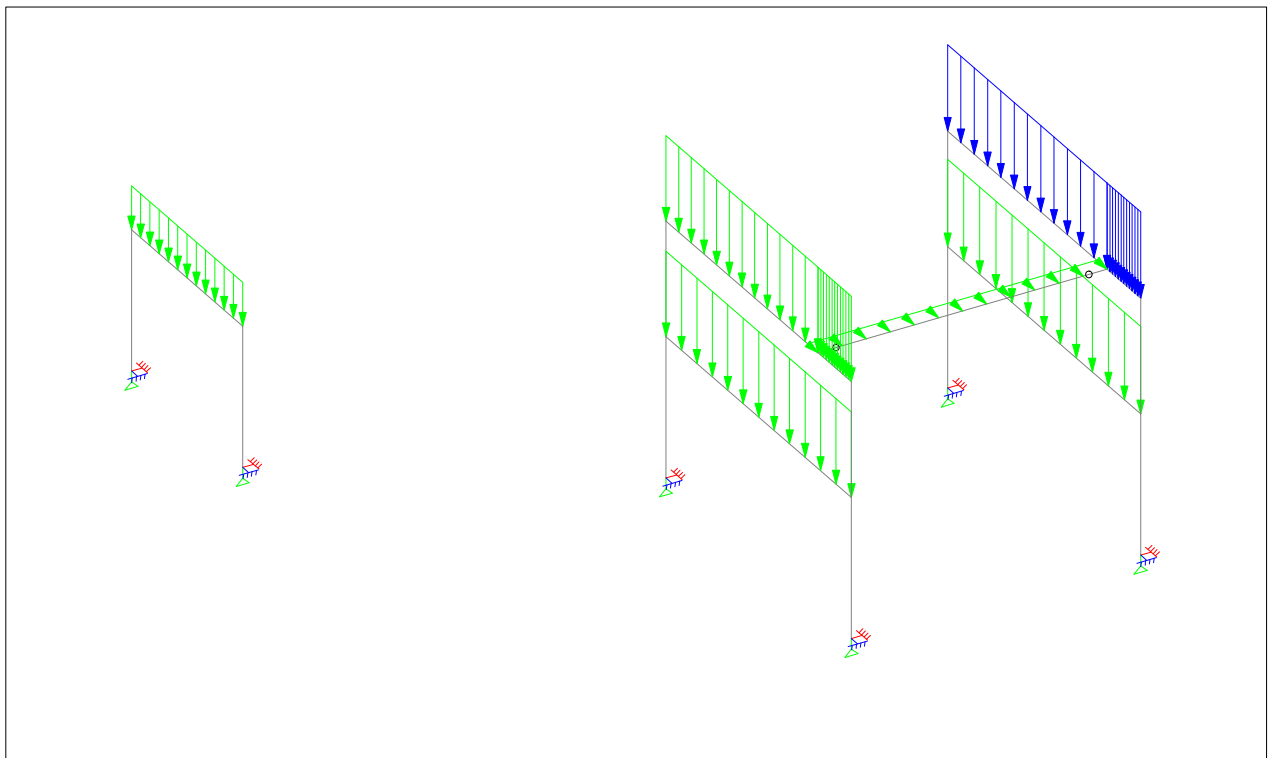
<u>BRACING</u>	<u>SPACING</u>			Maximum <u>KL/r</u>
	<u>12 in</u>	<u>16 in</u>	<u>24 in</u>	
NONE	14290	13702	12545	92
NONE	14290	13702	12545	92
NONE	14290	13702	12545	92
SHEATH 2 SIDES	N/A	N/A	N/A	53
DEFLECTION	L/2038	L/1528	L/1019	

Note: Axial loads for sheathing braced design are based on the North American Standard for Cold-Formed Steel Framing - Wall Stud Design, 2007 Edition with 1/2 inch gypsum sheathing and No. 6 fasteners max 12 inches on center

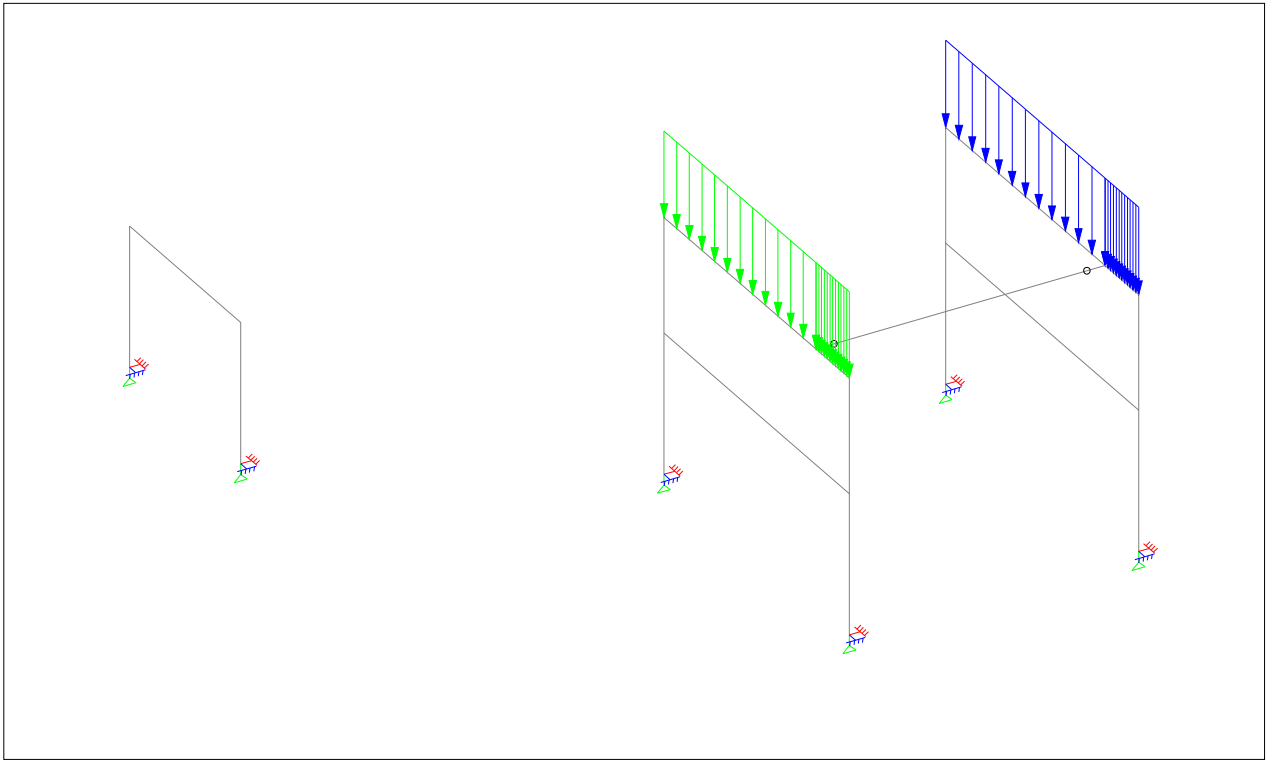
15 HRS FRAME DESIGN



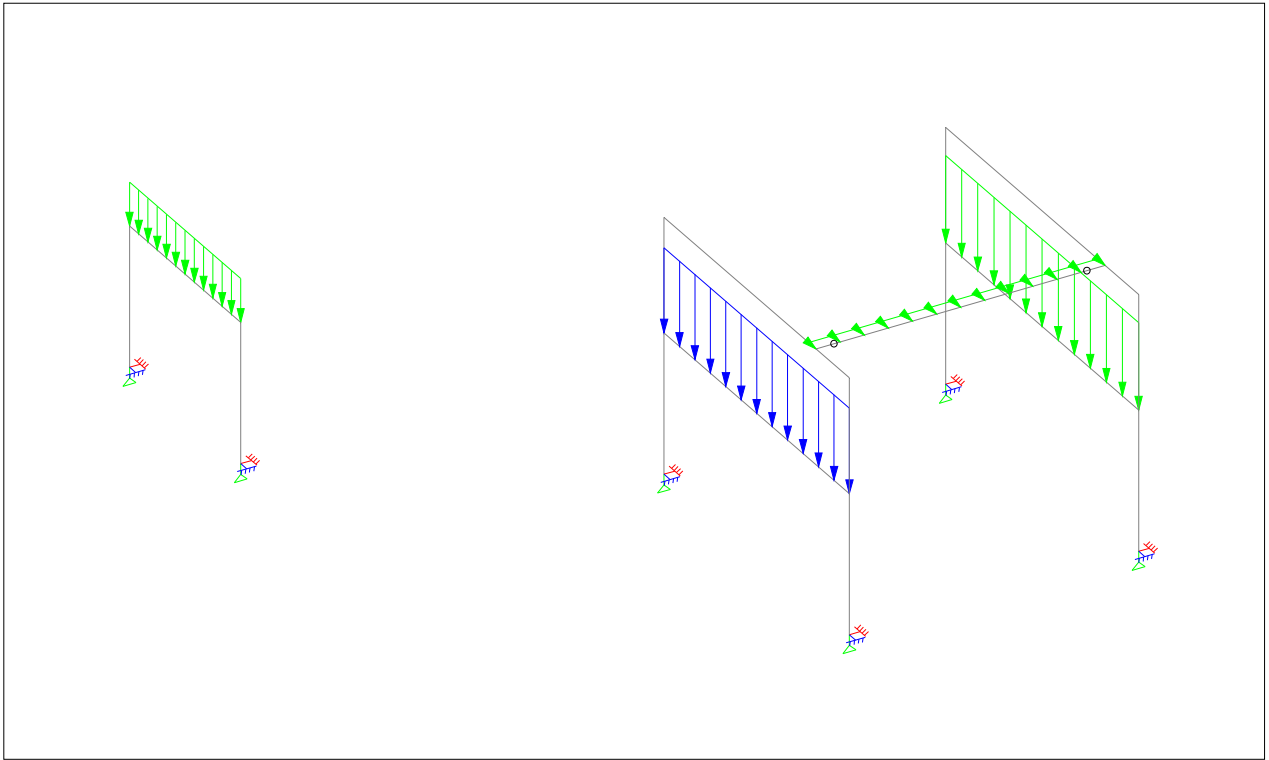
GEOMETRY



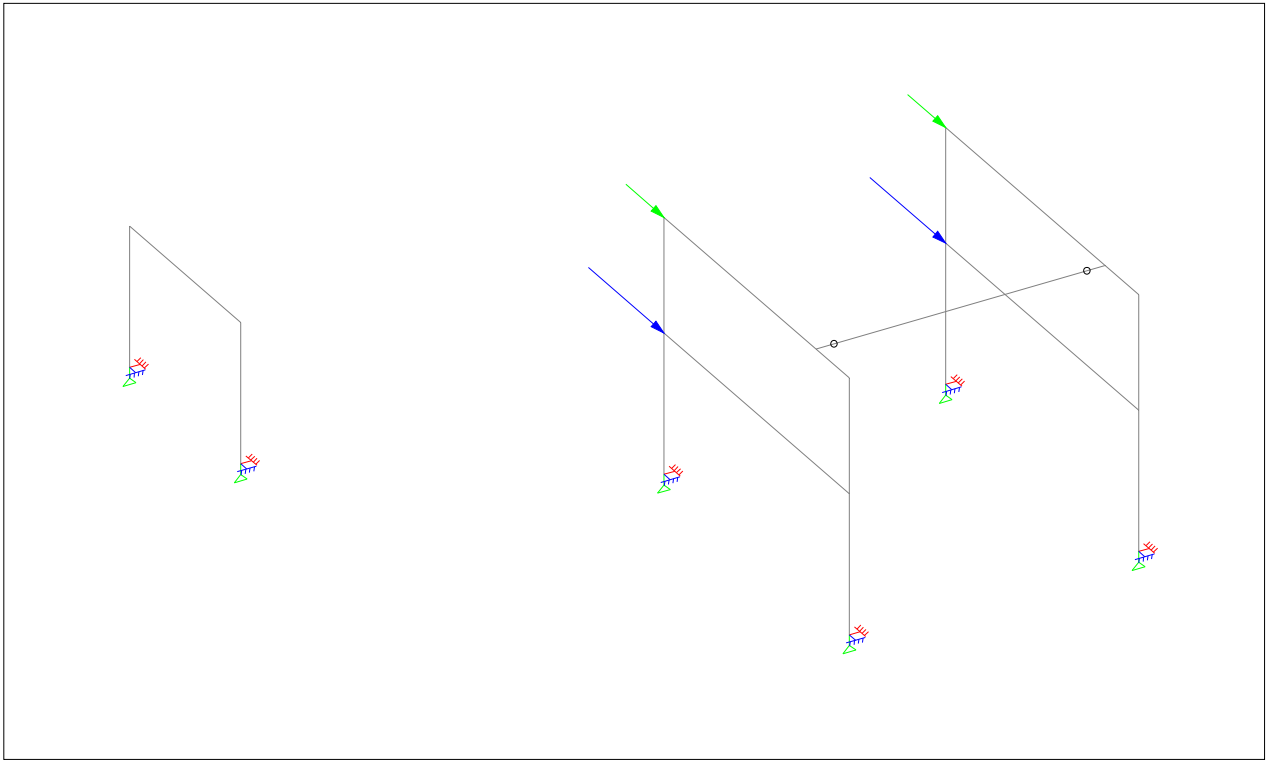
DEAD LOAD



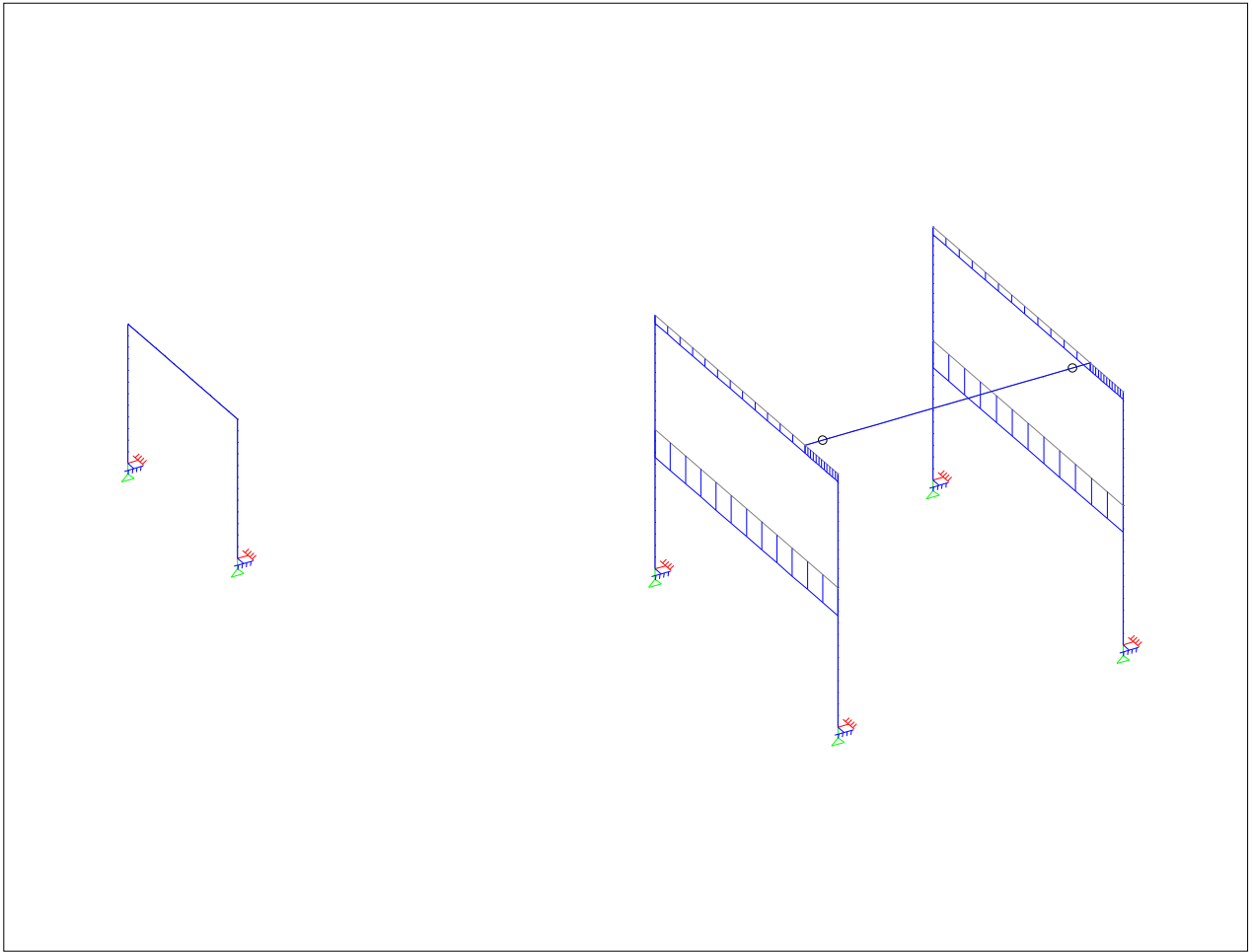
ROOF LIVE LOAD



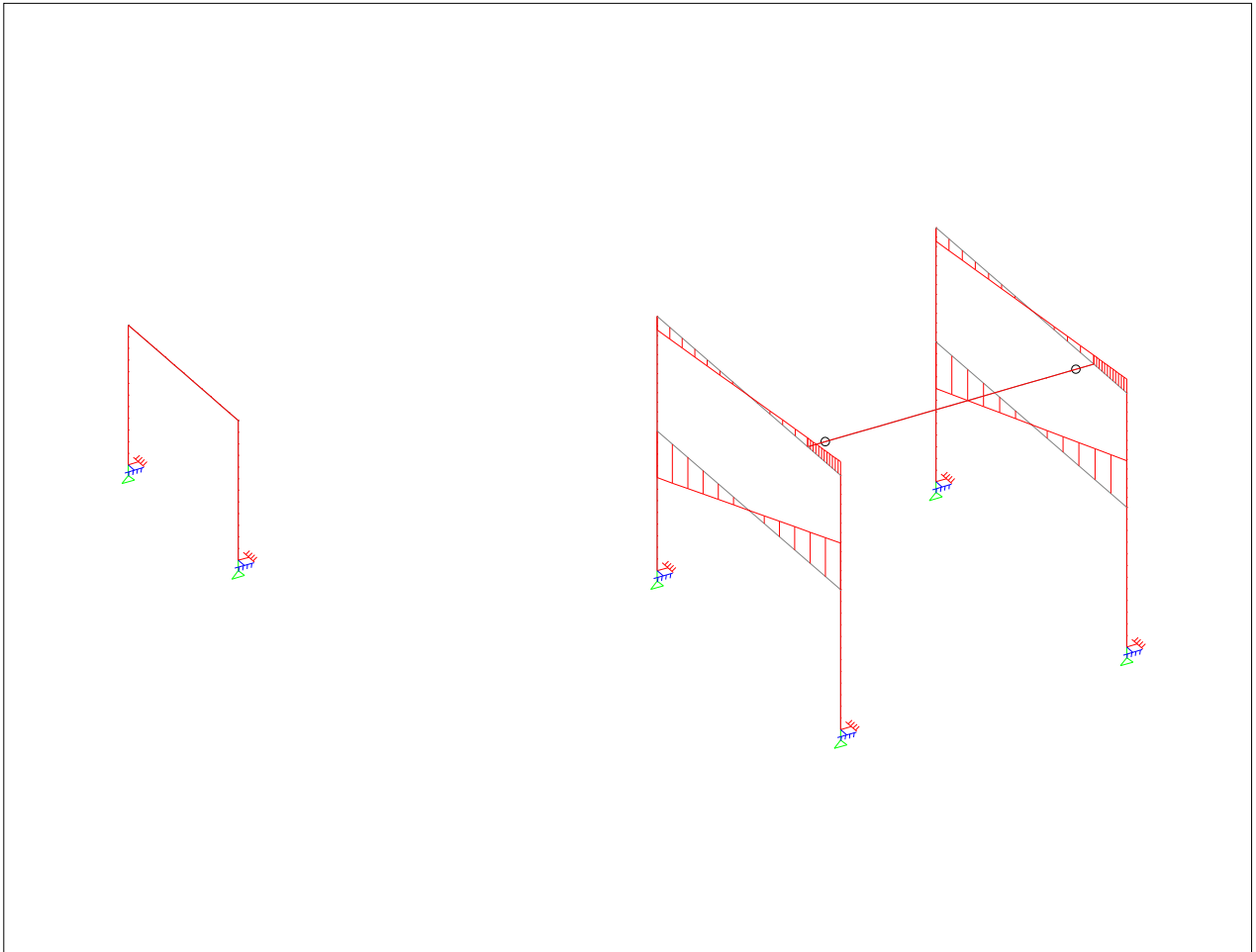
FLOOR LIVE LOAD



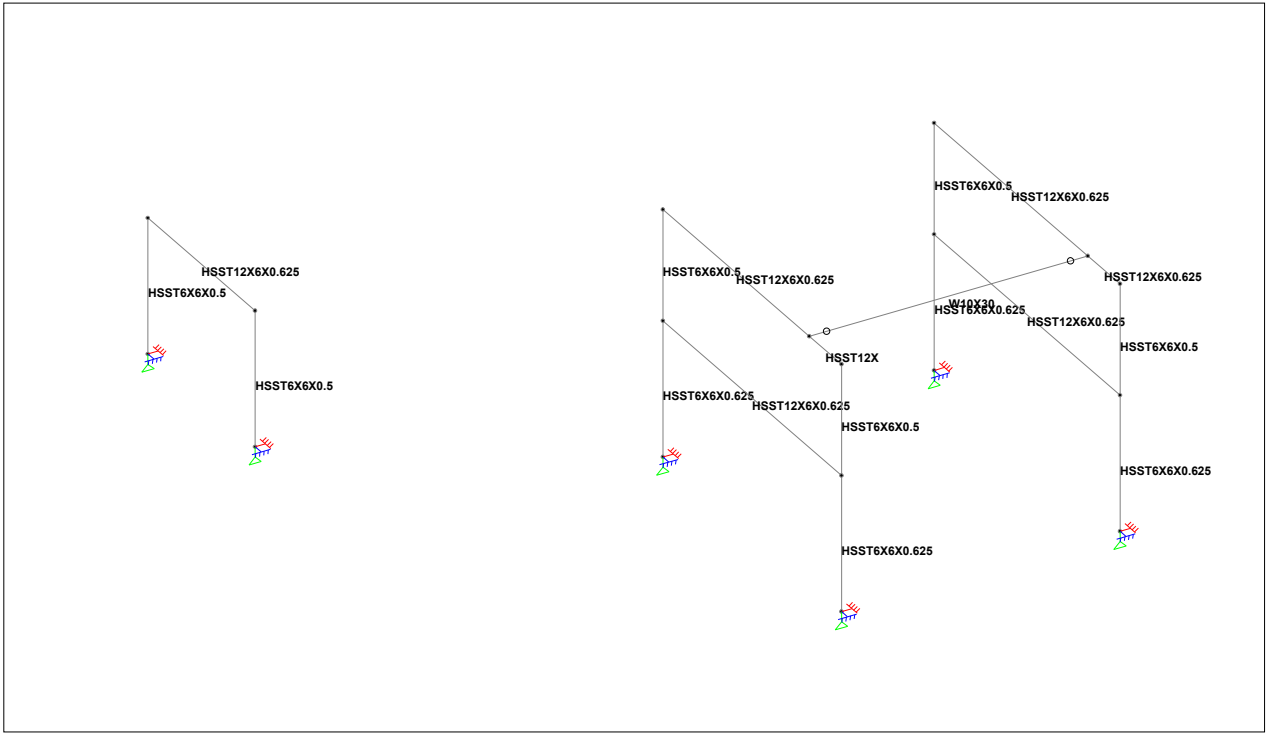
WIND LOAD



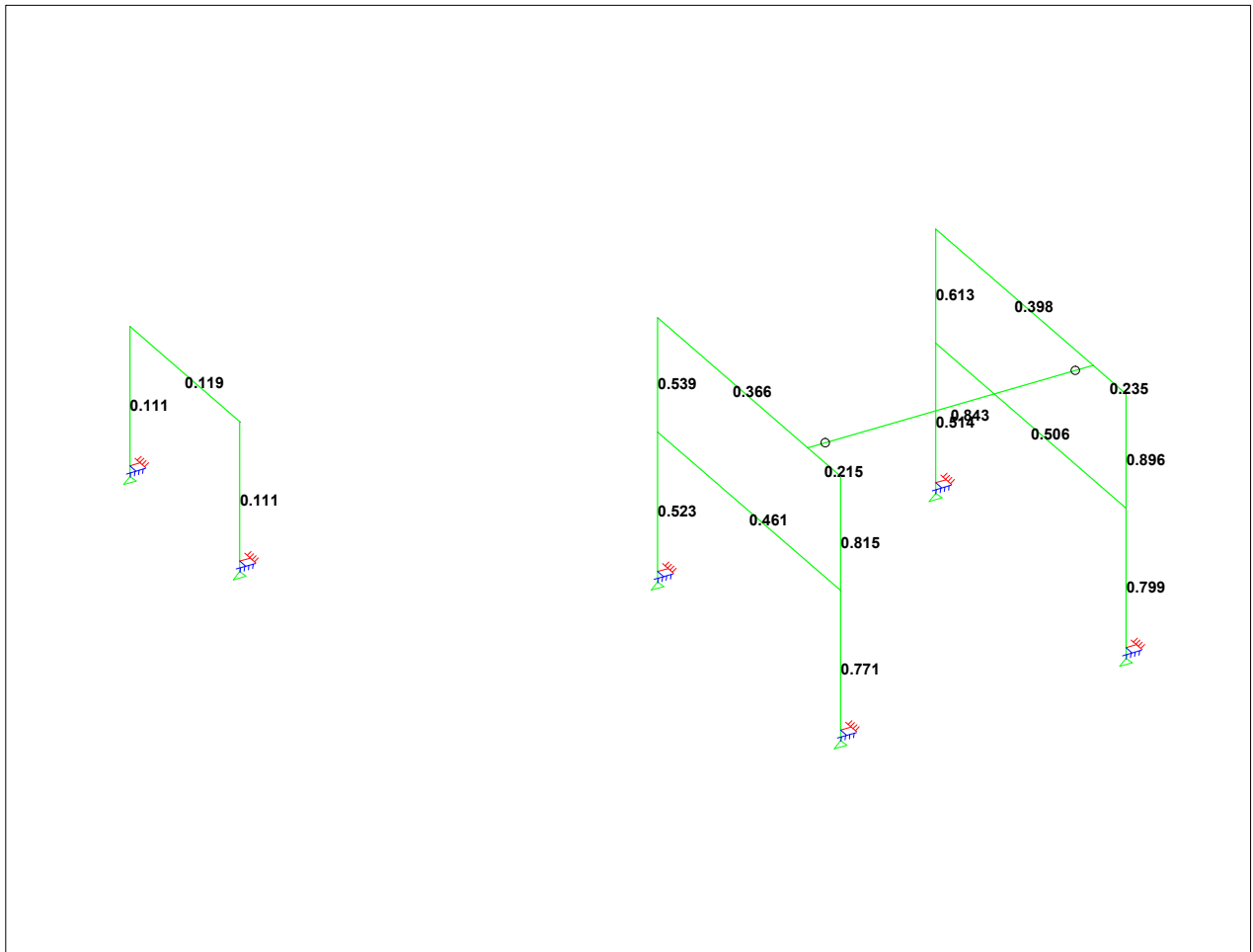
SHEAR FORCE DIAGRAM



MOMENT DIAGRAM



MEM - PROPERTY



UTILITY RATIO

1 DL : Beam Loads

Beam	Type	Direction	Fa	Da (ft)	Fb	Db	Ecc. (ft)
1	UNI kips/ft	GY	-0.512	-	-	-	-
9	UNI kips/ft	GY	-0.506	-	-	-	-
26	UNI kips/ft	GY	-0.500	-	-	-	-
28	UNI kips/ft	GY	-0.500	-	-	-	-
33	UNI kips/ft	GY	-0.256	-	-	-	-
37	UNI kips/ft	GY	-0.500	-	-	-	-
39	UNI kips/ft	GZ	0.105	-	-	-	-
40	UNI kips/ft	GY	-0.506	-	-	-	-

2 RLL : Beam Loads

Beam	Type	Direction	Fa	Da (ft)	Fb	Db	Ecc. (ft)
9	UNI kips/ft	GY	-0.405	-	-	-	-
28	UNI kips/ft	GY	-0.400	-	-	-	-
37	UNI kips/ft	GY	-0.400	-	-	-	-
40	UNI kips/ft	GY	-0.405	-	-	-	-

3 FLL : Beam Loads

Beam	Type	Direction	Fa	Da (ft)	Fb	Db	Ecc. (ft)
1	UNI kips/ft	GY	-0.820	-	-	-	-
26	UNI kips/ft	GY	-0.800	-	-	-	-
33	UNI kips/ft	GY	-0.410	-	-	-	-
39	UNI kips/ft	GZ	0.120	-	-	-	-

4 WL : Node Loads

Node	FX (lb)	FY (lb)	FZ (lb)	MX (kip·ft)	MY (kip·ft)	MZ (kip·ft)
8	-	-	7E+3	-	-	-
11	-	-	3.53E+3	-	-	-
24	-	-	7E+3	-	-	-
27	-	-	3.53E+3	-	-	-

Beams

Beam	Node A	Node B	Length (ft)	Property	β (degrees)
1	8	37	26.000	1	0
3	8	4	11.000	3	0
9	11	12	21.500	1	0
12	11	8	9.000	2	0
25	24	25	11.000	3	0
26	24	26	25.000	1	0
27	26	21	11.000	3	0
28	27	34	20.500	1	0
29	27	24	9.000	2	0
30	29	26	9.000	2	0
33	30	31	15.000	1	0
34	30	32	11.000	2	0
35	31	33	11.000	2	0
37	34	29	4.500	1	0
39	34	12	22.500	4	0
40	12	35	4.500	1	0
41	35	37	9.000	2	0
42	37	36	11.000	3	0

Nodes

Node	X (ft)	Y (ft)	Z (ft)
4	0	0	-25.500
8	0	11.000	-25.500
11	0	20.000	-25.500
12	0	20.000	-4.000
21	-22.500	0	0.500
24	-22.500	11.000	-24.500
25	-22.500	0	-24.500
26	-22.500	11.000	0.500
27	-22.500	20.000	-24.500
29	-22.500	20.000	0.500
30	-46.500	11.000	-55.000
31	-46.500	11.000	-40.000
32	-46.500	0	-55.000
33	-46.500	0	-40.000
34	-22.500	20.000	-4.000

Nodes Cont...

Node	X (ft)	Y (ft)	Z (ft)
35	0	20.000	0.500
36	0	0	0.500
37	0	11.000	0.500

Beam End Force Summary

The signs of the forces at end B of each beam have been reversed. For example: this means that the Min Fx entry gives the largest tension value for an beam.

	Beam	Node	L/C	Axial	Shear		Torsion	Bending	
				Fx (lb)	Fy (lb)	Fz (lb)	Mx (kip·ft)	My (kip·ft)	Mz (kip·ft)
Max Fx	42	37	6:ULC, 1.2 DE/	37.5E+3	0	5.99E+3	0	-39.604	0
Min Fx	1	8	8:ULC, 1.2 DE/	-5.36E+3	12.6E+3	0	0	0	31.924
Max Fy	1	8	6:ULC, 1.2 DE/	-3.46E+3	23.8E+3	0	0	0	53.968
Min Fy	1	37	6:ULC, 1.2 DE/	-3.46E+3	-26.3E+3	-0	-0	-0	86.411
Max Fz	41	35	6:ULC, 1.2 DE/	11.2E+3	0	9.45E+3	0	-38.211	0
Min Fz	12	11	8:ULC, 1.2 DE/	15.9E+3	0	-6.11E+3	0	31.133	0
Max Mx	39	34	9:ULC, 1.2 DE/	0	0.000	-1.42E+3	0.002	0	0
Min Mx	39	34	3:FLL	0	-0.000	-1.35E+3	-0.000	0	0
Max My	41	37	6:ULC, 1.2 DE/	11.2E+3	-0	9.45E+3	-0	46.807	-0
Min My	42	37	13:ULC, 1.2 DE/	25.6E+3	0	8.44E+3	0	-47.488	0
Max Mz	1	37	6:ULC, 1.2 DE/	-3.46E+3	-26.3E+3	-0	-0	-0	86.411
Min Mz	26	24	4:WL	3.49E+3	-2.65E+3	0	0	0	-33.072

Reaction Summary

	Node	L/C	Horizontal	Vertical	Horizontal	Moment		
			FX (lb)	FY (lb)	FZ (lb)	MX (kip·ft)	MY (kip·ft)	MZ (kip·ft)
Max FX	4	1:DL	0	12.6E+3	370.468	-0.107	0	0
Min FX	4	1:DL	0	12.6E+3	370.468	-0.107	0	0
Max FY	36	6:ULC, 1.2 DE/	0	37.5E+3	-5.99E+3	-26.248	0	0
Min FY	25	4:WL	0	-3.41E+3	-5.27E+3	-31.172	0	0
Max FZ	4	7:ULC, 1.2 DE/	0	31E+3	2.52E+3	4.736	0	0
Min FZ	36	13:ULC, 1.2 DE/	0	25.6E+3	-8.44E+3	-45.367	0	0
Max MX	4	7:ULC, 1.2 DE/	0	31E+3	2.52E+3	4.736	0	0
Min MX	36	13:ULC, 1.2 DE/	0	25.6E+3	-8.44E+3	-45.367	0	0
Max MY	4	1:DL	0	12.6E+3	370.468	-0.107	0	0
Min MY	4	1:DL	0	12.6E+3	370.468	-0.107	0	0
Max MZ	4	1:DL	0	12.6E+3	370.468	-0.107	0	0
Min MZ	4	1:DL	0	12.6E+3	370.468	-0.107	0	0

Utilization Ratio

Beam	Analysis Property	Design Property	Actual Ratio	Allowable Ratio	Ratio (Act./Allow.)	Clause	L/C	Ax (in ²)	Iz (in ⁴)	Iy (in ⁴)	Ix (in ⁴)
1	HSST12X6X	HSST12X6X	0.506	1.000	0.506	Eq.H1-1b	7	18.700	321.000	107.000	271.000
3	HSST6X6X0	HSST6X6X0	0.514	1.000	0.514	Eq.H1-1b	12	11.700	55.200	55.200	94.900
9	HSST12X6X	HSST12X6X	0.398	1.000	0.398	Eq.H1-1b	9	18.700	321.000	107.000	271.000
12	HSST6X6X0	HSST6X6X0	0.613	1.000	0.613	Eq.H1-1b	6	9.740	48.300	48.300	81.100
25	HSST6X6X0	HSST6X6X0	0.523	1.000	0.523	Eq.H1-1b	12	11.700	55.200	55.200	94.900
26	HSST12X6X	HSST12X6X	0.461	1.000	0.461	Eq.H1-1b	7	18.700	321.000	107.000	271.000
27	HSST6X6X0	HSST6X6X0	0.771	1.000	0.771	Eq.H1-1b	13	11.700	55.200	55.200	94.900
28	HSST12X6X	HSST12X6X	0.366	1.000	0.366	Eq.H1-1b	9	18.700	321.000	107.000	271.000
29	HSST6X6X0	HSST6X6X0	0.539	1.000	0.539	Eq.H1-1b	8	9.740	48.300	48.300	81.100
30	HSST6X6X0	HSST6X6X0	0.815	1.000	0.815	Eq.H1-1b	6	9.740	48.300	48.300	81.100
33	HSST12X6X	HSST12X6X	0.119	1.000	0.119	Eq.H1-1b	6	18.700	321.000	107.000	271.000
34	HSST6X6X0	HSST6X6X0	0.111	1.000	0.111	Eq.H1-1b	6	9.740	48.300	48.300	81.100
35	HSST6X6X0	HSST6X6X0	0.111	1.000	0.111	Eq.H1-1b	6	9.740	48.300	48.300	81.100
37	HSST12X6X	HSST12X6X	0.215	1.000	0.215	Eq.H1-1b	9	18.700	321.000	107.000	271.000
39	W10X30	W10X30	0.843	1.000	0.843	Cl.F6.1	6	8.840	170.000	16.700	0.622
40	HSST12X6X	HSST12X6X	0.235	1.000	0.235	Eq.H1-1b	9	18.700	321.000	107.000	271.000
41	HSST6X6X0	HSST6X6X0	0.896	1.000	0.896	Eq.H1-1b	6	9.740	48.300	48.300	81.100
42	HSST6X6X0	HSST6X6X0	0.799	1.000	0.799	Eq.H1-1b	13	11.700	55.200	55.200	94.900

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Specifier's comments:

1 Input data

Anchor type and diameter:

HIT-RE 500 V3 + HAS-E-55 (ASTM F1554 Gr.55) 7/8

Item number:

2198008 HAS-E-55 7/8"x13" (element) / 2123401 HIT-RE 500 V3 (adhesive)

Specification text:

Hilti $\varnothing$ 7/8 in HIT-RE 500 V3 + HAS-E-55 (ASTM F1554 Gr.55) with 10 in nominal embedment depth per ICC-ES ESR-3814 , Hammer drill bit installation per MPII,

Effective embedment depth:

$h_{ef,act} = 10.000$ in. ($h_{ef,limit} = -$ in.)

Material:

ASTM F1554 Grade 55

Evaluation Service Report:

ESR-3814

Issued | Valid:

1/1/2025 | 1/1/2027

Proof:

Design Method ACI 318-19 / Chem

Shear edge breakout verification:

Row closest to edge (Case 3 only from ACI 318-19 Fig. R.17.7.2.1b)

Stand-off installation:

$e_b = 0.000$ in. (no stand-off); $t = 0.250$ in.

Anchor plate^R :

$l_x \times l_y \times t = 3.000$ in. x 3.000 in. x 0.250 in.; (Recommended plate thickness: not calculated)

Profile:

no profile

Base material:

cracked concrete, 5000, $f'_c = 5,000$ psi; $h = 24.000$ in., Temp. short/long: 32/32 °F

Installation:

Hammer drilled hole, Installation condition: Dry

Reinforcement:

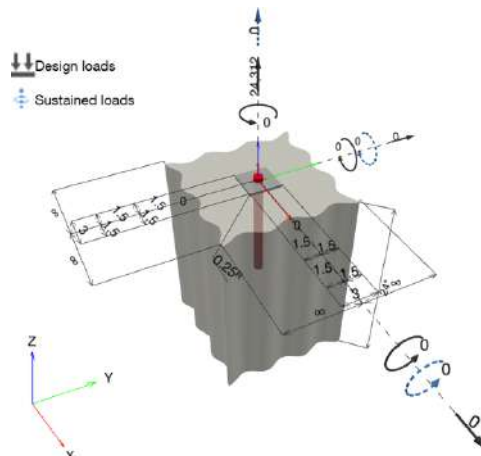
tension: not present, shear: not present; no supplemental splitting reinforcement present

edge reinforcement: none or < No. 4 bar



^R - The anchor calculation is based on a rigid anchor plate assumption.

Geometry [in.] & Loading [lb, in.lb]



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1.1 Design results

Case	Description	Forces [lb] / Moments [in.lb]	Seismic	Max. Util. Anchor [%]
1	Combination 1	N = 24,312; V _x = 0; V _y = 0; M _x = 0; M _y = 0; M _z = 0;	no	99

2 Load case/Resulting anchor forces

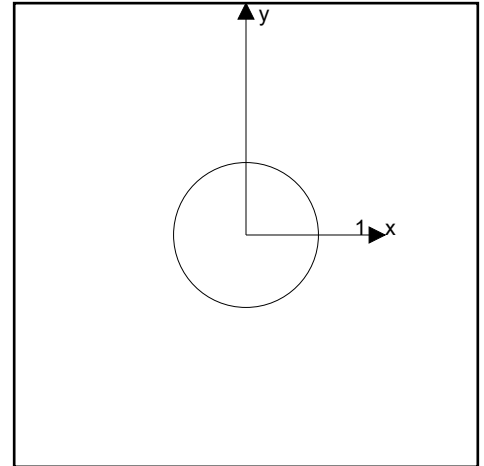
Anchor reactions [lb]

Tension force: (+Tension, -Compression)

Anchor	Tension force	Shear force	Shear force x	Shear force y
1	24,312	0	0	0

Max. concrete compressive strain: - [%]
Max. concrete compressive stress: - [psi]
Resulting tension force in (x/y)=(0.000/0.000): 24,312 [lb]
Resulting compression force in (x/y)=(-/-): 0 [lb]

Anchor forces are calculated based on the assumption of a rigid anchor plate.



3 Tension load

	Load N _{ua} [lb]	Capacity ϕN_n [lb]	Utilization $\beta_N = N_{ua} / \phi N_n$	Status
Steel Strength*	24,312	25,973	94	OK
Bond Strength**	24,312	24,584	99	OK
Sustained Tension Load Bond Strength*	N/A	N/A	N/A	N/A
Concrete Breakout Failure**	24,312	24,709	99	OK

* highest loaded anchor **anchor group (anchors in tension)

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3.1 Steel Strength

N_{sa} = ESR value refer to ICC-ES ESR-3814
 $\phi N_{sa} \geq N_{ua}$ ACI 318-19 Table 17.5.2

Variables

$A_{se,N}$ [in. ²]	f_{uta} [psi]
0.46	75,000

Calculations

N_{sa} [lb]
34,630

Results

N_{sa} [lb]	ϕ_{steel}	ϕN_{sa} [lb]	N_{ua} [lb]
34,630	0.750	25,973	24,312

3.2 Bond Strength

$N_a = \left(\frac{A_{Na}}{A_{Na0}} \right) \psi_{ed,Na} \psi_{cp,Na} N_{ba}$ ACI 318-19 Eq. (17.6.5.1a)

$\phi N_a \geq N_{ua}$ ACI 318-19 Table 17.5.2

A_{Na} see ACI 318-19, Section 17.6.5.1, Fig. R 17.6.5.1(b)

$A_{Na0} = (2 c_{Na})^2$ ACI 318-19 Eq. (17.6.5.1.2a)

$c_{Na} = 10 d_a \sqrt{\frac{\tau_{uncr}}{1100}}$ ACI 318-19 Eq. (17.6.5.1.2b)

$\psi_{ed,Na} = 0.7 + 0.3 \left(\frac{c_{a,min}}{c_{Na}} \right) \leq 1.0$ ACI 318-19 Eq. (17.6.5.4.1b)

$\psi_{cp,Na} = \text{MAX} \left(\frac{c_{a,min}}{c_{ac}}, \frac{c_{Na}}{c_{ac}} \right) \leq 1.0$ ACI 318-19 Eq. (17.6.5.5.1b)

$N_{ba} = \lambda_a \cdot \tau_{k,c} \cdot \pi \cdot d_a \cdot h_{ef}$ ACI 318-19 Eq. (17.6.5.2.1)

Variables

$\tau_{k,c,uncr}$ [psi]	d_a [in.]	h_{ef} [in.]	$c_{a,min}$ [in.]	$\alpha_{overhead}$	$\tau_{k,c}$ [psi]
2,426	0.875	10.000	∞	1.000	1,376
c_{ac} [in.]	λ_a				
17.487	1.000				

Calculations

c_{Na} [in.]	A_{Na} [in. ²]	A_{Na0} [in. ²]	$\psi_{ed,Na}$
12.936	669.33	669.33	1.000
$\psi_{cp,Na}$	N_{ba} [lb]		
1.000	37,821		

Results

N_a [lb]	ϕ_{bond}	ϕN_a [lb]	N_{ua} [lb]
37,821	0.650	24,584	24,312

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3.3 Concrete Breakout Failure

$$N_{cb} = \left(\frac{A_{Nc}}{A_{Nc0}} \right) \psi_{ed,N} \psi_{c,N} \psi_{cp,N} N_b \quad \text{ACI 318-19 Eq. (17.6.2.1a)}$$

$$\phi N_{cb} \geq N_{ua} \quad \text{ACI 318-19 Table 17.5.2}$$

$$A_{Nc} \text{ see ACI 318-19, Section 17.6.2.1, Fig. R 17.6.2.1(b)}$$

$$A_{Nc0} = 9 h_{ef}^2 \quad \text{ACI 318-19 Eq. (17.6.2.1.4)}$$

$$\psi_{ed,N} = 0.7 + 0.3 \left(\frac{c_{a,min}}{1.5 h_{ef}} \right) \leq 1.0 \quad \text{ACI 318-19 Eq. (17.6.2.4.1b)}$$

$$\psi_{cp,N} = \text{MAX} \left(\frac{c_{a,min}}{c_{ac}}, \frac{1.5 h_{ef}}{c_{ac}} \right) \leq 1.0 \quad \text{ACI 318-19 Eq. (17.6.2.6.1b)}$$

$$N_b = k_c \lambda_a \sqrt{f'_c} h_{ef}^{1.5} \quad \text{ACI 318-19 Eq. (17.6.2.2.1)}$$

Variables

h_{ef} [in.]	$c_{a,min}$ [in.]	$\psi_{c,N}$	c_{ac} [in.]	k_c	λ_a	f'_c [psi]
10.000	∞	1.000	17.487	17	1.000	5,000

Calculations

A_{Nc} [in. ²]	A_{Nc0} [in. ²]	$\psi_{ed,N}$	$\psi_{cp,N}$	N_b [lb]
900.00	900.00	1.000	1.000	38,013

Results

N_{cb} [lb]	$\phi_{concrete}$	ϕN_{cb} [lb]	N_{ua} [lb]
38,013	0.650	24,709	24,312

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4 Shear load

	Load V_{ua} [lb]	Capacity ϕV_n [lb]	Utilization $\beta_v = V_{ua} / \phi V_n$	Status
Steel Strength*	N/A	N/A	N/A	N/A
Steel failure (with lever arm)*	N/A	N/A	N/A	N/A
Pryout Strength (Bond Strength controls)*	N/A	N/A	N/A	N/A
Concrete edge failure in direction **	N/A	N/A	N/A	N/A

* highest loaded anchor **anchor group (relevant anchors)

5 Warnings

- The anchor design methods in PROFIS Engineering require rigid anchor plates per current regulations (AS 5216:2021, ETAG 001/Annex C, EOTA TR029 etc.). This means load re-distribution on the anchors due to elastic deformations of the anchor plate are not considered - the anchor plate is assumed to be sufficiently stiff, in order not to be deformed when subjected to the design loading. PROFIS Engineering calculates the minimum required anchor plate thickness with CBFEM to limit the stress of the anchor plate based on the assumptions explained above. The proof if the rigid anchor plate assumption is valid is not carried out by PROFIS Engineering. Input data and results must be checked for agreement with the existing conditions and for plausibility!
- The equations presented in this report are based on imperial units. When inputs are displayed in metric units, the user should be aware that the equations remain in their imperial format.
- Condition A applies where the potential concrete failure surfaces are crossed by supplementary reinforcement proportioned to tie the potential concrete failure prism into the structural member. Condition B applies where such supplementary reinforcement is not provided, or where pullout or pryout strength governs.
- Design Strengths of adhesive anchor systems are influenced by the cleaning method. Refer to the INSTRUCTIONS FOR USE given in the Evaluation Service Report for cleaning and installation instructions.
- For additional information about ACI 318 strength design provisions, please go to <https://viewer.joomag.com/profis-design-guide-us-en-summer-2021/0841849001625154758?short&/>
- Installation of Hilti adhesive anchor systems shall be performed by personnel trained to install Hilti adhesive anchors. Reference ACI 318-19, Section 26.7.

Fastening meets the design criteria!

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6 Installation data

Profile: no profile

Hole diameter in the fixture: $d_f = 0.938$ in.

Plate thickness (input): 0.250 in.

Recommended plate thickness: not calculated

Drilling method: Hammer drilled

Cleaning: Compressed air cleaning of the drilled hole according to instructions for use is required

Anchor type and diameter: HIT-RE 500 V3 + HAS-E-55 (ASTM F1554 Gr.55) 7/8

Item number: 2198008 HAS-E-55 7/8"x13" (element) / 2123401 HIT-RE 500 V3 (adhesive)

Maximum installation torque: 1,500 in.lb

Hole diameter in the base material: 1.000 in.

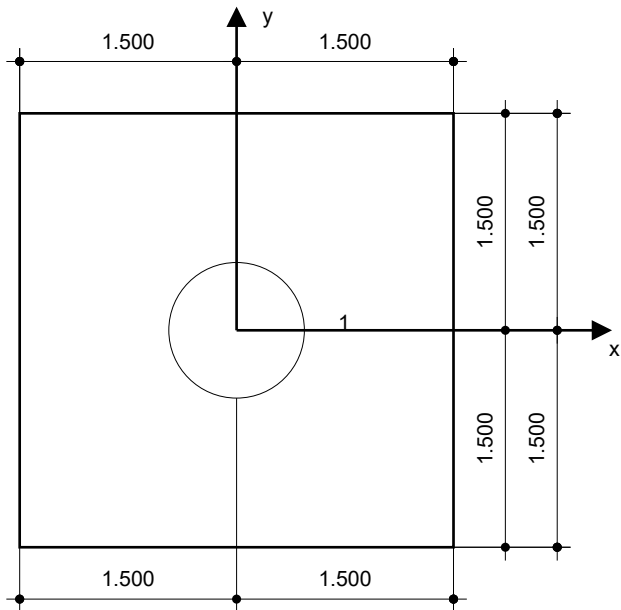
Hole depth in the base material: 10.000 in.

Minimum thickness of the base material: 12.000 in.

Hilti $\varnothing$ 7/8 in HIT-RE 500 V3 + HAS-E-55 (ASTM F1554 Gr.55) with 10 in nominal embedment depth per ICC-ES ESR-3814 , Hammer drill bit installation per MPII

6.1 Recommended accessories

Drilling	Cleaning	Setting
- Suitable Rotary Hammer - Properly sized drill bit	- Compressed air with required accessories to blow from the bottom of the hole - Proper diameter wire brush	- Dispenser including cassette and mixer - Torque wrench



Coordinates Anchor [in.]

Anchor	x	y	c _{-x}	c _{+x}	c _{-y}	c _{+y}
1	0.000	0.000	-	-	-	-



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7 Remarks; Your Cooperation Duties

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17.10 HSS BEAM TO COLUMN CONNECTION

Project:
Project no:
Author:



Project item HSS Column to beam

Design

Name: HSS Column to beam
Description:
Analysis: Stress, strain/ loads in equilibrium
Design code: AISC 360-22 (ASD) / ACI 318-14

Members

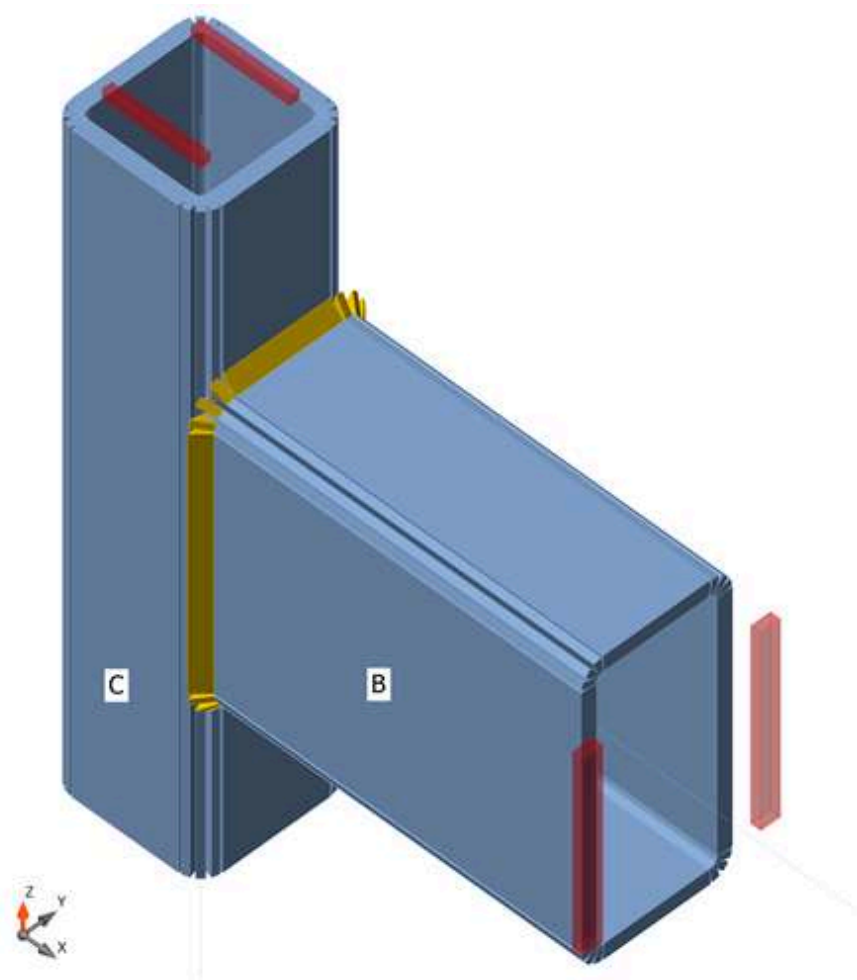
Geometry

Name	Cross-section	β - Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [in]	Offset ey [in]	Offset ez [in]
C	6 - HSS(Im)6X6X5/8	0.0	90.0	0.0	0"	0"	0"
B	5 - HSS(Im)12X6X5/8	0.0	0.0	0.0	0"	0"	0"

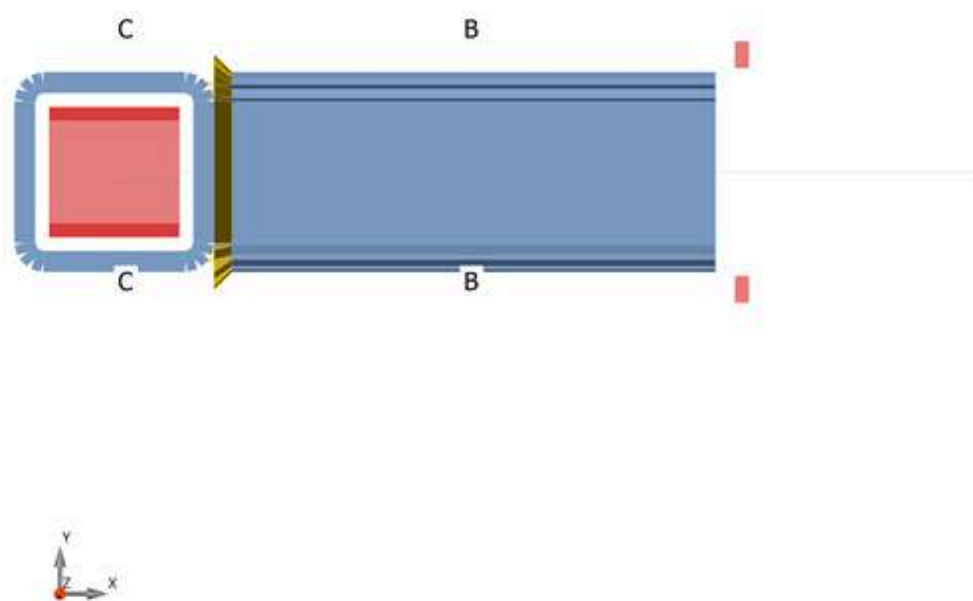
Supports and forces

Name	Support	Forces in	X [in]
C / begin	N-Vy-Vz-Mx-My-Mz	Bolts	0"
C / end	Vy-Mx-Mz	Bolts	0"
B / end	Vy-Mx-Mz	Bolts	0"

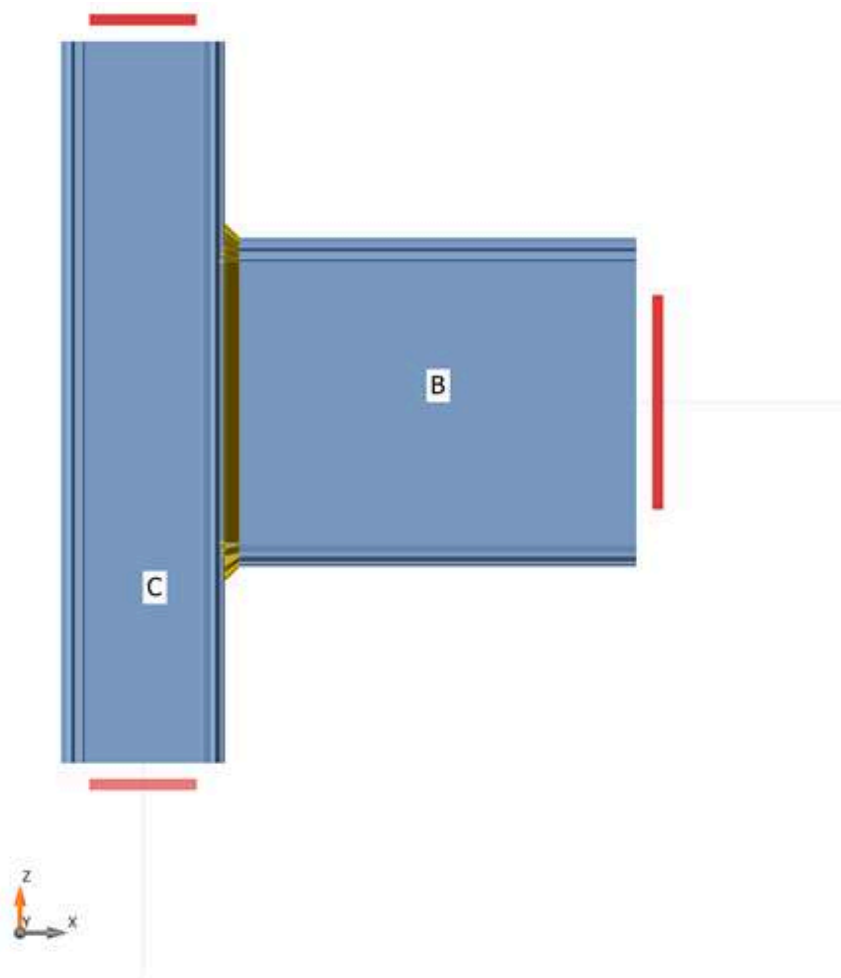
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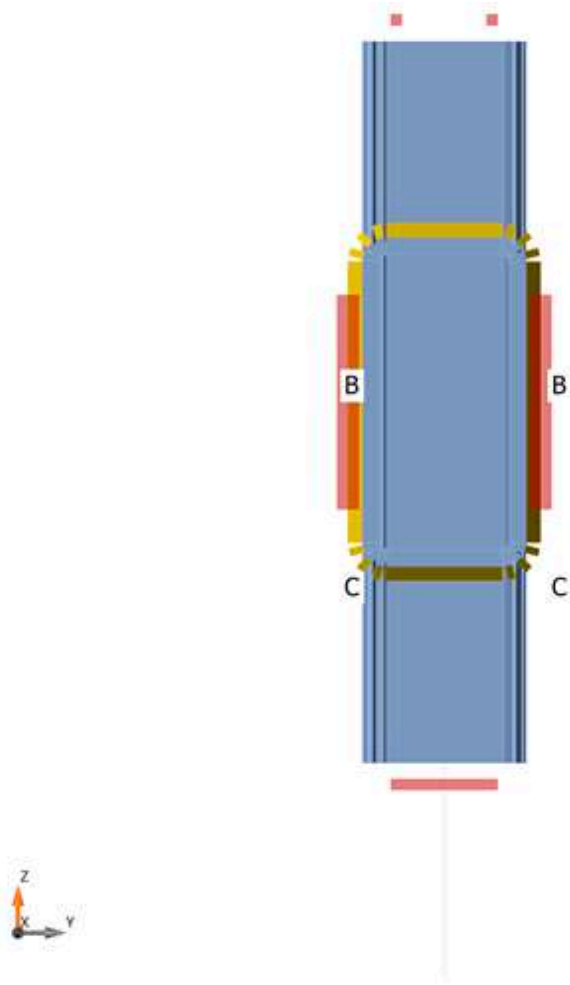
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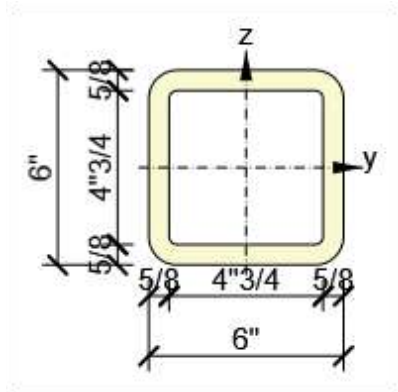
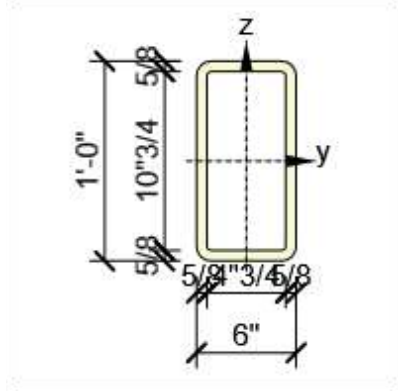


Cross-sections

Name	Material
6 - HSS(Imp)6X6X5/8	A500. Gr. C
5 - HSS(Imp)12X6X5/8	A500. Gr. C

Project:
Project no:
Author:

Cross-sections

Name	Material	Drawing
6 - HSS(Imp)6X6X5/8	A500. Gr. C	
5 - HSS(Imp)12X6X5/8	A500. Gr. C	

Load effects (forces in equilibrium)

Name	Member	N [kip]	Vy [kip]	Vz [kip]	Mx [kip.ft]	My [kip.ft]	Mz [kip.ft]
LE1	C / End	0.00	0.00	0.00	0.00	0.00	0.00
	B / End	3.10	0.00	-23.50	0.00	62.50	0.00
	C / Begin	0.00	0.00	0.00	0.00	0.00	0.00

Unbalanced forces

Name	X [kip]	Y [kip]	Z [kip]	Mx [kip.ft]	My [kip.ft]	Mz [kip.ft]
LE1	3.10	0.00	-23.50	0.00	62.50	0.00

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	1.9 < 5.0%	OK
Loc. deformation	0.3 < 3%	OK
Welds	88.8 < 100%	OK
Buckling	Not calculated	
GMNA	Calculated	

Project:
Project no:
Author:

Plates

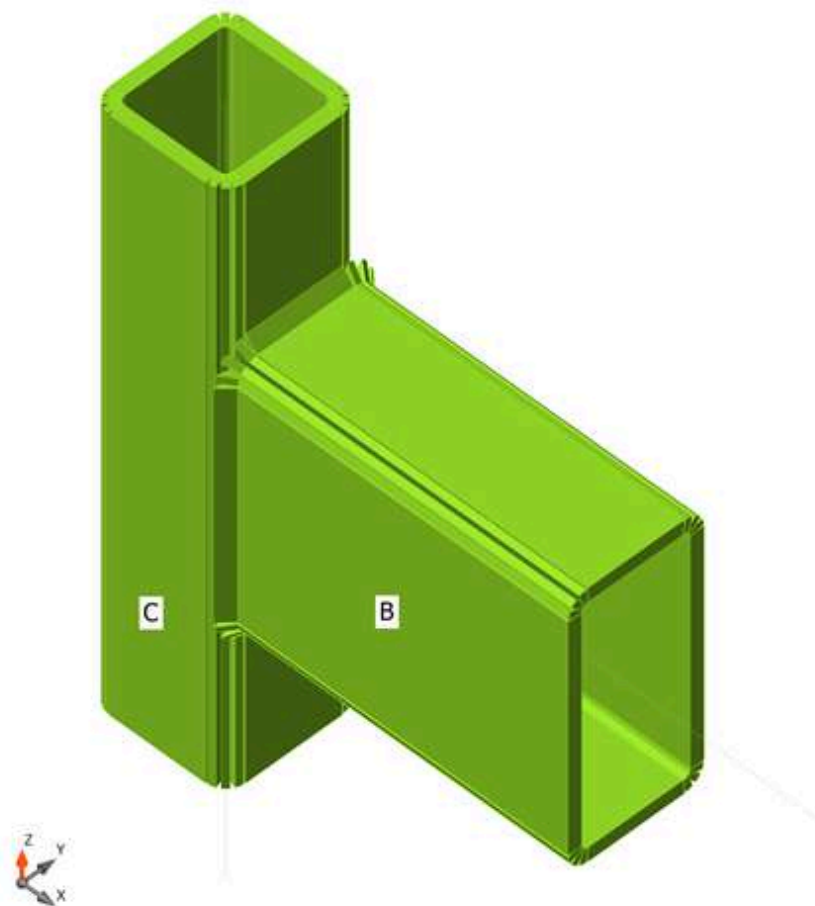
Name	t_p [in]	Loads	σ_{Ed} [kPa]	ϵ_{pl} [%]	$\sigma_{c,Ed}$ [kPa]	Status
C	5/8	LE1	210228.9	1.9	0.0	OK
B	5/8	LE1	185281.9	0.0	0.0	OK

Design data

Material	F_y [kPa]	ϵ_{lim} [%]
A500. Gr. C	344737.9	5.0

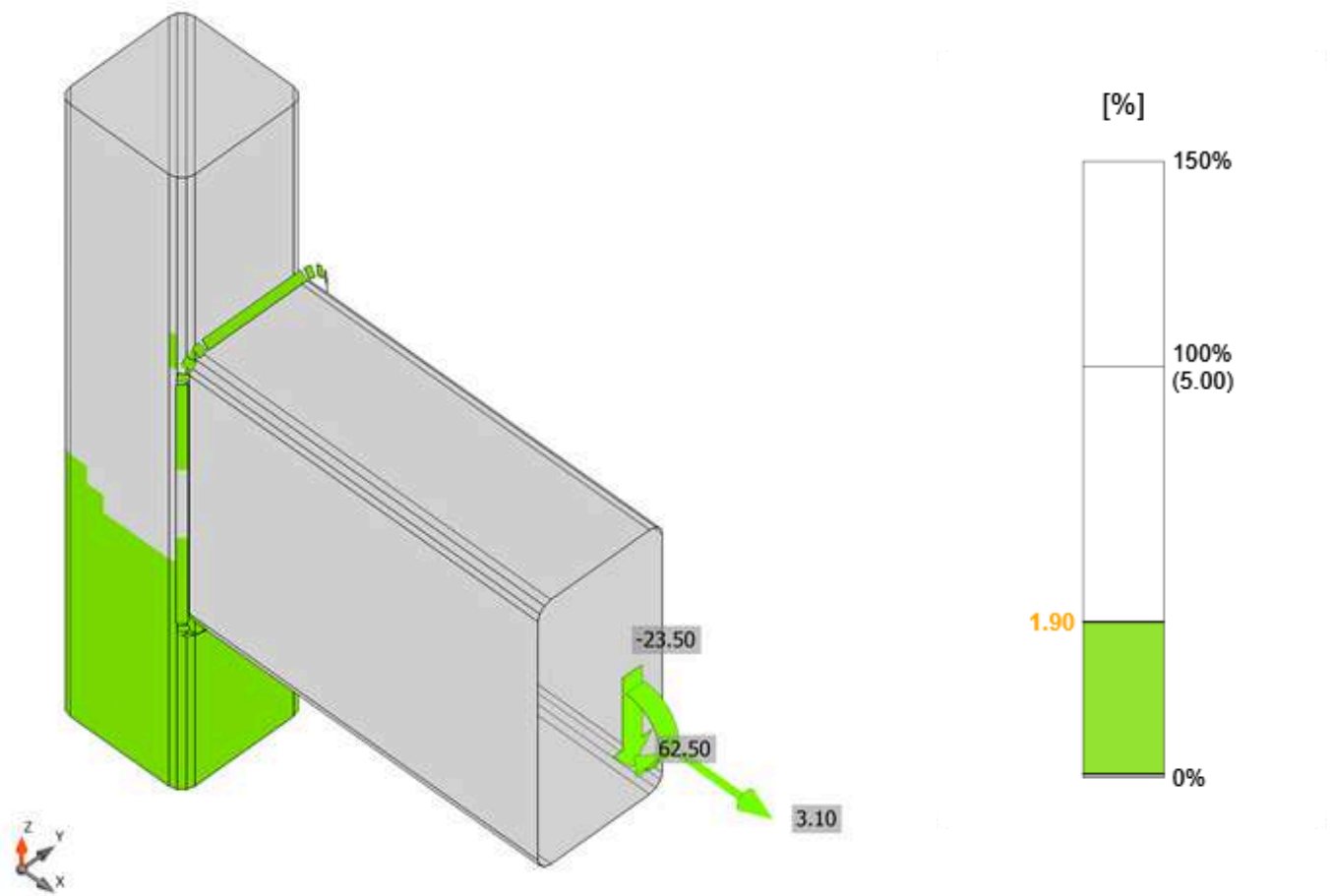
Loc. deformation

Name	d_0 [in]	Loads	δ [in]	δ_{lim} [in]	δ/d_0 [%]	Check status
C	6"	LE1	0"	3/16	0.3	OK
B	6"	LE1	0"	3/16	0.2	OK



Overall check, LE1

Project:
Project no:
Author:



Strain check, LE1

Welds

Item	Edge	Xu	t_w [in]	w [in]	L [in]	L_c [in]	Loads	F_n [kip]	R_n/Ω [kip]	Ut [%]	Ut _c [%]	Status
C-w 1	B	E70xx	▲ 3/8	▲ 1/2	2'-8"3/8	5/16	LE1	2.02	2.27	88.8	88.0	OK

Design data

Material	F_{exx} [kPa]
E70xx	482633.0

Buckling

Buckling analysis was not calculated.

17.11 BASE PLATE CONNECTION

Project:
Project no:
Author:

Project item COLUMN BASE PLATE

Design

Name COLUMN BASE PLATE
Description
Analysis Stress, strain/ loads in equilibrium
Design code AISC 360-16 (LRFD) / ACI 318-14

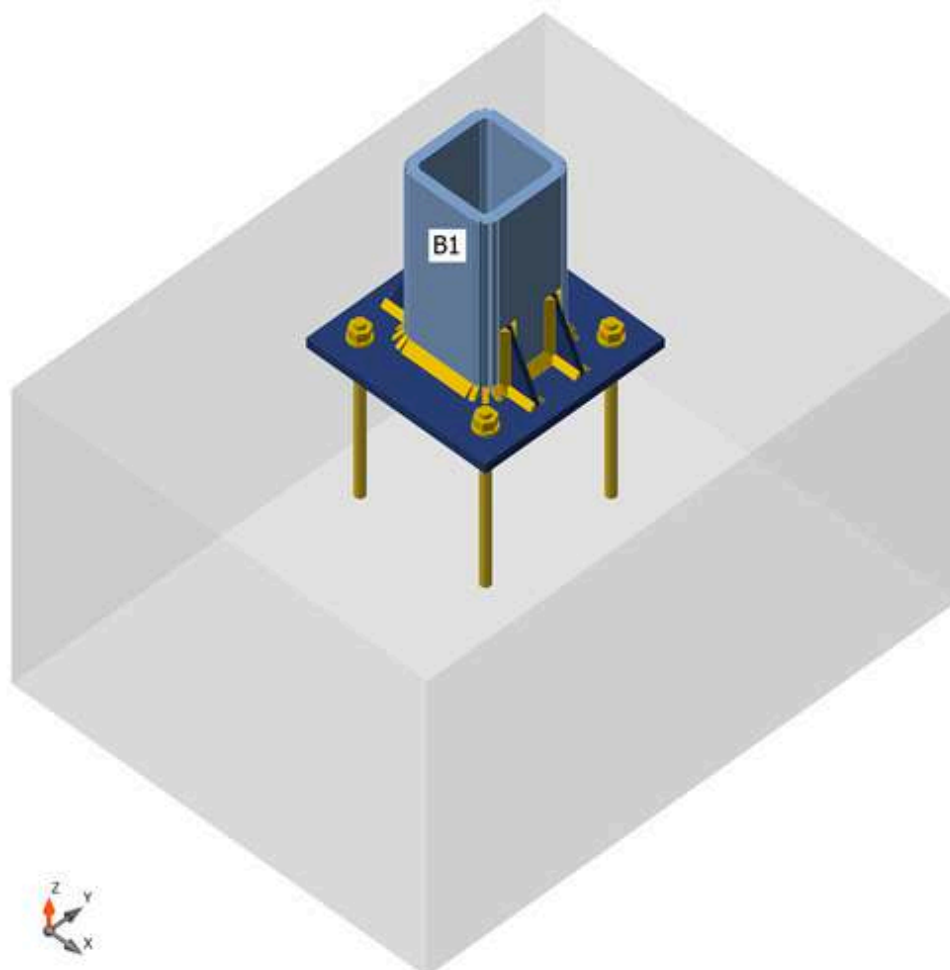
Members

Geometry

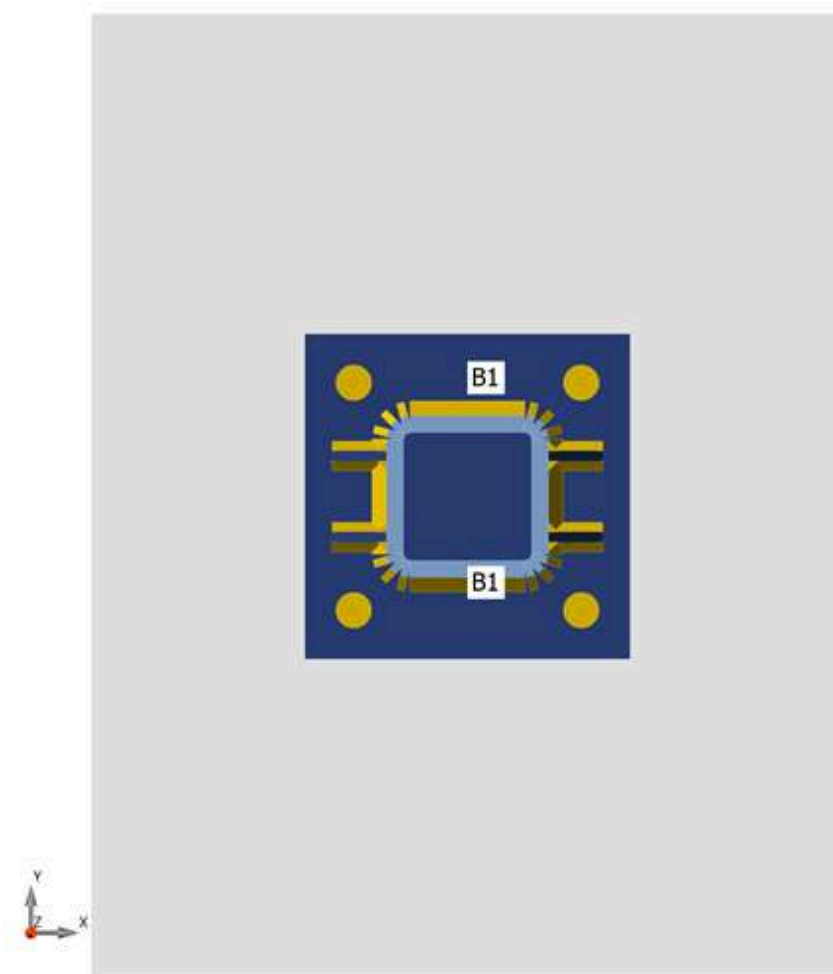
Name	Cross-section	β - Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [in]	Offset ey [in]	Offset ez [in]
B1	6 - HSS(Im)6X6X5/8	0.0	90.0	0.0	0"	0"	0"

Supports and forces

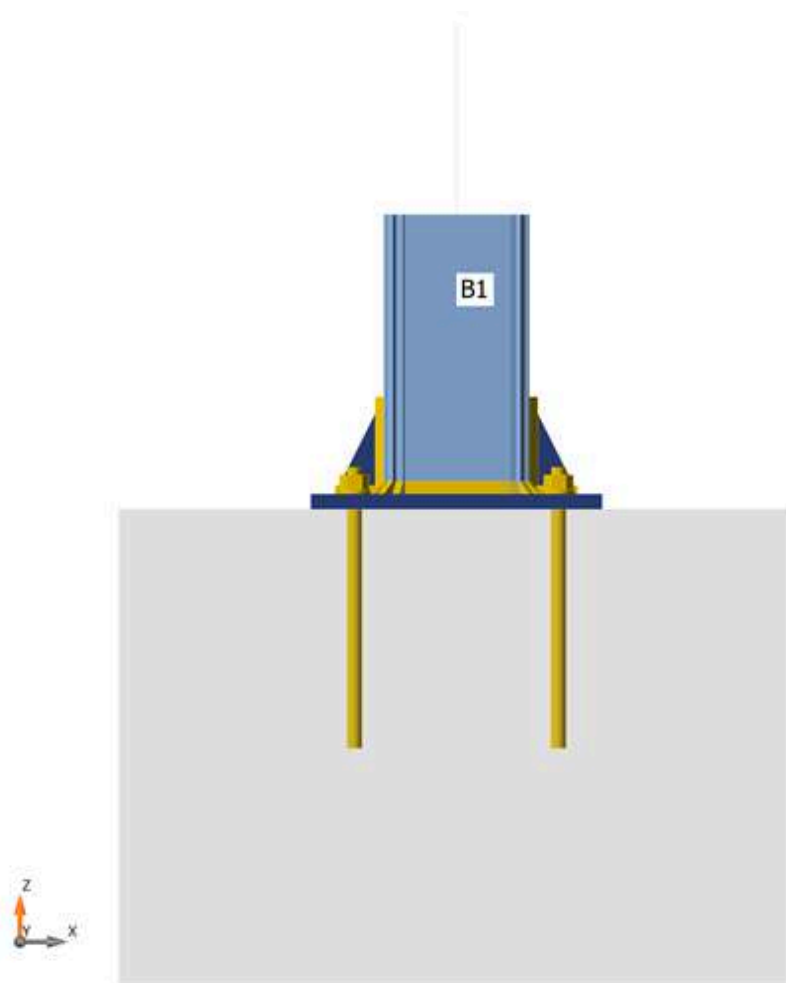
Name	Support	Forces in	X [in]
B1 / end		Node	0"



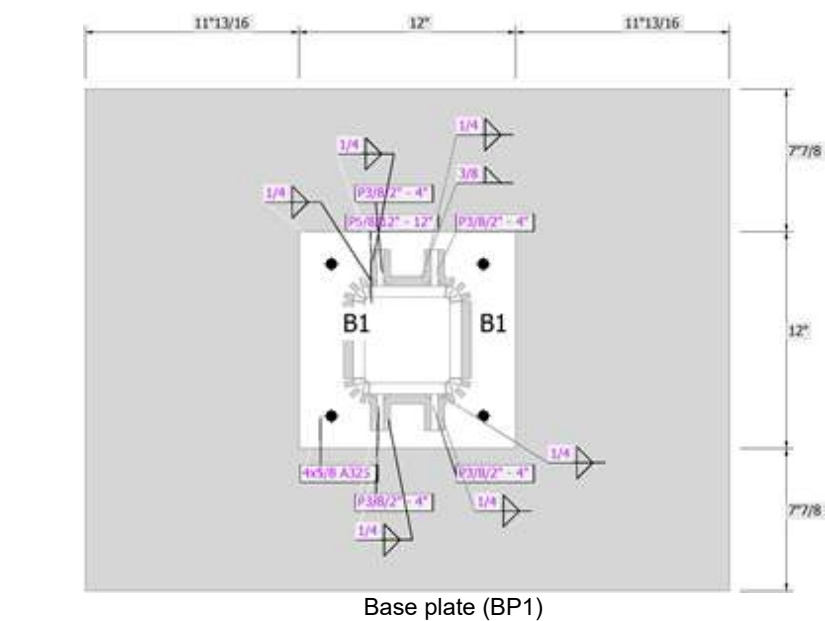
Project:
Project no:
Author:



Project:
Project no:
Author:



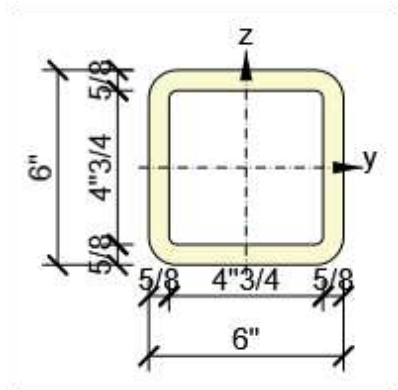
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Name	Material
6 - HSS(Imp)6X6X5/8	A500. Gr. C

Project:
Project no:
Author:

Cross-sections

Name	Material	Drawing
6 - HSS(Imp)6X6X5/8	A500. Gr. C	

Anchors

Name	Diameter [in]	f_y [kPa]	f_u [kPa]	Gross area [in ²]
5/8 A325	5/8	634317.7	827370.9	0.3068

Load effects (forces in equilibrium)

Name	Member	N [kip]	V _y [kip]	V _z [kip]	M _x [kip.ft]	M _y [kip.ft]	M _z [kip.ft]
LE1	B1 / End	3.40	0.00	-5.20	0.00	32.00	0.00

Unbalanced forces

Name	X [kip]	Y [kip]	Z [kip]	M _x [kip.ft]	M _y [kip.ft]	M _z [kip.ft]
LE1	5.20	0.00	3.40	0.00	32.00	0.00

Foundation block

Item	Value	Unit
CB 1		
Dimensions	2'-11"5/8 x 2'-3"3/4	in
Depth	1'-7"11/16	in
Anchor	5/8 A325	
Anchoring length	9"13/16	in
Shear force transfer	Anchors	

Project:
Project no:
Author:

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	$3.4 < 5.0\%$	OK
Loc. deformation	$0.1 < 3\%$	OK
Anchors	$273.0 > 100\%$	Not OK!
Welds	$92.3 < 100\%$	OK
Concrete block	$49.8 < 100\%$	OK
Buckling	Not calculated	

Plates

Name	Material	t_p [in]	Loads	σ_{Ed} [kPa]	ϵ_{pl} [%]	$\sigma_{c,Ed}$ [kPa]	Status
B1	A500. Gr. C	5/8	LE1	285488.7	0.0	0.0	OK
BP1	A572 Gr.50	5/8	LE1	312995.9	1.4	0.0	OK
RIB1a	A36	3/8	LE1	230132.1	3.4	0.0	OK
RIB1b	A36	3/8	LE1	230241.3	3.4	0.0	OK
RIB2a	A36	3/8	LE1	226796.8	1.7	0.0	OK
RIB2b	A36	3/8	LE1	226843.9	1.7	0.0	OK

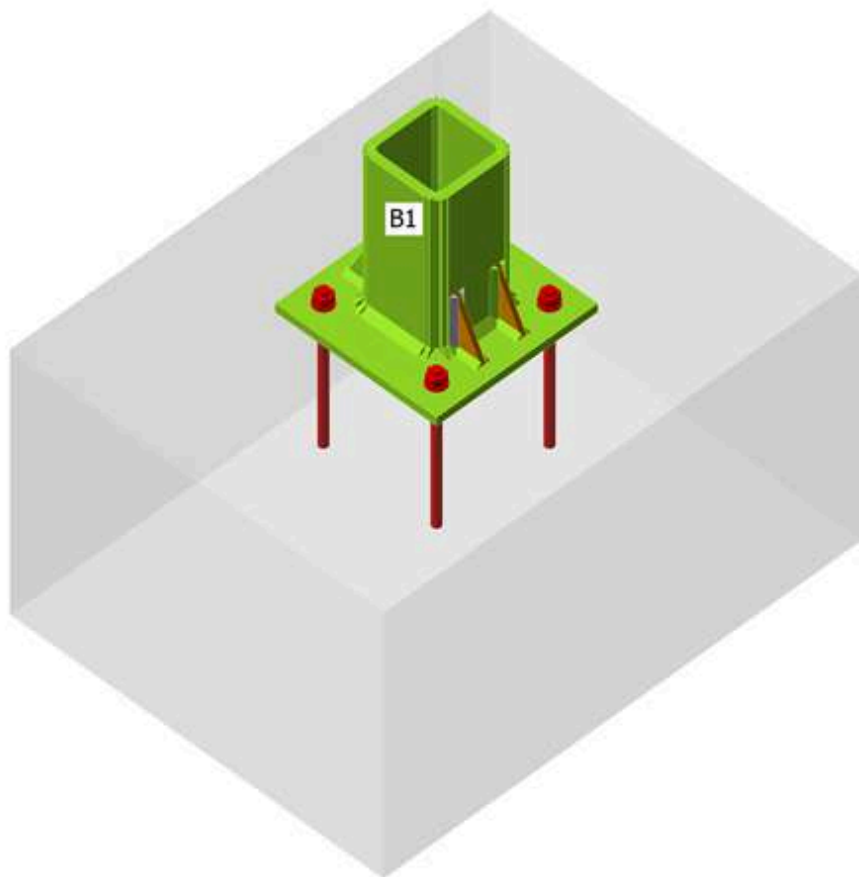
Design data

Material	F_y [kPa]	ϵ_{lim} [%]
A500. Gr. C	344737.9	5.0
A572 Gr.50	344740.0	5.0
A36	248210.0	5.0

Loc. deformation

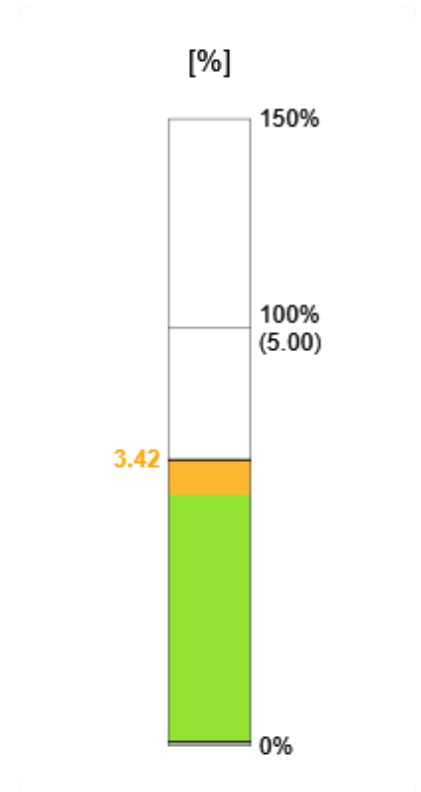
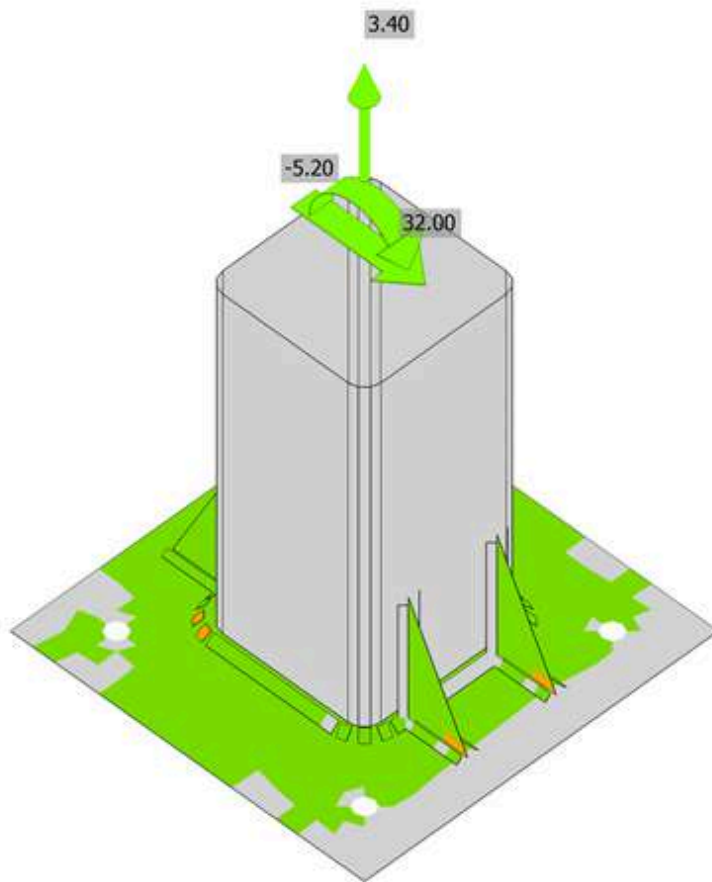
Name	d_0 [in]	Loads	δ [in]	δ_{lim} [in]	δ/d_0 [%]	Check status
B1	6"	LE1	0"	3/16	0.1	OK

Project:
Project no:
Author:



Overall check, LE1

Project:
Project no:
Author:



Strain check, LE1

Anchors

Shape	Item	Loads	N_f [kip]	V [kip]	ϕN_{cbg} [kip]	ϕV_{cbg} [kip]	ϕV_{cp} [kip]	U_{t_t} [%]	U_{t_s} [%]	$U_{t_{ts}}$ [%]	Status
	A5	LE1	20.83	1.37	29.39	0.00	65.30	173.2	12.9	253.1	Not OK!
	A6	LE1	20.76	1.38	29.39	0.00	65.30	173.2	13.0	253.2	Not OK!
	A7	LE1	4.66	1.29	29.39	12.50	65.30	173.2	41.6	273.0	Not OK!
	A8	LE1	4.66	1.29	29.39	12.50	65.30	173.2	41.6	273.0	Not OK!

Design data

Grade	ϕN_{sa} [kip]	ϕV_{sa} [kip]
5/8 A325 - 1	18.98	10.58

Project:
Project no:
Author:

Welds

Item	Edge	Xu	t_w [in]	w [in]	L [in]	L_c [in]	Loads	F_n [kip]	ϕR_n [kip]	Ut [%]	Ut _c [%]	Status
BP1	RIB1a	E70xx	▲ 1/4 ▼	▲ 3/8 ▼	2"	3/8	LE1	3.21	4.17	77.0	61.1	OK
		E70xx	▲ 1/4 ▼	▲ 3/8 ▼	2"	3/8	LE1	3.23	4.21	76.9	59.2	OK
B1-w 1	RIB1a	E70xx	▲ 1/4 ▼	▲ 3/8 ▼	4"	3/8	LE1	2.23	4.60	48.4	26.1	OK
		E70xx	▲ 1/4 ▼	▲ 3/8 ▼	4"	3/8	LE1	2.65	4.26	62.2	36.2	OK
BP1	RIB1b	E70xx	▲ 1/4 ▼	▲ 3/8 ▼	2"	3/8	LE1	3.22	4.20	76.8	56.0	OK
		E70xx	▲ 1/4 ▼	▲ 3/8 ▼	2"	3/8	LE1	3.20	4.16	76.9	58.1	OK
B1-w 1	RIB1b	E70xx	▲ 1/4 ▼	▲ 3/8 ▼	3"15/16	3/8	LE1	1.71	3.22	53.1	33.8	OK
		E70xx	▲ 1/4 ▼	▲ 3/8 ▼	4"	3/8	LE1	1.87	4.69	39.8	22.4	OK
BP1	RIB2a	E70xx	▲ 1/4 ▼	▲ 3/8 ▼	2"	3/8	LE1	3.54	4.68	75.6	46.8	OK
		E70xx	▲ 1/4 ▼	▲ 3/8 ▼	2"	3/8	LE1	3.03	3.86	78.6	68.2	OK
B1-w 3	RIB2a	E70xx	▲ 1/4 ▼	▲ 3/8 ▼	4"	3/8	LE1	2.19	4.30	50.9	27.7	OK
		E70xx	▲ 1/4 ▼	▲ 3/8 ▼	4"	3/8	LE1	2.25	4.54	49.7	28.4	OK
BP1	RIB2b	E70xx	▲ 1/4 ▼	▲ 3/8 ▼	2"	3/8	LE1	2.98	3.77	79.0	69.1	OK
		E70xx	▲ 1/4 ▼	▲ 3/8 ▼	2"	3/8	LE1	3.53	4.67	75.6	53.3	OK
B1-w 3	RIB2b	E70xx	▲ 1/4 ▼	▲ 3/8 ▼	4"	3/8	LE1	2.54	4.57	55.7	30.6	OK
		E70xx	▲ 1/4 ▼	▲ 3/8 ▼	4"	3/8	LE1	2.74	4.22	65.0	36.3	OK
BP1	B1	E70xx	▲ 3/8 ▼	▲ 1/2 ▼	1'-8"3/8	5/16	LE1	3.14	3.41	92.3	92.3	OK

Design data

Material	F_{exx} [kPa]
E70xx	482633.0

Concrete block

Item	Loads	A_1 [in ²]	A_2 [in ²]	σ [kPa]	$\phi f_{p,max}$ [kPa]	Ut [%]	Status
CB 1	LE1	22.3278	439.0826	15164.4	30474.8	49.8	OK

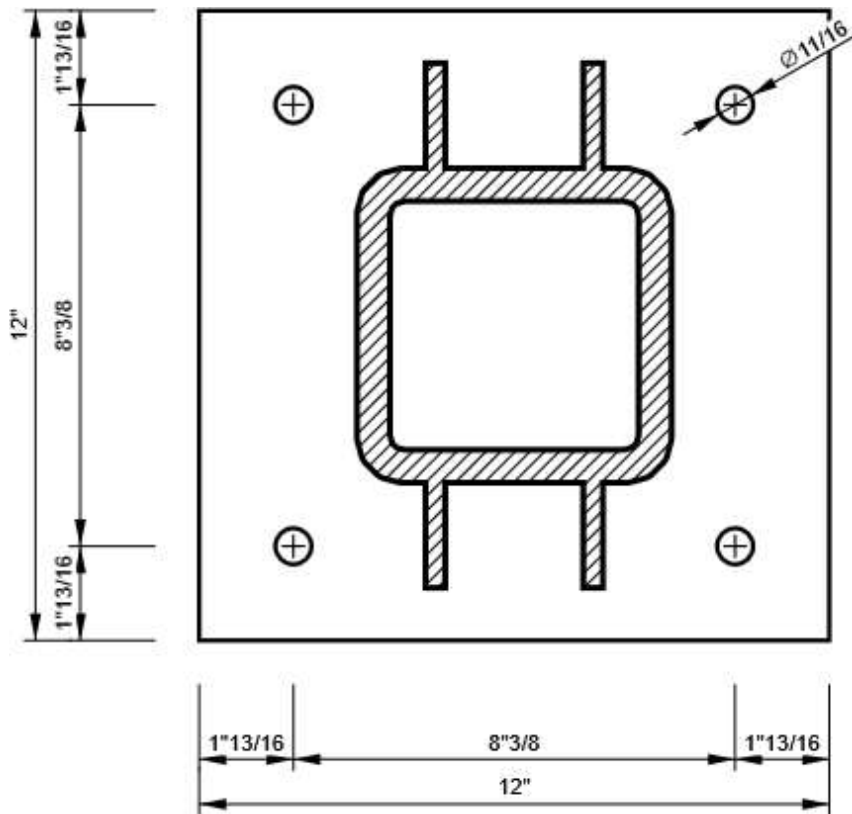
Buckling

Buckling analysis was not calculated.

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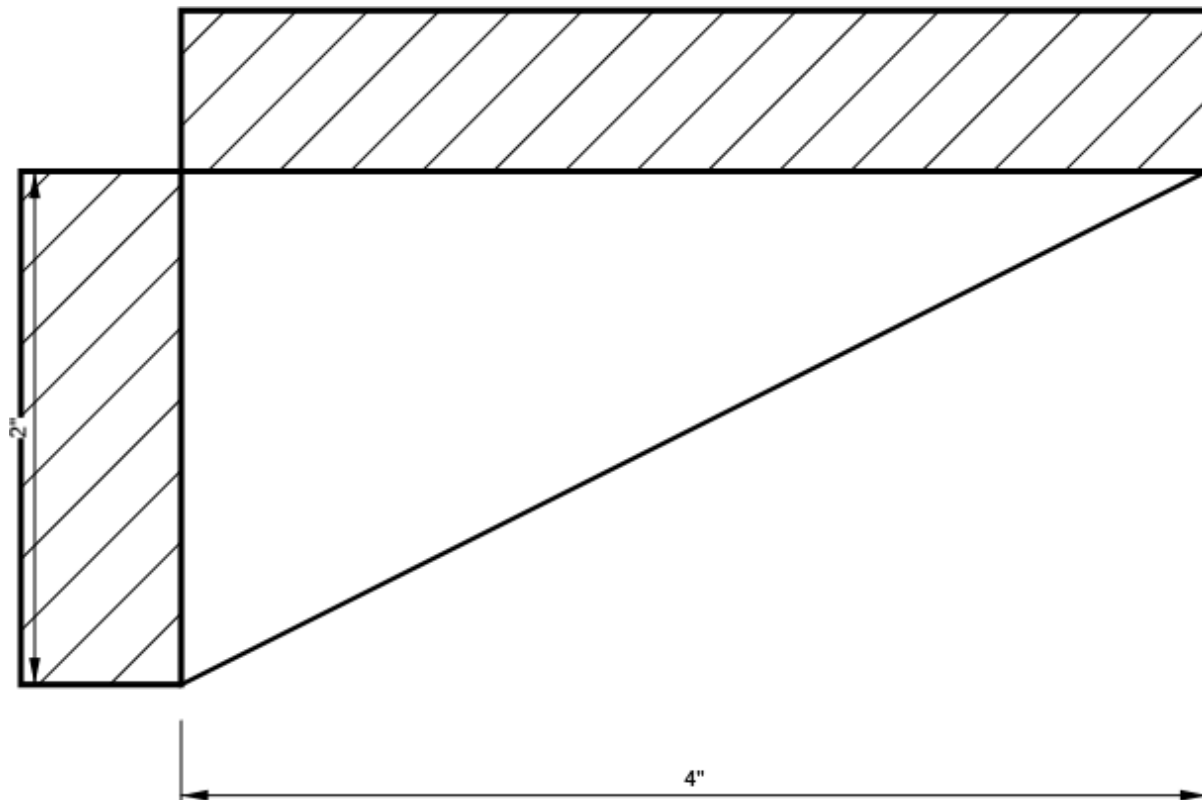
Project:
Project no:
Author:

P5/8x12"-12" (A572 Gr.50)



RIB1

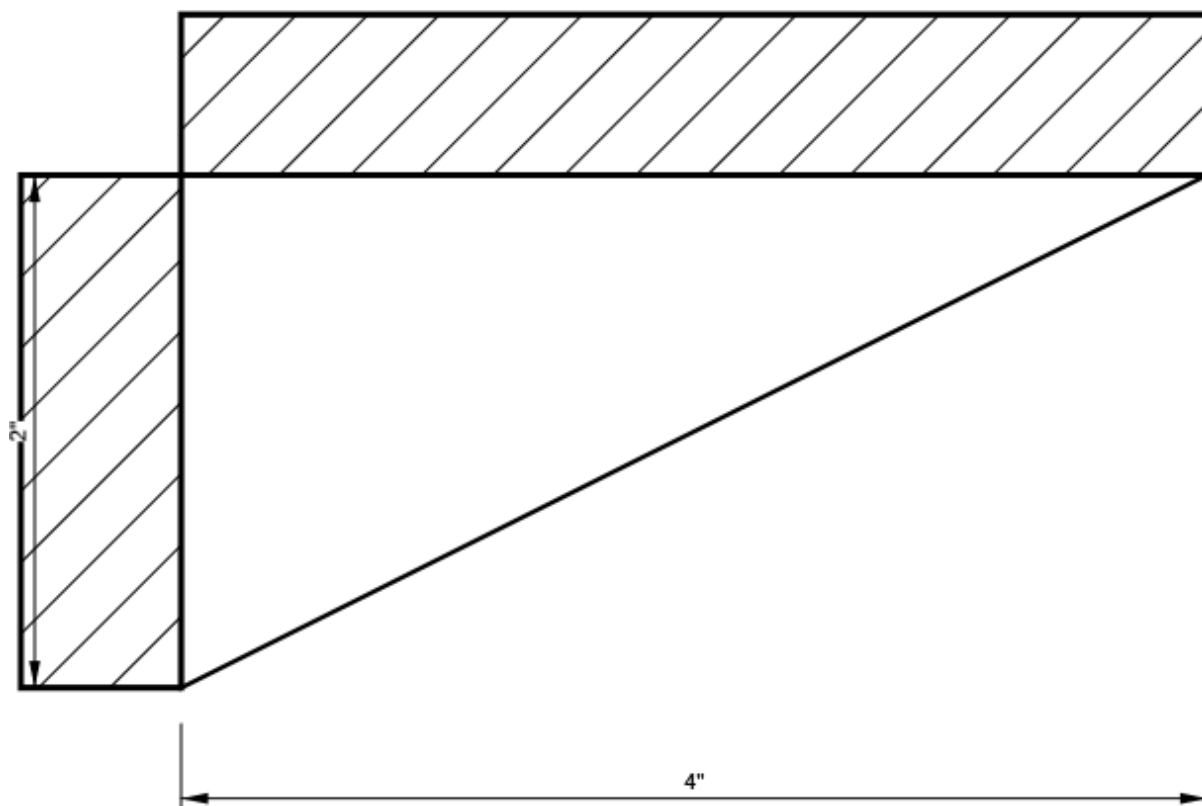
P3/8x4"-2" (A36)



Project:
Project no:
Author:

RIB2

P3/8x4"-2" (A36)



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Specifier's comments:

1 Input data

Anchor type and diameter:

Embed plate D1.1 GR. B 7/8

Item number:

not available

Specification text:

Ø 7/8 in Embed plate D1.1 GR. B with 12 in nominal embedment depth per Technical data, cast in place installation per MPII,

Effective embedment depth:

 $h_{ef} = 12.000$ in.

Material:

Evaluation Service Report:

Hilti Technical Data

Issued | Valid:

- | -

Proof:

Design Method ACI 318-19 / CIP

Shear edge breakout verification:

Row closest to edge (Case 3 only from ACI 318-19 Fig. R.17.7.2.1b)

Stand-off installation:

 $e_b = 0.000$ in. (no stand-off); $t = 0.375$ in.

Anchor plate^R:

 $l_x \times l_y \times t = 12.000$ in. x 12.000 in. x 0.375 in.; (Recommended plate thickness: not calculated)

Profile:

Square HSS (AISC), HSS6X6X.625; (L x W x T) = 6.000 in. x 6.000 in. x 0.625 in.

Base material:

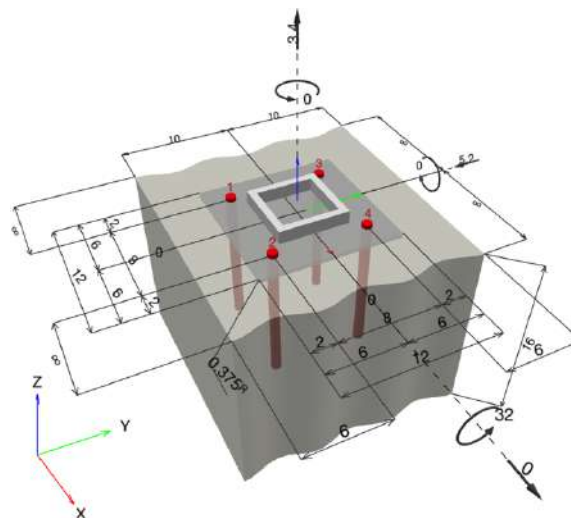
cracked concrete, 5000, $f'_c = 5,000$ psi; $h = 16.000$ in.

Reinforcement:

tension: present, shear: not present; anchor reinforcement: tension
edge reinforcement: none or < No. 4 bar

^R - The anchor calculation is based on a rigid anchor plate assumption.

Geometry [in.] & Loading [kip, ft.kip]



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1.1 Design results

Case	Description	Forces [kip] / Moments [ft.kip]	Seismic	Max. Util. Anchor [%]
1	Combination 1	N = 3.400; $V_x = 0.000$; $V_y = -5.200$; $M_x = 32.00000$; $M_y = 0.00000$; $M_z = 0.00000$;	no	92

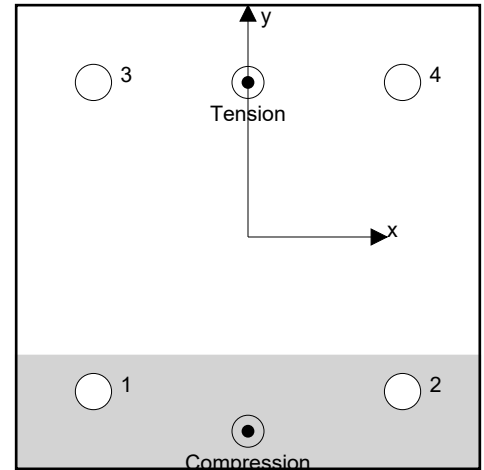
2 Load case/Resulting anchor forces

Anchor reactions [kip]

Tension force: (+Tension, -Compression)

Anchor	Tension force	Shear force	Shear force x	Shear force y
1	0.000	1.300	0.000	-1.300
2	0.000	1.300	0.000	-1.300
3	22.218	1.300	0.000	-1.300
4	22.218	1.300	0.000	-1.300

Max. concrete compressive strain: 0.54 [‰]
 Max. concrete compressive stress: 2,342 [psi]
 Resulting tension force in (x/y)=(0.000/4.000): 44.435 [kip]
 Resulting compression force in (x/y)=(0.000/-5.026): 41.035 [kip]



Anchor forces are calculated based on the assumption of a rigid anchor plate.

3 Tension load

	Load N_{ua} [kip]	Capacity ϕN_n [kip]	Utilization $\beta_N = N_{ua} / \phi N_n$	Status
Steel Strength*	22.218	29.299	76	OK
Pullout Strength*	22.218	24.752	90	OK
Concrete Breakout Failure** ¹	N/A	N/A	N/A	N/A
Concrete Side-Face Blowout, direction **	N/A	N/A	N/A	N/A

* highest loaded anchor **anchor group (anchors in tension)

¹ Tension Anchor Reinforcement has been selected!

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3.1 Steel Strength

$$N_{sa} = A_{se,N} f_{uta} \quad \text{ACI 318-19 Eq. (17.6.1.2)}$$

$$\phi N_{sa} \geq N_{ua} \quad \text{ACI 318-19 Table 17.5.2}$$

Variables

$A_{se,N} [\text{in.}^2]$	$f_{uta} [\text{psi}]$
0.59	65,000

Calculations

$N_{sa} [\text{kip}]$
39.065

Results

$N_{sa} [\text{kip}]$	ϕ_{steel}	$\phi N_{sa} [\text{kip}]$	$N_{ua} [\text{kip}]$
39.065	0.750	29.299	22.218

3.2 Pullout Strength

$$N_{pN} = \psi_{c,p} N_p \quad \text{ACI 318-19 Eq. (17.6.3.1)}$$

$$N_p = 8 A_{brg} f'_c \quad \text{ACI 318-19 Eq. (17.6.3.2.2a)}$$

$$\phi N_{pN} \geq N_{ua} \quad \text{ACI 318-19 Table 17.5.2}$$

Variables

$\psi_{c,p}$	$A_{brg} [\text{in.}^2]$	λ_a	$f'_c [\text{psi}]$
1.000	0.88	1.000	5,000

Calculations

$N_p [\text{kip}]$
35.360

Results

$N_{pn} [\text{kip}]$	ϕ_{concrete}	$\phi N_{pn} [\text{kip}]$	$N_{ua} [\text{kip}]$
35.360	0.700	24.752	22.218

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4 Shear load

	Load V_{ua} [kip]	Capacity ϕV_n [kip]	Utilization $\beta_v = V_{ua}/\phi V_n$	Status
Steel Strength*	1.300	25.392	6	OK
Steel failure (with lever arm)*	N/A	N/A	N/A	N/A
Pryout Strength**	5.200	54.117	10	OK
Concrete edge failure in direction y-**	5.200	24.249	22	OK

* highest loaded anchor **anchor group (relevant anchors)

When the input edge distance is set to "infinity", edge breakout verification is not performed in that direction

4.1 Steel Strength

$$V_{sa} = A_{se,V} f_{uta} \quad \text{ACI 318-19 Eq. (17.7.1.2a)}$$

$$\phi V_{steel} \geq V_{ua} \quad \text{ACI 318-19 Table 17.5.2}$$

Variables

$A_{se,V}$ [in. ²]	f_{uta} [psi]
0.59	65,000

Calculations

V_{sa} [kip]
39.065

Results

V_{sa} [kip]	ϕ_{steel}	ϕV_{sa} [kip]	V_{ua} [kip]
39.065	0.650	25.392	1.300

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4.2 Pryout Strength

$$V_{cp} = k_{cp} \left[\left(\frac{A_{Nc}}{A_{Nc0}} \right) \psi_{ec,N} \psi_{ed,N} \psi_{c,N} \psi_{cp,N} N_b \right] \quad \text{ACI 318-19 Eq. (17.7.3.1b)}$$

$$\phi V_{cp} \geq V_{ua} \quad \text{ACI 318-19 Table 17.5.2}$$

$$A_{Nc} \text{ see ACI 318-19, Section 17.6.2.1, Fig. R 17.6.2.1(b)}$$

$$A_{Nc0} = 9 h_{ef}^2 \quad \text{ACI 318-19 Eq. (17.6.2.1.4)}$$

$$\psi_{ec,N} = \left(\frac{1}{1 + \frac{2 e_N}{3 h_{ef}}} \right) \leq 1.0 \quad \text{ACI 318-19 Eq. (17.6.2.3.1)}$$

$$\psi_{ed,N} = 0.7 + 0.3 \left(\frac{c_{a,min}}{1.5 h_{ef}} \right) \leq 1.0 \quad \text{ACI 318-19 Eq. (17.6.2.4.1b)}$$

$$\psi_{cp,N} = \text{MAX} \left(\frac{c_{a,min}}{c_{ac}}, \frac{1.5 h_{ef}}{c_{ac}} \right) \leq 1.0 \quad \text{ACI 318-19 Eq. (17.6.2.6.1b)}$$

$$N_b = 16 \lambda_a \sqrt{f'_c} h_{ef}^{5/3} \quad \text{ACI 318-19 Eq. (17.6.2.2.3)}$$

Variables

k_{cp}	h_{ef} [in.]	$e_{c1,N}$ [in.]	$e_{c2,N}$ [in.]	$c_{a,min}$ [in.]
2	12.000	0.000	0.000	6.000
$\psi_{c,N}$	c_{ac} [in.]	k_c	λ_a	f'_c [psi]
1.000	-	16	1.000	5,000

Calculations

A_{Nc} [in. ²]	A_{Nc0} [in. ²]	$\psi_{ec1,N}$	$\psi_{ec2,N}$	$\psi_{ed,N}$	$\psi_{cp,N}$	N_b [kip]
880.00	1,296.00	1.000	1.000	0.800	1.000	71.161

Results

V_{cp} [kip]	$\phi_{concrete}$	ϕV_{cp} [kip]	V_{ua} [kip]
77.310	0.700	54.117	5.200

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4.3 Concrete edge failure in direction y-

$$V_{cbg} = \left(\frac{A_{Vc}}{A_{Vc0}} \right) \psi_{ec,V} \psi_{ed,V} \psi_{c,V} \psi_{h,V} \psi_{parallel,V} V_b \quad \text{ACI 318-19 Eq. (17.7.2.1b)}$$

$$\phi V_{cbg} \geq V_{ua} \quad \text{ACI 318-19 Table 17.5.2}$$

$$A_{Vc} \text{ see ACI 318-19, Section 17.7.2.1, Fig. R 17.7.2.1(b)*}$$

$$A_{Vc0} = 4.5 c_{a1}^2 \quad \text{ACI 318-19 Eq. (17.7.2.1.3)}$$

$$\psi_{ec,V} = \left(\frac{1}{1 + \frac{e_v}{1.5c_{a1}}} \right) \leq 1.0 \quad \text{ACI 318-19 Eq. (17.7.2.3.1)}$$

$$\psi_{ed,V} = 0.7 + 0.3 \left(\frac{c_{a2}}{1.5c_{a1}} \right) \leq 1.0 \quad \text{ACI 318-19 Eq. (17.7.2.4.1b)}$$

$$\psi_{h,V} = \sqrt{\frac{1.5c_{a1}}{h_a}} \geq 1.0 \quad \text{ACI 318-19 Eq. (17.7.2.6.1)}$$

$$V_b = 9 \lambda_a \sqrt{f_c} c_{a1}^{1.5} \quad \text{ACI 318-19 Eq. (17.7.2.2.1b)}$$

Variables

c_{a1} [in.]	c_{a2} [in.]	e_{cV} [in.]	$\psi_{c,V}$	h_a [in.]
14.000	-	0.000	1.000	16.000
l_e [in.]	λ_a	d_a [in.]	f_c [psi]	$\psi_{parallel,V}$
7.000	1.000	0.875	5,000	1.000

Calculations

A_{Vc} [in. ²]	A_{Vc0} [in. ²]	$\psi_{ec,V}$	$\psi_{ed,V}$	$\psi_{h,V}$	V_b [kip]
800.00	882.00	1.000	1.000	1.146	33.336

Results

V_{cbg} [kip]	$\phi_{concrete}$	ϕV_{cbg} [kip]	V_{ua} [kip]
34.641	0.700	24.249	5.200

*Anchor row defined by: Anchor 3, 4; Case 3 controls

When the input edge distance is set to "infinity", edge breakout verification is not performed in that direction

5 Combined tension and shear loads, per ACI 318-19 section 17.8

β_N	β_V	ζ	Utilization β_{NV} [%]	Status
0.898	0.214	5/3	92	OK

$$\beta_{NV} = \beta_N^\zeta + \beta_V^\zeta \leq 1$$



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6 Warnings

- The anchor design methods in PROFIS Engineering require rigid anchor plates per current regulations (AS 5216:2021, ETAG 001/Annex C, EOTA TR029 etc.). This means load re-distribution on the anchors due to elastic deformations of the anchor plate are not considered - the anchor plate is assumed to be sufficiently stiff, in order not to be deformed when subjected to the design loading. PROFIS Engineering calculates the minimum required anchor plate thickness with CBFEM to limit the stress of the anchor plate based on the assumptions explained above. The proof if the rigid anchor plate assumption is valid is not carried out by PROFIS Engineering. Input data and results must be checked for agreement with the existing conditions and for plausibility!
- The equations presented in this report are based on imperial units. When inputs are displayed in metric units, the user should be aware that the equations remain in their imperial format.
- Condition A applies where the potential concrete failure surfaces are crossed by supplementary reinforcement proportioned to tie the potential concrete failure prism into the structural member. Condition B applies where such supplementary reinforcement is not provided, or where pullout or pryout strength governs.
- For additional information about ACI 318 strength design provisions, please go to <https://viewer.joomag.com/profis-design-guide-us-en-summer-2021/0841849001625154758?short&/>
- The design of Anchor Reinforcement is beyond the scope of PROFIS Engineering. Refer to ACI 318-19, Section 17.5.2.1 (a) for information about Anchor Reinforcement.
- Anchor Reinforcement has been selected as a design option, calculations should be compared with PROFIS Engineering calculations.

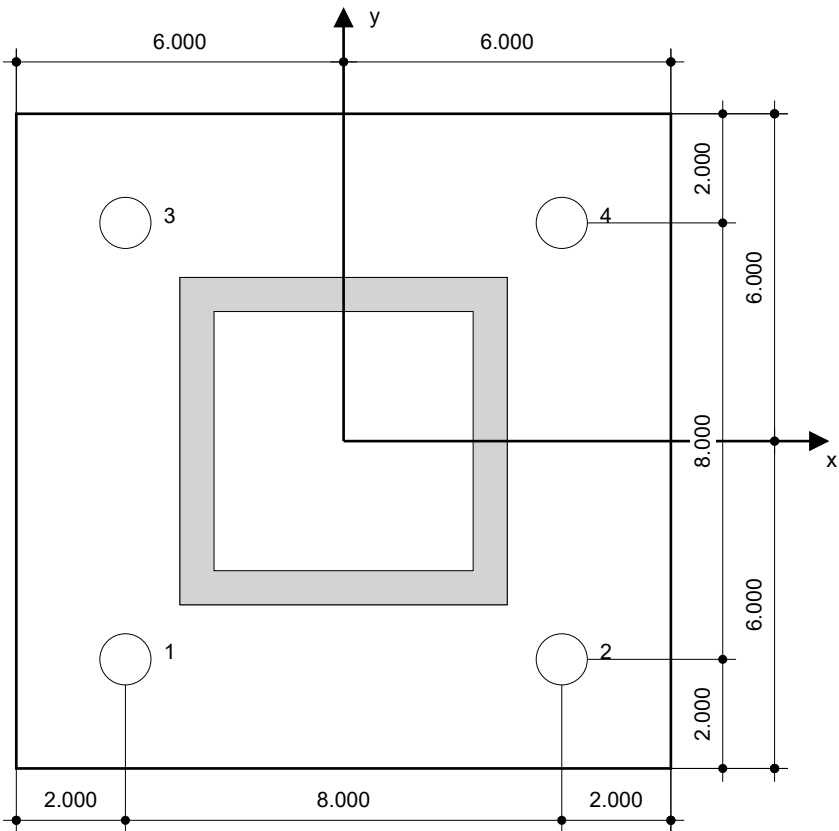
Fastening meets the design criteria!

7 Installation data

Profile: Square HSS (AISC), HSS6X6X.625; (L x W x T) = 6.000 in. x 6.000 in. x 0.625 in.
 Hole diameter in the fixture: $d_f = 0.938$ in.
 Plate thickness (input): 0.375 in.
 Recommended plate thickness: not calculated

Anchor type and diameter: Embed plate D1.1 GR. B 7/8
 Item number: not available
 Maximum installation torque: -
 Hole diameter in the base material: - in.
 Hole depth in the base material: 12.000 in.
 Minimum thickness of the base material: 12.875 in.

Ø 7/8 in Embed plate D1.1 GR. B with 12 in nominal embedment depth per Technical data , cast in place installation per MPII



Coordinates Anchor [in.]

Anchor	x	y	C _{-x}	C _{+x}	C _{-y}	C _{+y}
1	-4.000	-4.000	-	-	6.000	14.000
2	4.000	-4.000	-	-	6.000	14.000
3	-4.000	4.000	-	-	14.000	6.000
4	4.000	4.000	-	-	14.000	6.000



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Address:		Specifier:	
Phone Fax:		E-Mail:	
Design:	Concrete - Jul 3, 2025	Date:	7/3/2025
Fastening point:			

8 Remarks; Your Cooperation Duties

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17.12 Beam to Beam Connection

Project:
Project no:
Author:

Project item Beam to Beam

Design

Name Beam to Beam
Description
Analysis Stress, strain/ loads in equilibrium
Design code AISC 360-16 (LRFD) / ACI 318-14

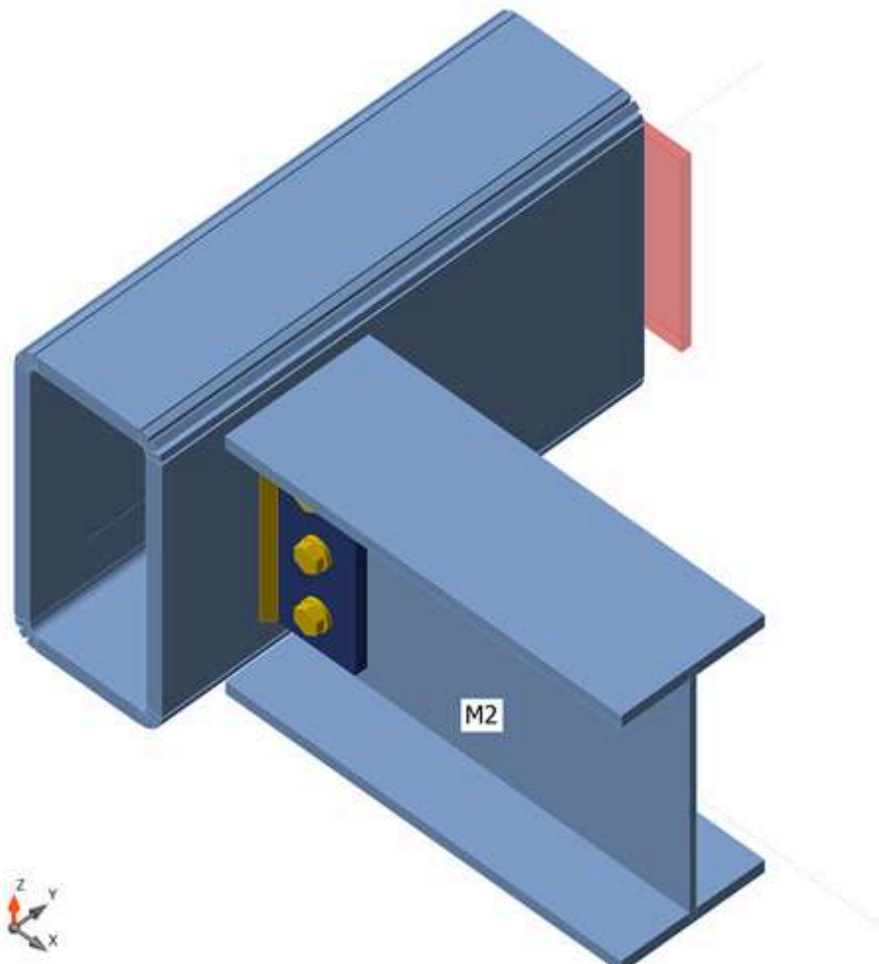
Members

Geometry

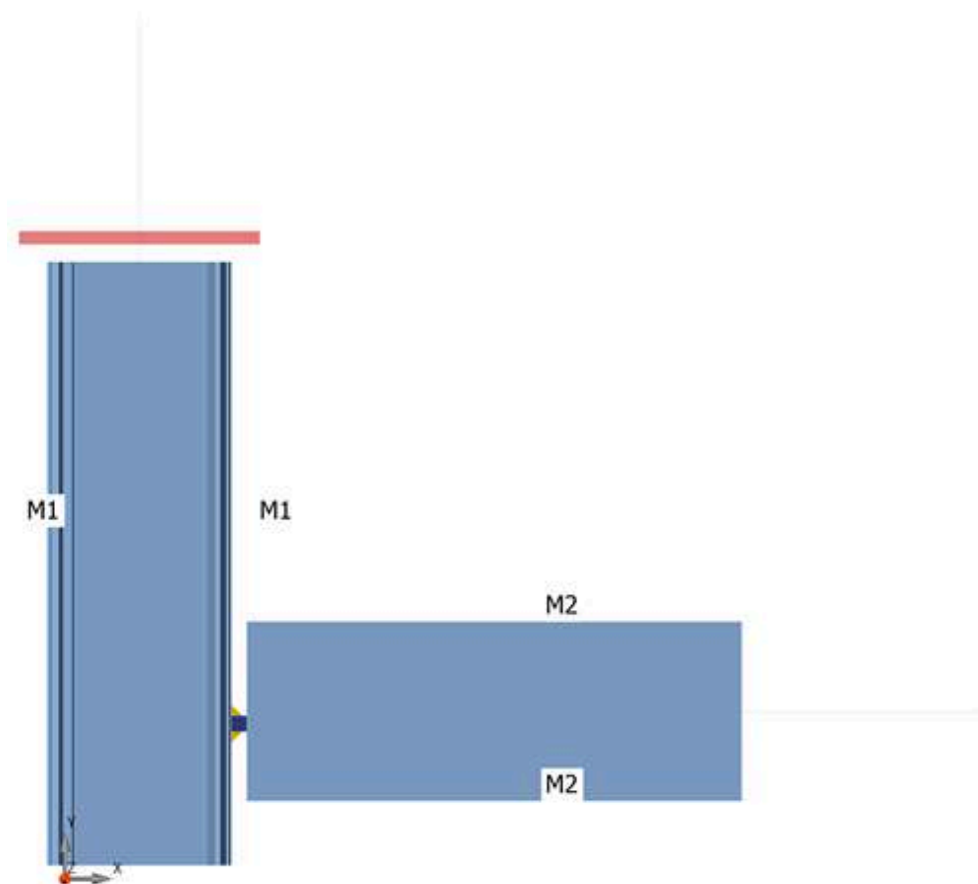
Name	Cross-section	β - Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [in]	Offset ey [in]	Offset ez [in]
M1	14 - HSS(Im)12X6X5/8	90.0	0.0	0.0	0"	0"	0"
M2	13 - W(Im)10X30	0.0	0.0	0.0	3"1/2	5"	0"

Supports and forces

Name	Support	Forces in	X [in]
M1 / end	N-Vy-Vz-Mx-My-Mz	Node	0"
M2 / end		Node	0"



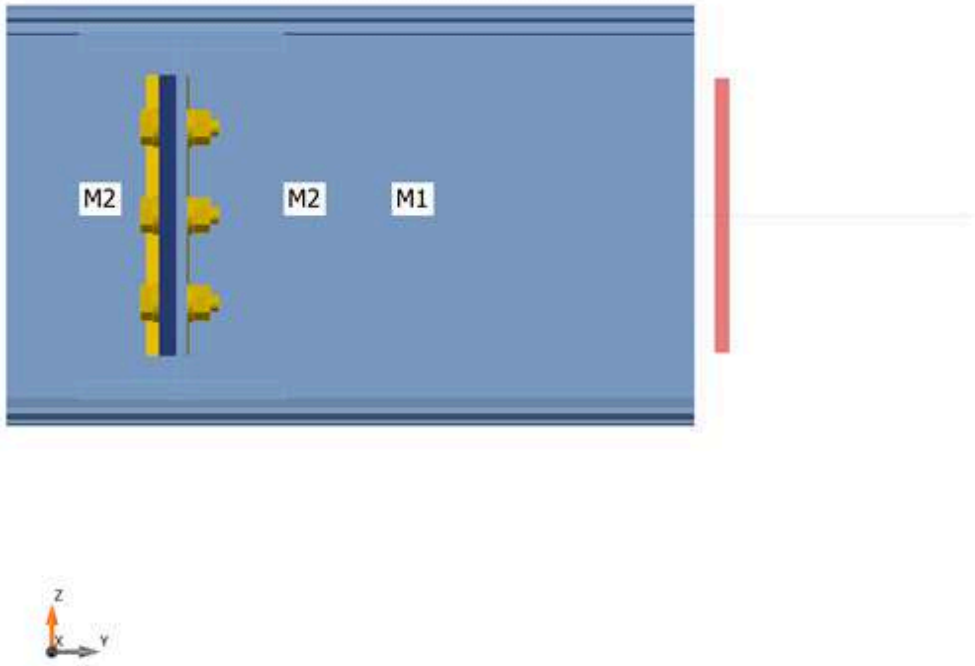
Project:
Project no:
Author:



Project:
Project no:
Author:



Project:
Project no:
Author:



Cross-sections

Name	Material
14 - HSS(Imp)12X6X5/8	A500. Gr. C
13 - W(Imp)10X30	A572 Gr.50

Project:
Project no:
Author:

Cross-sections

Name	Material	Drawing
14 - HSS(Imp)12X6X5/8	A500. Gr. C	
13 - W(Imp)10X30	A572 Gr.50	

Bolts

Name	Diameter [in]	f_y [kPa]	f_u [kPa]	Gross area [in ²]
1/2 A325	1/2	634317.7	827370.9	0.1964

Load effects (forces in equilibrium)

Name	Member	N [kip]	Vy [kip]	Vz [kip]	Mx [kip.ft]	My [kip.ft]	Mz [kip.ft]
LE1	M1 / End	0.00	0.00	0.00	0.00	0.00	0.00
	M2 / End	0.00	-3.50	0.00	0.00	0.00	0.00

Unbalanced forces

Name	X [kip]	Y [kip]	Z [kip]	Mx [kip.ft]	My [kip.ft]	Mz [kip.ft]
LE1	0.00	-3.50	0.00	0.00	0.00	-1.02

Project:
Project no:
Author:

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	5.3 < 100%	OK
Welds	21.8 < 100%	OK
Buckling	Not calculated	
GMNA	Calculated	

Plates

Name	Material	t_p [in]	Loads	σ_{Ed} [kPa]	ϵ_{pl} [%]	$\sigma_{c,Ed}$ [kPa]	Status
M1	A500. Gr. C	5/8	LE1	30647.9	0.0	0.0	OK
M2-bfl 1	A572 Gr.50	1/2	LE1	49552.5	0.0	0.0	OK
M2-tfl 1	A572 Gr.50	1/2	LE1	49561.6	0.0	0.0	OK
M2-w 1	A572 Gr.50	5/16	LE1	165528.5	0.0	13679.8	OK
SP1	A36	1/2	LE1	128682.1	0.0	68402.1	OK

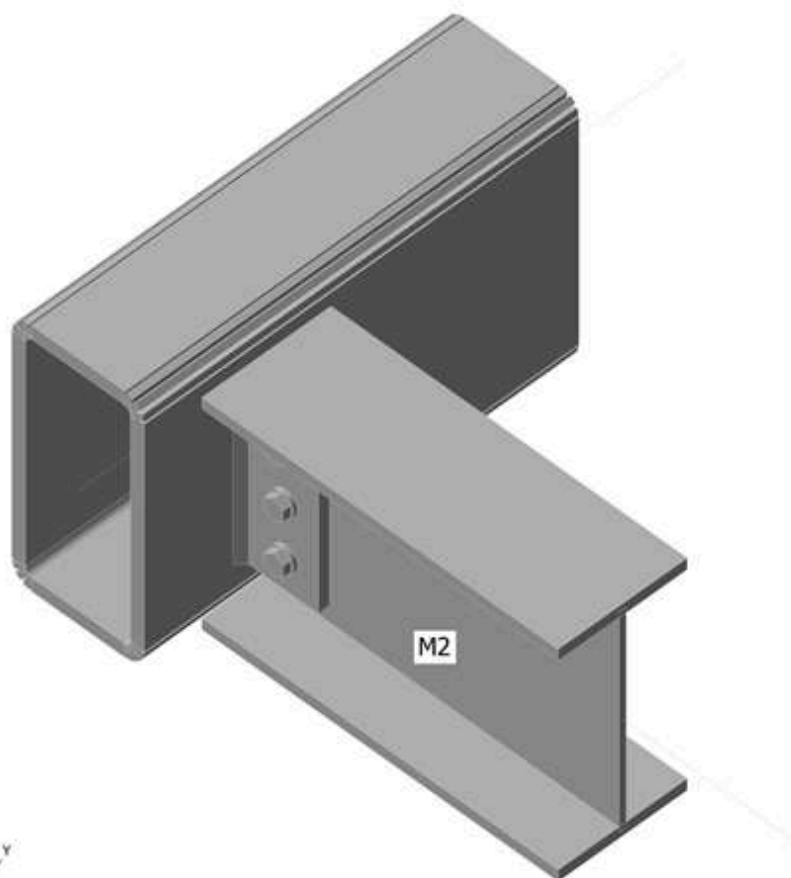
Design data

Material	F_y [kPa]	ϵ_{lim} [%]
A500. Gr. C	344737.9	5.0
A572 Gr.50	344740.0	5.0
A36	248210.0	5.0

Loc. deformation

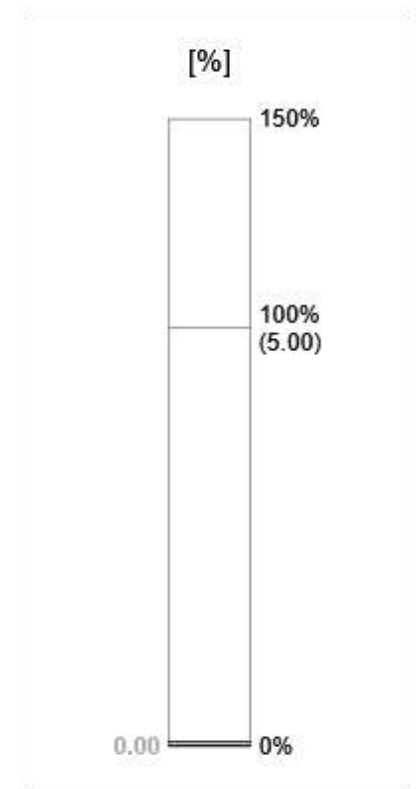
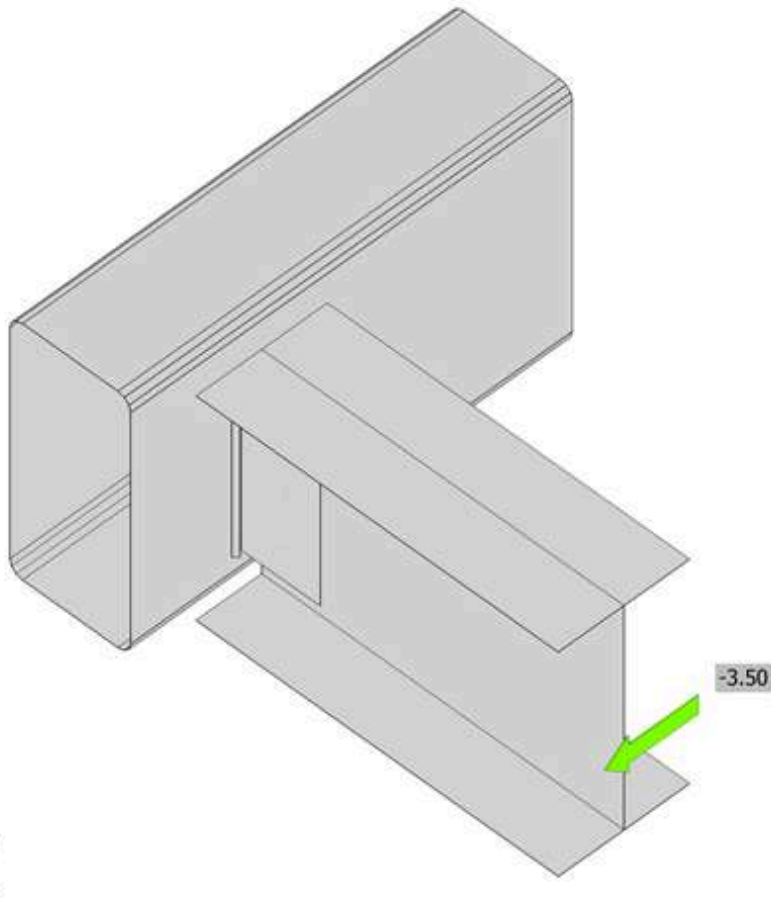
Name	d_0 [in]	Loads	δ [in]	δ_{lim} [in]	δ/d_0 [%]	Check status
M1	6"	LE1	0"	3/16	0.0	OK

Project:
Project no:
Author:



Overall check, LE1

Project:
Project no:
Author:



Strain check, LE1

Bolts

Shape	Item	Grade	Loads	F_t [kip]	V [kip]	$\phi R_{n,bearing}$ [kip]	U_{t_t} [%]	U_{t_s} [%]	$U_{t_{ts}}$ [%]	Status
	B4	1/2 A325 - 1	LE1	0.19	0.02	17.50	1.4	0.3	0.0	OK
	B5	1/2 A325 - 1	LE1	0.70	0.07	17.50	5.3	0.9	0.0	OK
	B6	1/2 A325 - 1	LE1	0.70	0.07	17.50	5.3	0.9	0.0	OK

Design data

Grade	$\phi R_{n,tension}$ [kip]	$\phi R_{n,shear}$ [kip]
1/2 A325 - 1	13.24	7.95

Welds

Item	Edge	Xu	t_w [in]	w [in]	L [in]	L_c [in]	Loads	F_n [kip]	ϕR_n [kip]	U_t [%]	U_c [%]	Status
M1-w 4	SP1	E70xx	▲ 1/4 ▼	▲ 3/8 ▼	7"15/16	1/2	LE1	1.24	5.73	21.7	9.5	OK
		E70xx	▲ 1/4 ▼	▲ 3/8 ▼	7"15/16	1/2	LE1	1.24	5.71	21.8	9.5	OK

Project:
Project no:
Author:

Design data

Material	F_{exx} [kPa]
E70xx	482633.0

Buckling

Buckling analysis was not calculated.

Bill of material

Manufacturing operations

Name	Plates [in]	Shape	Nr.	Welds [in]	Length [in]	Bolts	Nr.
SP1	P1/2x3"1/2-8" (A36)		1			1/2 A325	3

Welds

Type	Material	Throat thickness [in]	Leg size [in]	Length [in]
Double fillet	E70xx	1/4	3/8	8"

Bolts

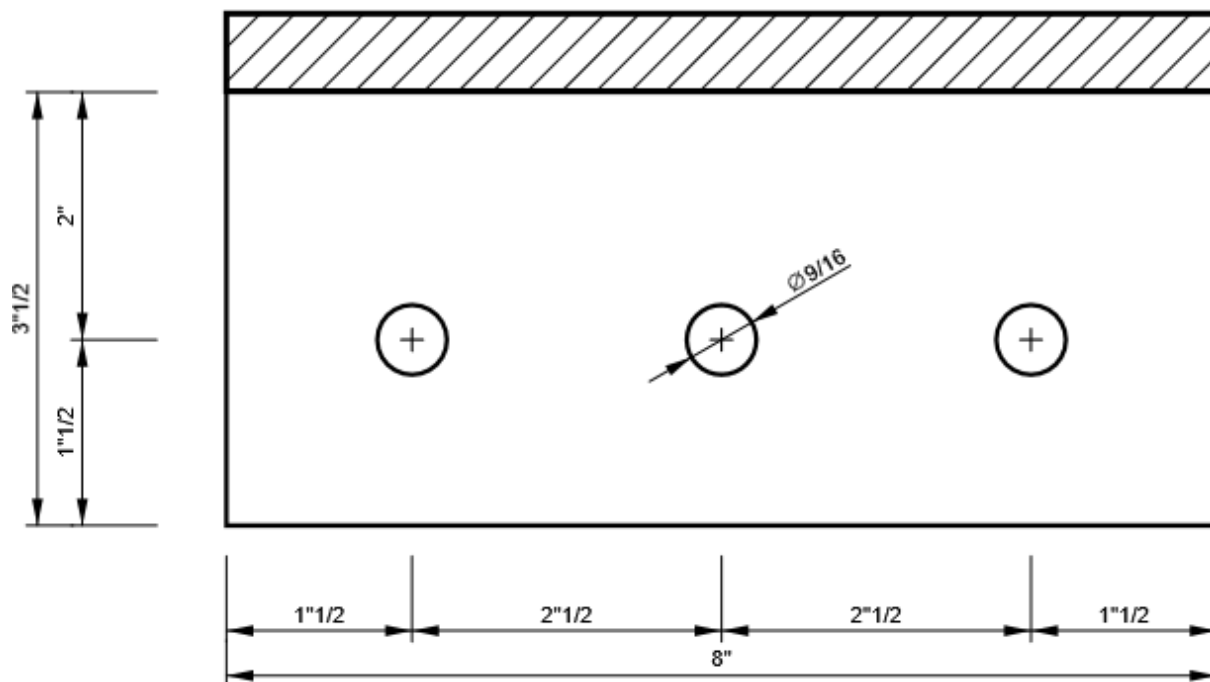
Name	Grip length [in]	Count
1/2 A325	13/16	3

Drawing

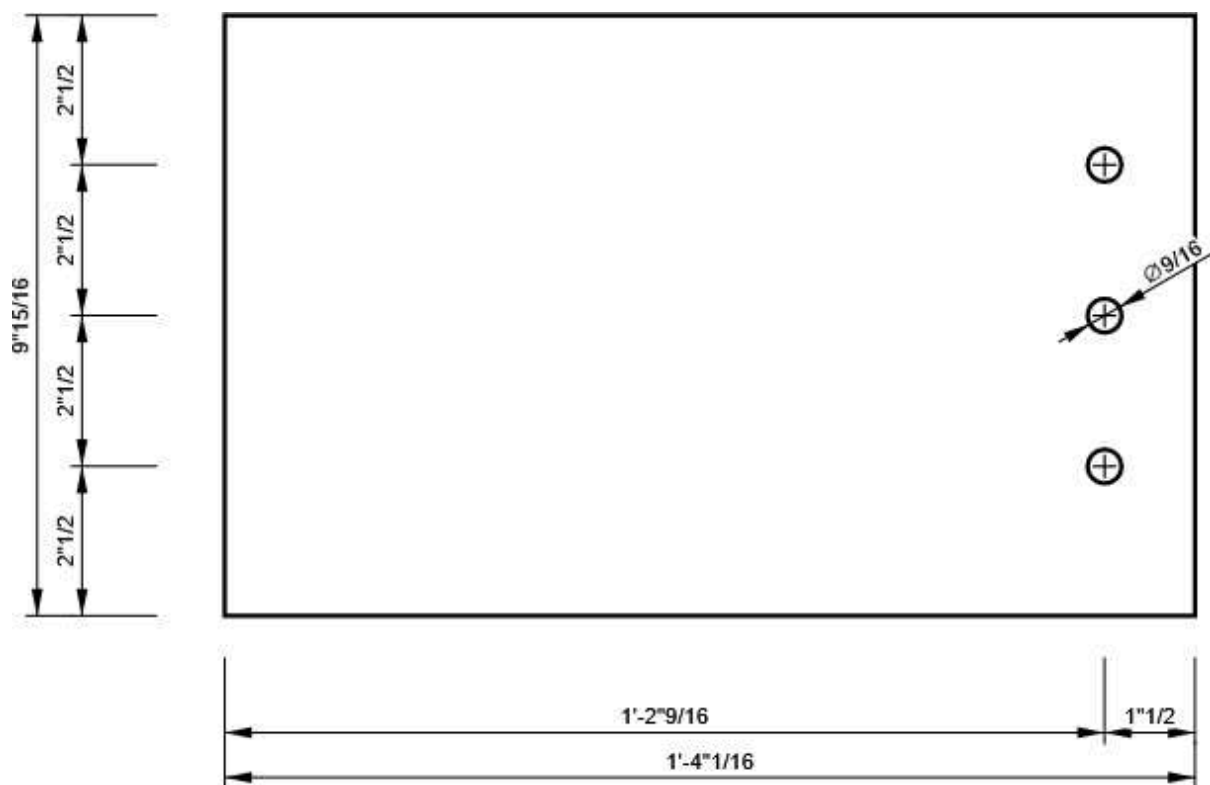
SP1

Project:
Project no:
Author:

P1/2x8"-3"1/2 (A36)



M2, W(lmp)10X30 - Web 1:



17.13 HSS column to beam connection

Project:
Project no:
Author:

Project item Beam to Column

Design

Name Beam to Column
Description
Analysis Stress, strain/ loads in equilibrium
Design code AISC 360-16 (LRFD) / ACI 318-14

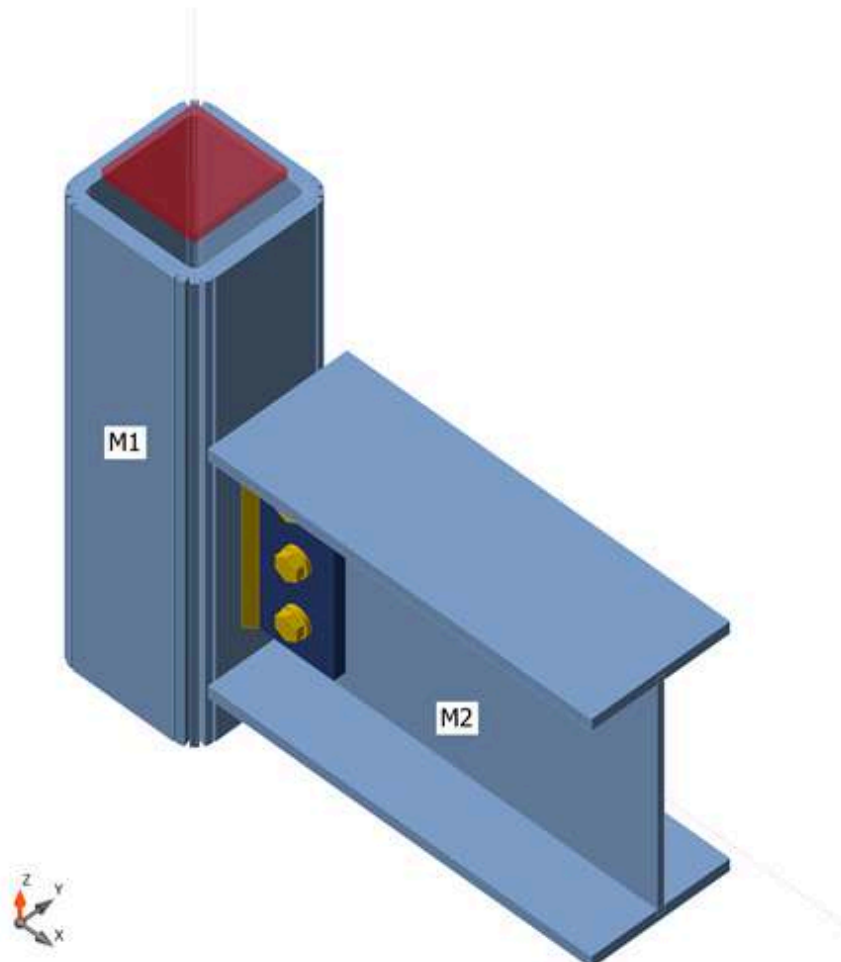
Members

Geometry

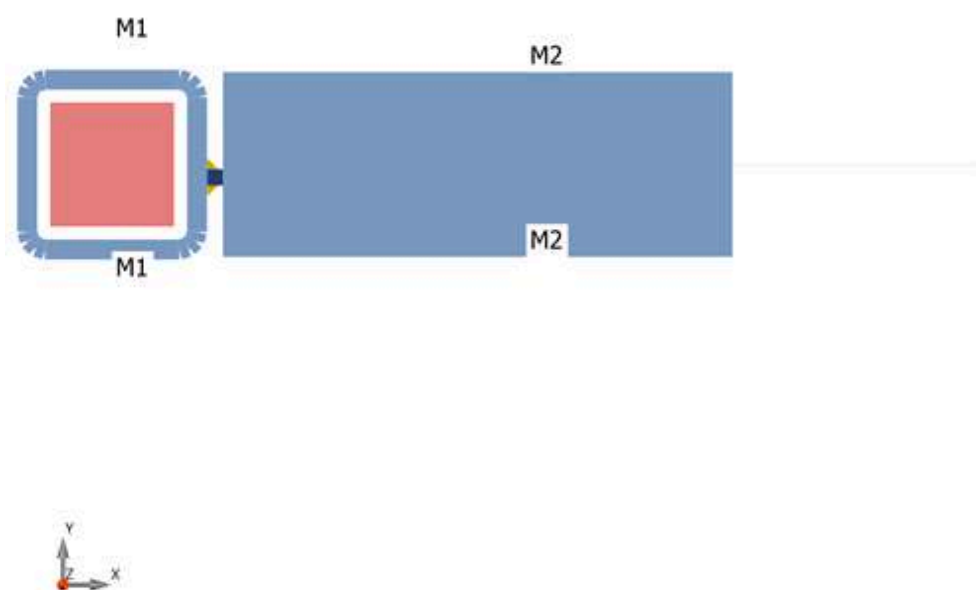
Name	Cross-section	β - Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [in]	Offset ey [in]	Offset ez [in]
M1	12 - HSS(Im)6X6X5/8	0.0	90.0	0.0	-3"	5"	0"
M2	13 - W(Im)10X30	0.0	0.0	0.0	3"1/2	5"	5"

Supports and forces

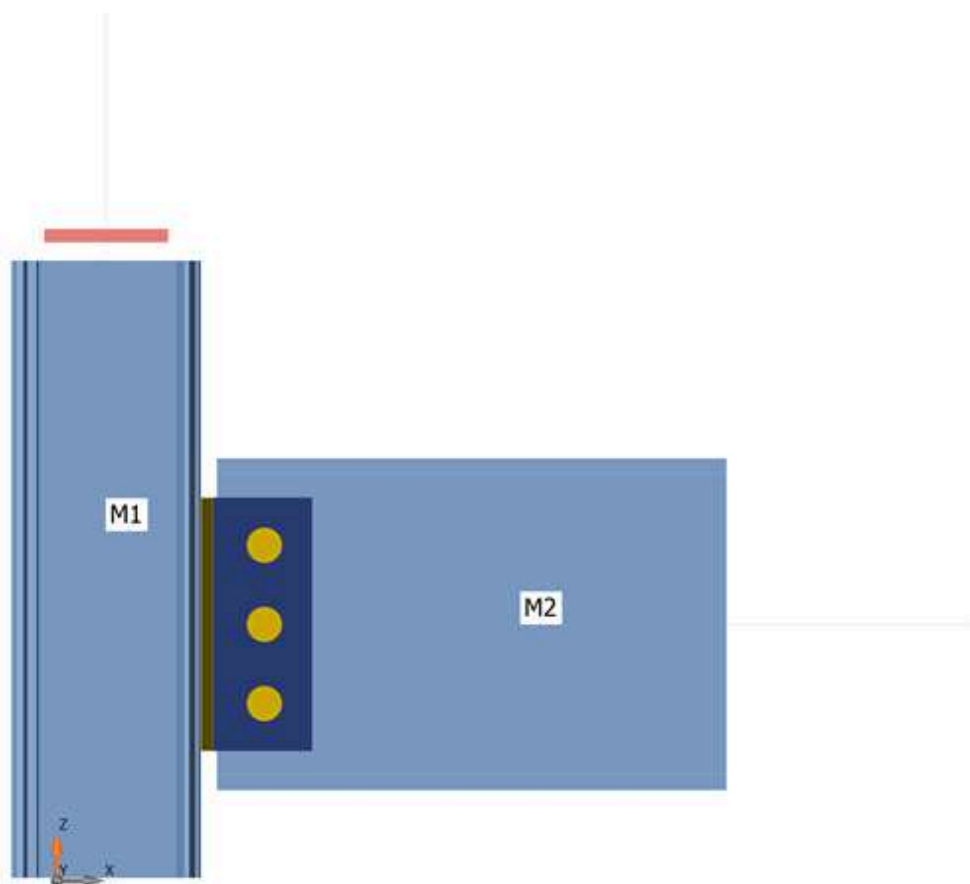
Name	Support	Forces in	X [in]
M1 / end	N-Vy-Vz-Mx-My-Mz	Node	0"
M2 / end		Node	0"



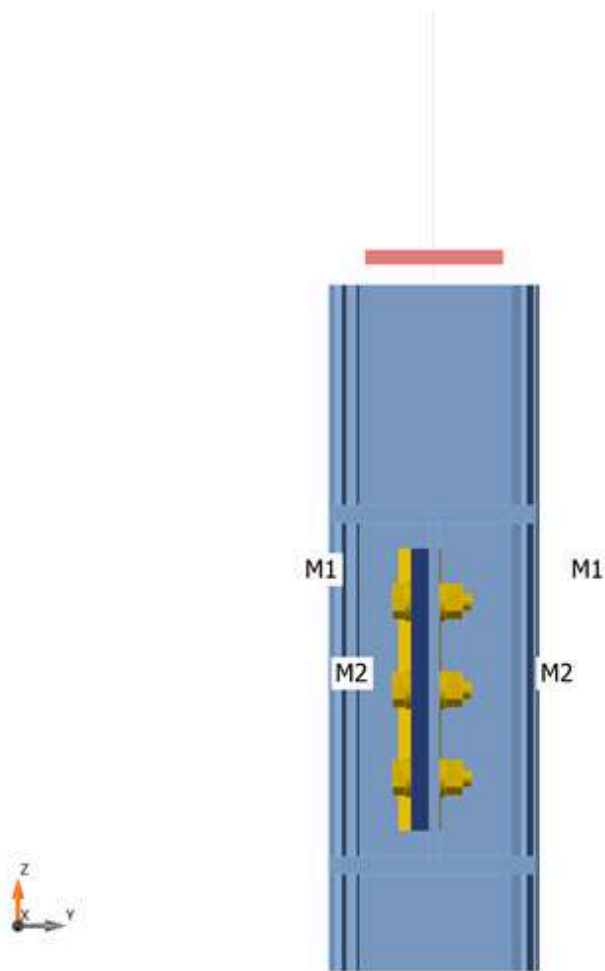
Project:
Project no:
Author:



Project:
Project no:
Author:



Project:
Project no:
Author:

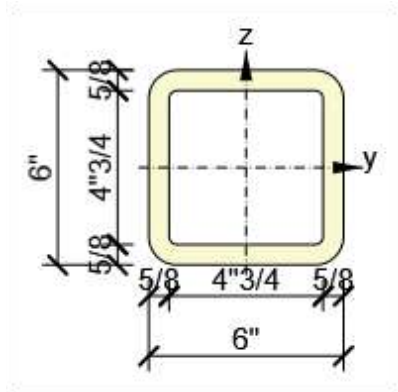
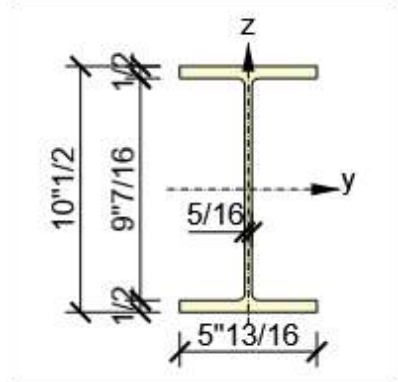


Cross-sections

Name	Material
12 - HSS(Imp)6X6X5/8	A500. Gr. C
13 - W(Imp)10X30	A572 Gr.50

Project:
Project no:
Author:

Cross-sections

Name	Material	Drawing
12 - HSS(Imp)6X6X5/8	A500. Gr. C	
13 - W(Imp)10X30	A572 Gr.50	

Bolts

Name	Diameter [in]	f_y [kPa]	f_u [kPa]	Gross area [in ²]
1/2 A325	1/2	634317.7	827370.9	0.1964

Load effects (forces in equilibrium)

Name	Member	N [kip]	Vy [kip]	Vz [kip]	Mx [kip.ft]	My [kip.ft]	Mz [kip.ft]
LE1	M1 / End	0.00	0.00	0.00	0.00	0.00	0.00
	M2 / End	0.00	-3.50	0.00	0.00	0.00	0.00

Unbalanced forces

Name	X [kip]	Y [kip]	Z [kip]	Mx [kip.ft]	My [kip.ft]	Mz [kip.ft]
LE1	0.00	-3.50	0.00	1.46	0.00	-1.02

Project:
Project no:
Author:

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	5.3 < 100%	OK
Welds	23.4 < 100%	OK
Buckling	Not calculated	
GMNA	Calculated	

Plates

Name	Material	t_p [in]	Loads	σ_{Ed} [kPa]	ϵ_{pl} [%]	$\sigma_{c,Ed}$ [kPa]	Status
M1	A500. Gr. C	5/8	LE1	42524.8	0.0	0.0	OK
M2-bfl 1	A572 Gr.50	1/2	LE1	49657.6	0.0	0.0	OK
M2-tfl 1	A572 Gr.50	1/2	LE1	49520.4	0.0	0.0	OK
M2-w 1	A572 Gr.50	5/16	LE1	165550.9	0.0	13707.5	OK
SP1	A36	1/2	LE1	123826.8	0.0	68443.7	OK

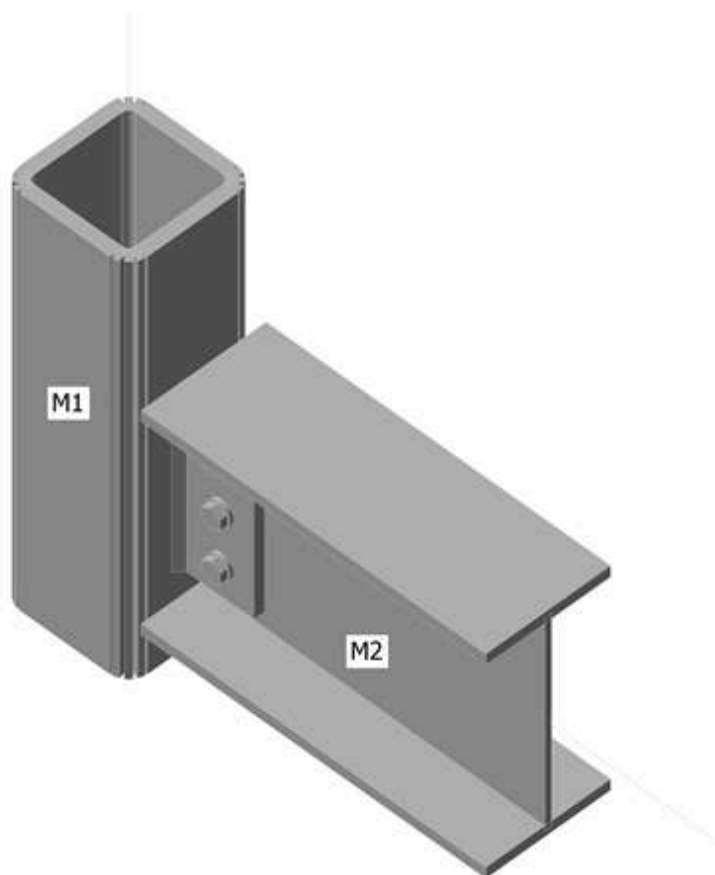
Design data

Material	F_y [kPa]	ϵ_{lim} [%]
A500. Gr. C	344737.9	5.0
A572 Gr.50	344740.0	5.0
A36	248210.0	5.0

Loc. deformation

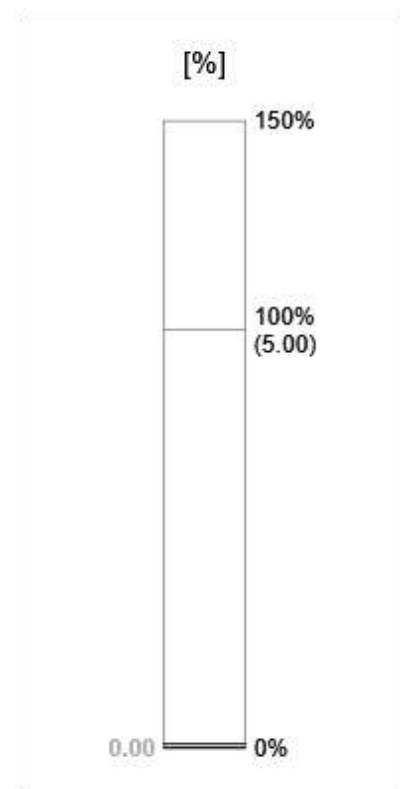
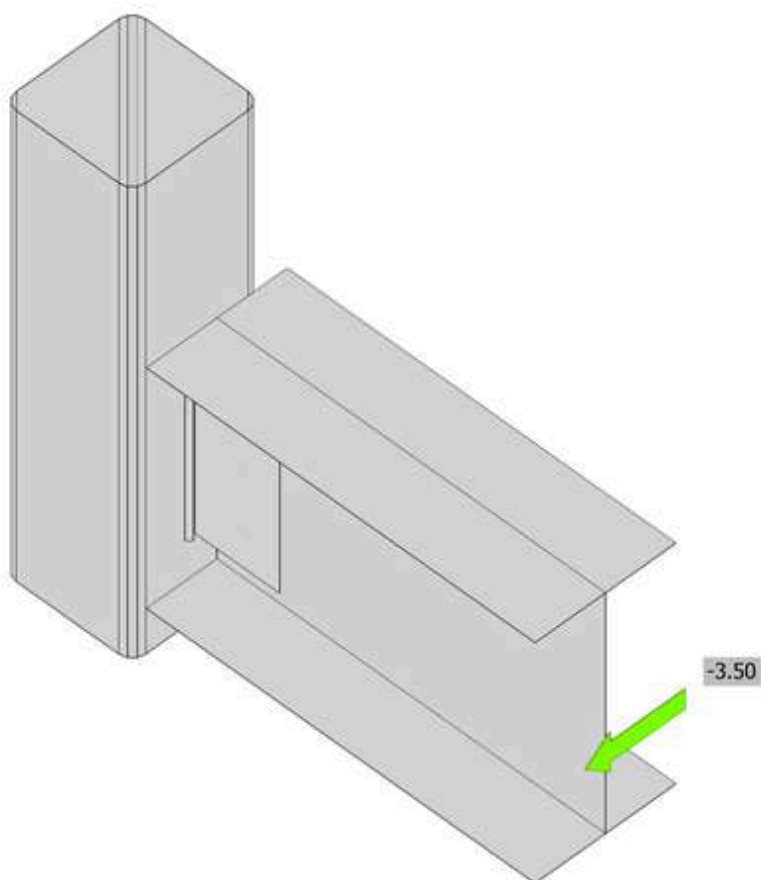
Name	d_0 [in]	Loads	δ [in]	δ_{lim} [in]	δ/d_0 [%]	Check status
M1	6"	LE1	0"	3/16	0.0	OK

Project:
Project no:
Author:



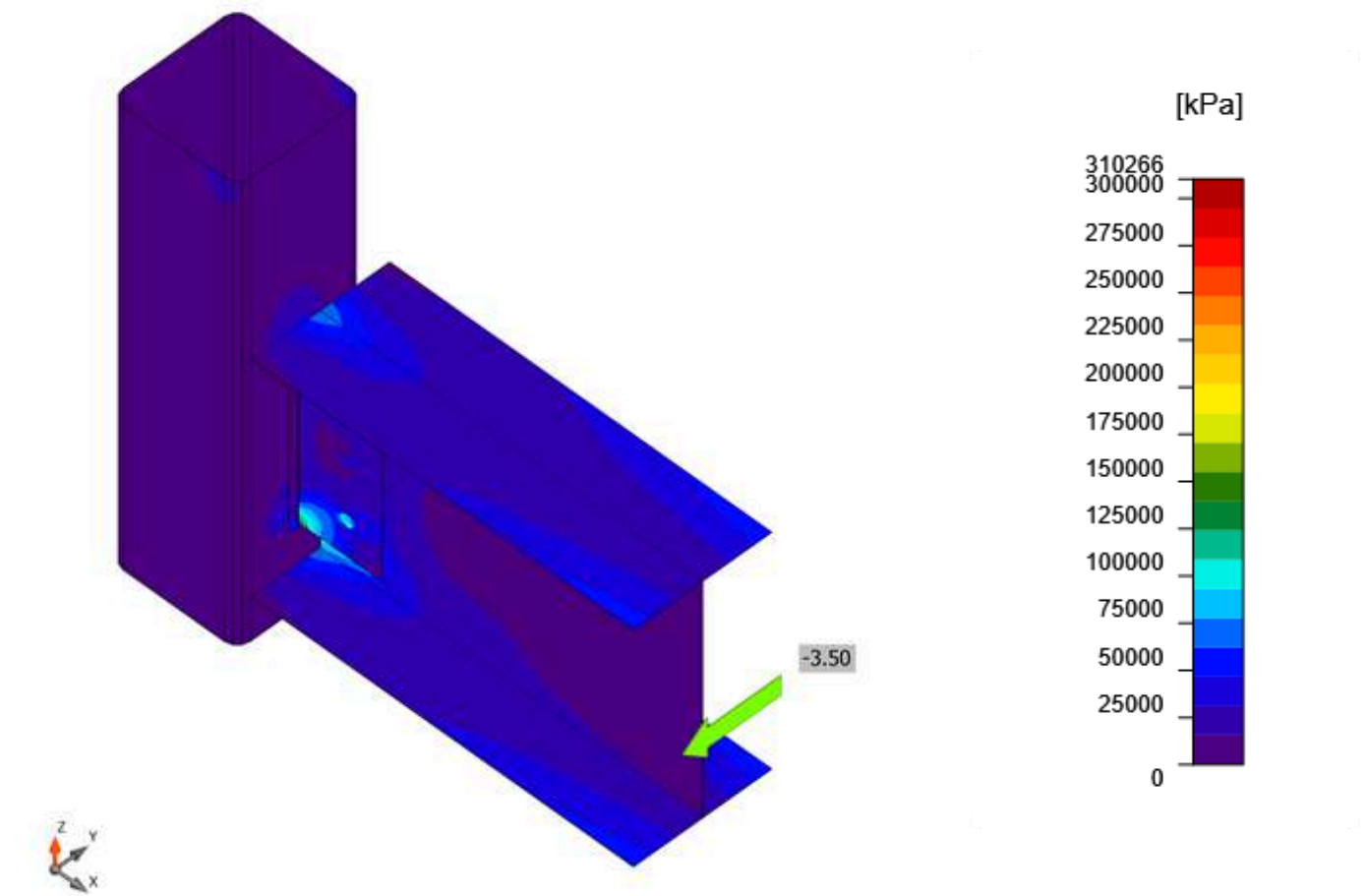
Overall check, LE1

Project:
Project no:
Author:



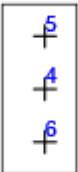
Strain check, LE1

Project:
Project no:
Author:



Equivalent stress, LE1

Bolts

Shape	Item	Grade	Loads	F_t [kip]	V [kip]	$\phi R_{n,bearing}$ [kip]	U_{t_t} [%]	U_{t_s} [%]	$U_{t_{ts}}$ [%]	Status
	B4	1/2 A325 - 1	LE1	0.19	0.02	17.50	1.5	0.3	0.0	OK
	B5	1/2 A325 - 1	LE1	0.70	0.07	17.50	5.3	0.9	0.0	OK
	B6	1/2 A325 - 1	LE1	0.70	0.08	17.50	5.3	1.0	0.0	OK

Design data

Grade	$\phi R_{n,tension}$ [kip]	$\phi R_{n,shear}$ [kip]
1/2 A325 - 1	13.24	7.95

Welds

Item	Edge	Xu	t_w [in]	w [in]	L [in]	L_c [in]	Loads	F_n [kip]	ϕR_n [kip]	U_t [%]	U_{t_c} [%]	Status
M1-w 1	SP1	E70xx	▲ 1/4 ▼	▲ 3/8 ▼	7"15/16	3/8	LE1	1.05	4.51	23.3	10.3	OK
		E70xx	▲ 1/4 ▼	▲ 3/8 ▼	7"15/16	3/8	LE1	1.05	4.51	23.4	10.3	OK

Project:
Project no:
Author:

Design data

Material	F_{exx} [kPa]
E70xx	482633.0

Buckling

Buckling analysis was not calculated.

Bill of material

Manufacturing operations

Name	Plates [in]	Shape	Nr.	Welds [in]	Length [in]	Bolts	Nr.
SP1	P1/2x3"1/2-8" (A36)		1			1/2 A325	3

Welds

Type	Material	Throat thickness [in]	Leg size [in]	Length [in]
Double fillet	E70xx	1/4	3/8	8"

Bolts

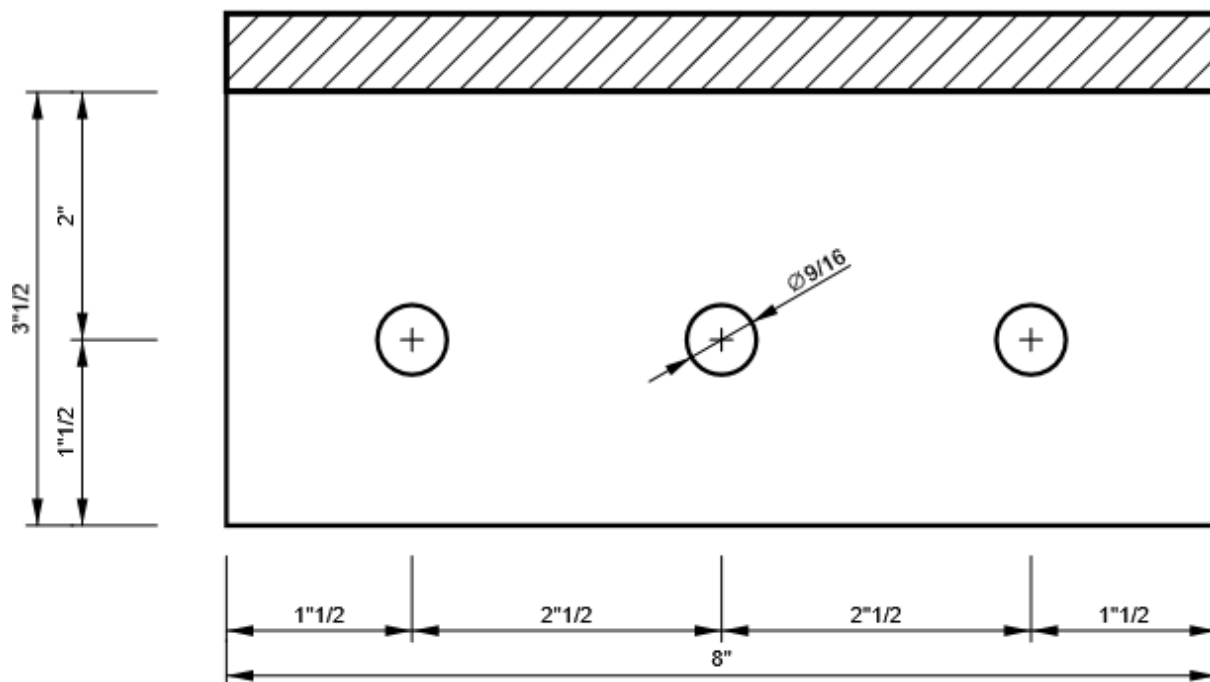
Name	Grip length [in]	Count
1/2 A325	13/16	3

Drawing

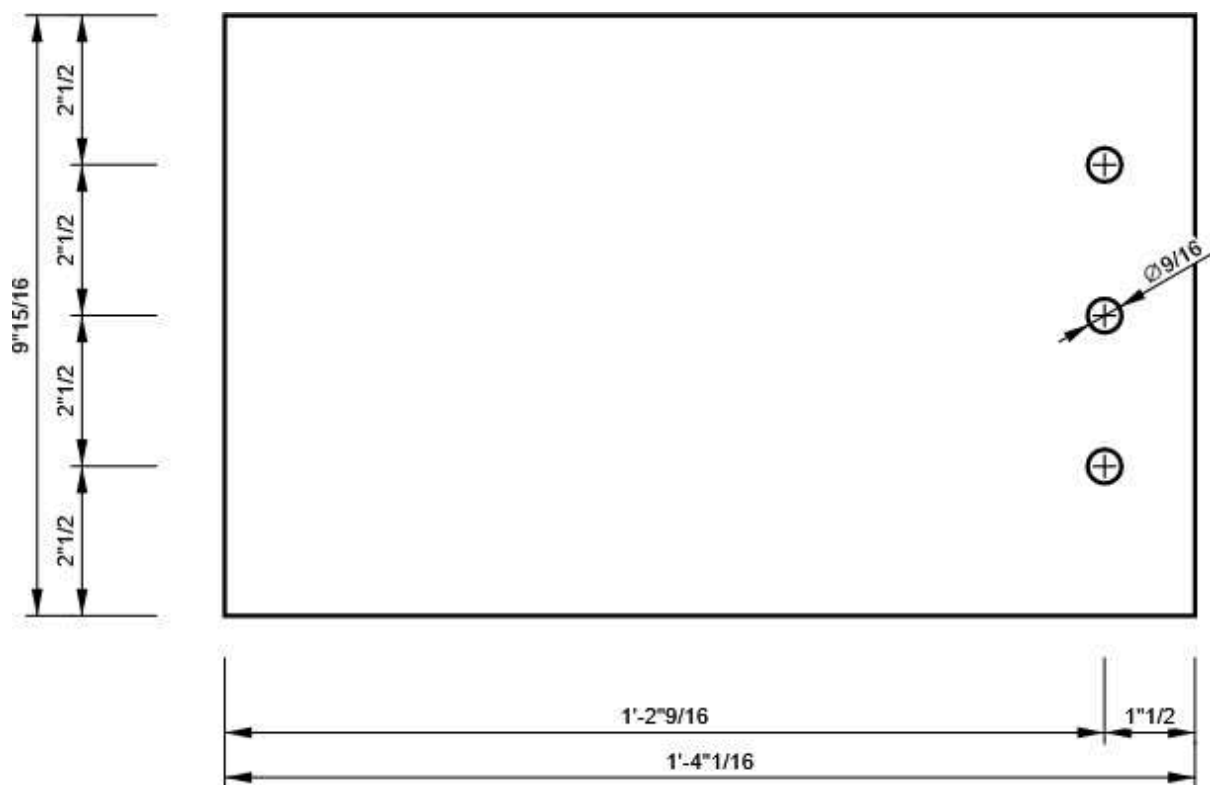
SP1

Project:
Project no:
Author:

P1/2x8"-3"1/2 (A36)



M2, W(lmp)10X30 - Web 1:



16. Lateral Design

Wind Anyalysis and Design

Primary Structure design analysis:

Max Leeward and Windward Interior = 98.97 Psf Max

Vertical Shear Force Calculations for Right Building 1												
Floor	Length (E-W)	Width (N-S)	Floor Height	Wind Pressure	Wind Pressure	Wind Force (E-W)	Wind Force (N-S)	LOAD FACTOR	Cumulative Wind Force (E-W), Kips	Cumulative Wind Force(N-S), Kips		
	Ft	Ft	Ft	Psf	Psf	lb	lb		kips	kips		
2	60	80	9.00	98.97	98.97	53444	71258	1.0	53	71		
1	60	80	11.00	98.97	98.97	65320	87094	1.0	119	158		
Total						53444	71258		53	71		

LATERAL DESIGN

PROJECT TITLE				
ROOF DEAD LOAD	30	ROOF LIVE LOAD		20
FLOOR DEAD LOAD	25	FLOOR LIVE LOAD		40
WALL LOAD	30			

X-Strap capacity

6x43 MIL

8 Kips

X-Strap capacity

6x54 MIL

10 Kips

X-Strap capacity

8x54 MIL

14 Kips

E	29000 ksi						
#12 Screw capacity	2.41 kips						

2ND FLOOR (SHEAR LINE)	TRIBUTARY HEIGHT (ft)	TRIBUTARY WIDTH (ft)	LOAD DISTRIBUTION ON RATIO (%)	HEIGHT OF SHEAR WALL (ft)	SHEAR WALL MIN WIDTH(ft)	SHEAR FORCE DUE TO WIND (KIPS)	SHEAR FORCE DUE TO EQ. (KIPS)	ANGLE	NUMBER OF SHEAR WALLS PER PANEL	WL SHEAR FORCE PER PANEL (KIPS)	NUMBER OF STRAPS ADDED	STRAP TYPE	STRAP/SHEATING TYPE	WIND STRAP CAPACITY(KIPS)	UTILIZATION % (WIND)	DUE TO WIND VERTICAL FORCE T = C	LOADING TRIB.	GRAVITY FORCE (kips)	COLLECTOR GRAVITY/ STUD (kip) WL	HOLD DOWN (KIPS) WL	HOLD DOWN CAPACITY (KIPS)	UTILIZATION % (WL)	STUD COMPRESSION (KIPS)	BOUNDARY POST CAPACITY(KIPS)	UTILIZATION BOUNDARY POST	BOUNDARY POST SPECIFICATION
A	7.00	19.33	0.32	9.00	10	22.77	0	0	4	5.69	4	SINGLE	6x54 MIL	7.80	72.95%	5.12	2	0.09	5.21	5.03	6.97 HTT5	72.20%	5.21	16.00	32.58%	(2)600S16254
F	7.00	29.00	0.48	9.00	8	34.16	0	0	5	6.83	8	DOUBLE	6x54 MIL	13.95	48.98%	7.69	2	0.09	7.78	7.60	13.98 HD8S	54.33%	7.78	16.00	48.60%	(2)600S16254
G	7.00	12.17	0.20	9.00	9	14.33	0	0	2	7.17	4	DOUBLE	6x43 MIL	14.85	48.28%	7.17	2	0.09	7.26	7.08	13.98 HD8S	50.62%	7.26	16.00	45.36%	(2)600S16254
423.50			1.00	27		71.26	0.00																			

2ND FLOOR (SHEAR LINE)	TRIBUTARY HEIGHT (ft)	TRIBUTARY WIDTH (ft)	LOAD DISTRIBUTION ON RATIO (%)	HEIGHT OF SHEAR WALL (ft)	SHEAR WALL MIN WIDTH(ft)	SHEAR FORCE DUE TO WIND (KIPS)	SHEAR FORCE DUE TO EQ. (KIPS)	ANGLE	NUMBER OF SHEAR WALLS PER PANEL	WL SHEAR FORCE PER PANEL (KIPS)	NUMBER OF STRAPS ADDED	STRAP TYPE	STRAP/SHEATING TYPE	WIND STRAP CAPACITY(KIPS)	UTILIZATION % (WIND)	DUE TO WIND VERTICAL FORCE T = C	LOADING TRIB.	GRAVITY FORCE (kips)	COLLECTOR GRAVITY/ STUD (kip) WL	HOLD DOWN (KIPS) WL	HOLD DOWN CAPACITY NOMINAL (KIPS) EQ	UTILIZATION % (WL)	STUD COMPRESSION (KIPS)	BOUNDARY POST CAPACITY(KIPS)	UTILIZATION BOUNDARY POST	BOUNDARY POST SPECIFICATION
1	7.00	9	0.15	9.00	7	8.20	0	0	2	4.10	2	SINGLE	6x54 MIL	6.44	63.64%	5.27	2	0.09	5.36	5.18	6.97 HTTS	74.36%	5.36	9.00	59.59%	(2)600S16243
3	7.00	22	0.36	9.00	6	19.18	0	0	3	6.39	3	SHEAR WALL + MOMENT FRAME														
4	7.00	20	0.33	9.00	10	17.78	0	0	2	8.89	2	SHEAR WALL + MOMENT FRAME														
5	7.00	10	0.15	9.00	9	8.28	0	0	3	2.76	3	SINGLE	6x43 MIL	5.91	46.70%	2.76	2	0.09	2.85	2.67	6.97 HTTS	38.31%	2.85	9.00	31.67%	(2)600S16243
429.17			1.00	32		53.44	0.00																			

1ST FLOOR (SHEAR LINE)	TRIBUTARY HEIGHT (ft)	TRIBUTARY WIDTH (ft)	LOAD DISTRIBUTION ON RATIO (%)	HEIGHT OF SHEAR WALL (ft)	SHEAR WALL MIN WIDTH(ft)	SHEAR FORCE DUE TO WIND (KIPS)	SHEAR FORCE DUE TO EQ. (KIPS)	ANGLE	NUMBER OF SHEAR WALLS PER PANEL	WL SHEAR FORCE PER PANEL (KIPS)	NUMBER OF STRAPS ADDED	STRAP TYPE	STRAP/SHEATING TYPE	WIND STRAP CAPACITY(KIPS)	UTILIZATION % (WIND)	DUE TO WIND VERTICAL FORCE T = C	LOADING TRIB.	GRAVITY FORCE (kips)	COLLECTOR GRAVITY/ STUD (kip) WL	HOLD DOWN (KIPS) WL	HOLD DOWN CAPACITY NOMINAL (KIPS) EQ	UTILIZATION % (WL)	STUD COMPRESSION (KIPS)	BOUNDARY POST CAPACITY(KIPS)	UTILIZATION BOUNDARY POST	BOUNDARY POST SPECIFICATION
A	12.50	19.33	0.32	11.00	10	50.59	0	0	4	12.65	8	DOUBLE	6x54 MIL	14.12	89.56%	13.91	2	0.09	14.00	18.86	31.00 S/HD10S	60.83%	19.22	30.50	63.00%	2X(2)600S16254
F	12.50	29.00	0.48	11.00	8	75.90	0	0	5	15.18	8	DOUBLE	8x54 MIL	16.47	92.20%	20.87	2	0.09	20.96	28.38	31.00 S/HD10S	91.55%	28.74	30.50	94.23%	2X(2)600S16254
G	12.50	12.17	0.20	11.00	9	31.85	0	0	2	15.93	4	DOUBLE	8x54 MIL	17.73	89.85%	19.47	2	0.09	19.56	26.45	31.00 S/HD10S	85.33%	26.81	30.50	87.91%	2X(2)600S16254
756.25			1.00	27		158.35	0.00																			

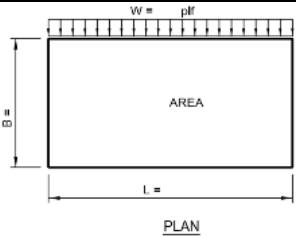
1ST FLOOR (SHEAR LINE)	TRIBUTARY HEIGHT (ft)	TRIBUTARY WIDTH (ft)	LOAD DISTRIBUTION ON RATIO (%)	HEIGHT OF SHEAR WALL (ft)	SHEAR WALL MIN WIDTH(ft)	SHEAR FORCE DUE TO WIND (KIPS)	SHEAR FORCE DUE TO EQ. (KIPS)	ANGLE	NUMBER OF SHEAR WALLS PER PANEL	WL SHEAR FORCE PER PANEL (KIPS)	NUMBER OF STRAPS ADDED	STRAP TYPE	STRAP/SHEATING TYPE	WIND STRAP CAPACITY(KIPS)	UTILIZATION % (WIND)	DUE TO WIND VERTICAL FORCE T = C	LOADING TRIB.	GRAVITY FORCE (kips)	COLLECTOR GRAVITY/ STUD (kip) WL	HOLD DOWN (KIPS) WL	HOLD DOWN CAPACITY NOMINAL (KIPS) EQ	UTILIZATION % (WL)	STUD COMPRESSION (KIPS)	BOUNDARY POST CAPACITY(KIPS)	UTILIZATION BOUNDARY POST	BOUNDARY POST SPECIFICATION
1	12.50	9	0.15	11.00	7	24.30	0	0	2	12.15	4	DOUBLE	8x54 MIL	15.03	80.86%	19.10	5	0.225	19.32	24.05	31.00 S/HD10S	77.59%	24.68	30.50	80.93%	2X(2)600S16254
3	12.50	22	0.36	11.00	6	56.82	0	0	3	18.94	6	SHEAR WALL + MOMENT FRAME														
4	12.50	20	0.33	11.00	10	52.69	0	0	2	26.34	4	SHEAR WALL + MOMENT FRAME														
5	12.50	10	0.15	11.00	9	24.54	0	0	3	8.18	6	DOUBLE	6x43 MIL	10.59	77.26%	10.00	5	0.225	10.22	12.44	31.00 S/HD10S	78.45%	10.22	16.00	63.88%	(2)600S16254
		766.38		1.00				23		133.82		0.00														

Diaphragm Analysis and Design

Diaphragm Design

Maximum size of roof/floor diaphragm is 20 ft x 20 ft

Floor Spacing
2.00 ft
Width
20.00 ft



Floor Spacing	Diaphragm Length
2.00 ft	20.00 ft

20

Uniform wind load (w) (N-S)	989.7 Plf	(Governing force in EW direction)
Uniform wind load (w) (E-W)	989.7 Plf	(Governing force in L-R direction)

Daiphragm design:

Using 9/16" MMD FORM DECK AT ROOF AND FLOOR

Floor Deck Capacity = 2225.0 Plf Strength

2225 plf > 989.7 plf, Hence O.K.

Fastener design:

Maximum Diaphragm Forces = 990 plf
Maximum shear force along edges = 495 lb/ft

Using min #12 screws per feet
Shear capacity of #12 screws = 854 lb

854 lb > 495 lb, Hence O.K.